

News Shocks and Inflation: Theory and Evidence from Large Oil Discoveries ^{*}

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Abstract

We study how news shocks affect inflation in open economies, using large oil and gas discoveries with realized production lags of at least five years as observable news about future output. Panel local projections show that discoveries raise core consumer prices over the subsequent decade, with about half the increase occurring prior to extraction. The response is driven by nontradables, while tradable and producer prices barely respond. The real exchange rate appreciates persistently and real activity expands ahead of extraction, while the current account first deteriorates and then improves as new oil comes online. The inflationary response is concentrated under inflexible exchange-rate regimes but muted under flexible ones. A three-sector New Keynesian small open-economy model rationalizes these findings through competing demand and exchange-rate channels, with exchange-rate policy determining which force dominates.

Keywords: News shocks, inflation, oil discoveries, New Keynesian model, small open economy, exchange-rate regimes

JEL: E31, E32, E52, F41, Q32

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1 Introduction

How does news about future fundamentals affect inflation today? In the canonical New Keynesian model, inflation is forward-looking: with sticky prices, firms set prices in response not only to current marginal costs, but also to expected future demand and cost conditions (Rotemberg, 1982; Calvo, 1983). News that shifts expectations about future fundamentals can therefore move inflation before those fundamentals are realized. Since Beaudry and Portier (2006), a large literature on news-driven business cycles has shown that news shocks can generate sizable movements in real activity—including output, consumption, investment, employment, and external balances (Jaimovich and Rebelo, 2009; Schmitt-Grohé and Uribe, 2012; Mertens and Ravn, 2012; Arezki et al., 2017). Much less is known about their inflationary effects. Existing theory gives ambiguous predictions, and empirical evidence remains mixed (Lorenzoni, 2009; Barsky et al., 2015). The key challenge in the literature is the identification problem: news shocks are rarely observed directly, and existing studies must infer them from identifying restrictions or structural models, so the results are inherently sensitive to those assumptions.

This paper provides theory and empirical evidence on how news shocks affect inflation in open economies. Our empirical strategy exploits plausibly exogenous variation in the timing of large oil and gas discoveries around the world as a directly observable measure of news about higher future output. A discovery reveals information about future resource income and output, but production usually begins only after a substantial development lag. Moreover, the timing of discoveries is shaped by geological uncertainty, making discoveries plausibly exogenous to contemporaneous macroeconomic conditions (Arezki et al., 2017). These features make oil and gas discoveries a useful laboratory for studying news shocks.

We construct a new dataset of 1,309 large oil and gas discoveries, defined as fields with reserves of at least 100 million barrels of oil equivalent and a realized production lag of at least five years, covering 79 economies over 1980–2019. The data are drawn from the Rystad Energy upstream database and substantially expand the coverage of oil and gas discoveries used in the existing literature.¹ An important advantage of the Rystad Energy database is

¹Arezki et al. (2017) use the giant oil and gas discoveries compiled by Horn (2014), defined as discoveries with ultimate recoverable reserves of at least 500 million barrels of oil equivalent. Under this definition, there

that it reports both the discovery year and the production-start year for each field. This allows us to observe field-specific realized production lags and to distinguish genuine news shocks from discoveries that quickly translate into current production. In our baseline analysis, we therefore exclude discoveries with realized production lags shorter than five years. This restriction ensures that the estimated inflation responses in the pre-production window capture the effects of news about future resource income rather than contemporaneous increases in productive capacity.

We measure the economic size of each discovery by its net present value (NPV) as a share of GDP, and aggregate to the country-year level. We combine these data with core consumer and producer price indexes (CPI and PPI) from the World Bank’s Global Database of Inflation (GDI), sectoral price deflators and bilateral real exchange rates from the United Nations National Accounts Statistics (UNNAS), and macroeconomic aggregates from standard international sources. We estimate dynamic responses using panel local projections with country and year fixed effects ([Jordà, 2005](#)), which absorb time-invariant country characteristics and common global shocks.

The empirical analysis delivers three main findings. First, oil-discovery news raises consumer prices but has little effect on producer prices. A median-sized discovery, with an estimated NPV equal to 5 percent of GDP, raises core CPI by about 1.13 percent cumulatively over the decade following the discovery, and about half of this increase—roughly 0.57 percent—materializes within four years, before new oil comes online. In contrast, the response of the PPI is small and statistically insignificant. This divergence reflects the sectoral composition of the two indices. Consumer prices place substantial weight on nontradable goods and services, whose prices are determined largely by domestic demand and costs. Producer prices are more concentrated in tradable goods and intermediate inputs, whose prices are disciplined by world markets.

Consistent with this interpretation, we find that nontradable prices rise strongly after discoveries while tradable prices remain broadly stable, with roughly 60 percent of the decade-long increase in nontradable prices also occurring within the first four years. The resulting increase in the relative price of nontradables is mirrored in an appreciation of the

were 343 giant oil and gas discoveries during 1980–2012. Our dataset expands this coverage by including fields above 100 million barrels of oil equivalent and extending the sample through 2019.

bilateral real exchange rate, roughly two-thirds of which materializes before new oil extraction begins—consistent with the wealth effect associated with an anticipated future resource windfall ([Harding et al., 2020](#)).

Second, the real-side responses indicate that discoveries initially operate as aggregate demand shocks, with the expansion concentrated in the pre-production window. For a median-sized discovery, consumption and investment rise cumulatively by about 0.80 percent and 0.37 percent within four years of the discovery, and real GDP by about 0.14 percent, with output continuing to build thereafter as newly discovered oil comes online. Because no oil discovery in the baseline sample has begun producing by horizon $h = 4$, these responses isolate the anticipation effect of the news shock. The responses of the external sector reinforce the demand interpretation: imports expand immediately, whereas exports respond positively and significantly only after several years, when new oil production begins. The current account therefore deteriorates initially and improves as resource exports expand and the import response recedes. These dynamics are also consistent with the intertemporal approach to the current account: in response to news of higher future income, economies increase absorption today by borrowing more or saving less ([Obstfeld and Rogoff, 1995b](#)).

Third, the inflation response depends systematically on the exchange-rate regime. Using the classification in [Ilzetzi et al. \(2019\)](#), we group pegs and narrow bands as inflexible regimes and wider bands and floating arrangements as flexible regimes. The increase in core CPI is concentrated in inflexible regimes. Under flexible exchange rates, the inflation response is small and statistically insignificant. This pattern points to the exchange rate as a central margin of adjustment to news about future resource income.

The results are robust to a wide range of alternative specifications. They are similar when we restrict the sample to countries with at least one large discovery, relax or tighten the realized production-lag threshold that defines the sample, vary assumptions about oil prices and the expected production lag used to value discoveries, use alternative definitions of tradable and nontradable sectors, control for exploration expenditures, account for heterogeneous exposure to U.S. monetary policy shocks, select lags using the Bayesian Information Criterion, and correct for dynamic panel bias using the Split-Panel Jackknife procedure ([Mei et al., 2026](#)). These exercises indicate that the findings are not driven by sample compo-

sition, the realized production-lag cutoff, shock measurement, sectoral classification, global financial conditions, lag choices, or Nickell bias.

To interpret the evidence, we develop a small open-economy New Keynesian model with oil, tradable goods, and nontradable goods. Oil discoveries follow [Arezki et al. \(2017\)](#): a discovery reveals future oil production and export income, but new reserves become productive only after a time-to-connect delay. Households are forward-looking and consume and invest using both tradable and nontradable goods. Nontradable and home tradable producers set prices subject to nominal rigidities. Final tradable goods combine domestic tradable inputs and imported inputs, so tradable inflation depends partly on the exchange rate. Monetary policy follows a generalized Taylor rule that allows the central bank to respond to inflation, output, and changes in the nominal exchange rate; the response to the exchange rate governs the sensitivity of monetary policy to exchange-rate movements.

The model highlights two opposing forces. The first is an inflationary domestic-demand channel. A discovery raises expected future resource income, leading households to increase consumption and investment before production begins. Because part of this additional demand falls on nontradables, whose supply is domestically constrained and whose prices are sticky, marginal costs and nontradable inflation rise. This pushes up CPI inflation. The second is a disinflationary exchange-rate channel. Anticipated future oil exports generate an immediate nominal appreciation, which lowers the domestic-currency price of imported inputs and dampens tradable-goods inflation. This channel helps explain why producer prices respond weakly relative to consumer prices.

Exchange-rate policy determines which force dominates. Under an inflexible regime, nominal appreciation is limited, so the decline in imported-input prices is muted. The domestic-demand channel therefore dominates, making discovery news inflationary. Under a flexible regime, nominal appreciation absorbs part of the shock by reducing import prices and easing pressure on tradable-goods inflation. As a result, the same news shock generates a much weaker inflation response. The calibrated model generates cumulative impulse responses that are broadly consistent with the empirical results. Thus, the model rationalizes the main empirical findings: oil-discovery news raises absorption, appreciates the real exchange rate, worsens the current account initially, increases nontradable prices,

and raises CPI inflation primarily under inflexible exchange-rate regimes.

First, the paper contributes to the literature on news-driven business cycles. Existing work has focused mainly on real aggregates and has reached mixed conclusions about inflation. In standard New Keynesian models, favorable TFP news can be disinflationary because firms anticipate lower future marginal costs (Barsky et al., 2015). In contrast, noisy signals or optimism about future productivity can generate demand booms and raise inflation (Lorenzoni, 2009). Empirical studies based on structural VARs often find that positive TFP news lowers inflation (Barsky and Sims, 2011; Barsky et al., 2015; Kurmann and Sims, 2021; Miranda-Agrippino et al., 2026), while Cascaldi-Garcia and Vukotić (2022) find inflationary effects using patent-based innovation news. A central challenge is identification: latent news shocks are typically recovered using recursive orderings and max-share restrictions.² Evidence from estimated DSGE models is also inherently model-dependent, as both the recovered news shocks and their transmission mechanisms depend on the assumed structure.³ These approaches are informative but can be sensitive to identifying assumptions and model structure.

Taken together, the literature points to a fundamental empirical challenge: because news shocks are typically latent, existing estimates must infer them through identifying restrictions or structural models, making it difficult to directly measure their inflationary effects. We contribute by using observable oil-discovery news as a direct source of variation in expected future output, thereby providing new evidence on the inflationary consequences of news shocks. This setting offers a unique advantage: oil discoveries are observed directly, so our estimates do not depend on VAR identifying restrictions or on the latent shock structure of an estimated DSGE model.⁴

²The VAR evidence relies on either the standard Cholesky triangularization or the max-share restrictions to recover TFP news shocks (Miranda-Agrippino et al., 2026; Francis et al., 2014; Kurmann and Sims, 2021). The first approach relies on an untested assumption about the ordering of VAR variables, while Dou et al. (2025) shows that the second approach—max-share identification—is valid only under restrictive necessary and sufficient conditions that are difficult to satisfy in practice, and that the identified TFP news shocks can be contaminated by other structural disturbances.

³For example, Görtz and Tsoukalas (2017) show that financial intermediation can amplify TFP news and turn it into a demand-like expansion that raises inflation, while Görtz et al. (2022) show that, even with financial frictions, the sign of the inflation response depends on the expected path of marginal costs and the horizon at which the news materializes.

⁴Our work is closely related to Arezki et al. (2017) and Sheng and Zhao (2024). Whereas they study the real effects of giant oil-discovery news in a setting without a central role for nominal rigidities, we focus on the nominal and inflationary consequences of large discoveries and develop a New Keynesian framework in which monetary policy and exchange-rate regimes shape the transmission of news shocks. Moreover, our identification for news shocks is cleaner as we use large oil discoveries with at least a five-year production lag,

Second, this paper contributes to the New Open Economy Macroeconomics literature (NOEM) (Obstfeld and Rogoff, 1995a; Bergin, 2006; Itskhoki and Mukhin, 2021), and in particular to small open-economy New Keynesian frameworks (Galí and Monacelli, 2005; Schmitt-Grohé and Uribe, 2003). A defining feature of this framework, relative to the classic Mundell–Fleming model, is its intertemporal nature: forward-looking agents respond today to anticipated changes in future fundamentals. We show that following large oil discoveries, economies begin to adjust well before new oil production comes online. Domestic demand rises, nontradable prices and the core CPI increase, the real exchange rate appreciates, and the current account initially deteriorates. The evidence therefore shows that news about future fundamentals can trigger immediate nominal, real, and external adjustment. Thus, this paper provides direct evidence for the intertemporal nature at the heart of NOEM models by using large oil discoveries with sufficient production lags as directly observed news shocks.

Finally, the paper identifies exchange-rate regimes as a key determinant of the inflationary effects of news in open economies. News about future output is not uniformly inflationary; its effect depends on the adjustment of the nominal exchange rate. When the exchange rate is constrained, higher expected income primarily raises domestic demand and nontradable prices. When the exchange rate is flexible, nominal appreciation lowers the domestic-currency cost of imports and attenuates the CPI response. Thus, the same news shock can have different nominal consequences depending on the monetary and exchange-rate regime. This open-economy dimension has received limited attention in the news-shock literature, which has focused primarily on closed-economy settings. We fill this gap by providing both empirical evidence and a theoretical rationale for the exchange-rate channel.

The rest of the paper is organized as follows. Section 2 describes the empirical strategy and data. Section 3 presents the empirical results and an extensive set of robustness checks. Section 4 develops the three-sector New Keynesian model of a small open economy with oil. Section 5 presents the calibration and the simulated impulse responses, which reproduce the main empirical patterns and illustrate the underlying mechanisms. Section 6 concludes.

while their data does not allow them to do so. Lastly, our empirical methodology differs as well: we adopt panel local projections, which are more flexible and robust than their dynamic panel distributed-lag model.

2 Empirical Strategy

This section lays out our empirical methodology and data. Section 2.1 presents the baseline econometric specification and identification strategy. Section 2.2 details the large discovery data and the construction of the news shocks, while Section 2.3 introduces the macroeconomic aggregates and price indices. Finally, Section 2.4 extends the analysis to a dyadic panel framework to examine bilateral real exchange rates.

2.1 Baseline Specification and Identification Strategy

We estimate the dynamic effects of oil-discovery news using panel local projections in a cross-country panel dataset (Jordà, 2005). The baseline specification is:

$$y_{i,t+h} - y_{i,t-1} = \beta^h \text{Disc}_{i,t} + \sum_{j=1}^5 \delta_j^h \text{Disc}_{i,t-j} + \sum_{k=1}^5 \zeta_k^h \Delta y_{i,t-k} + \alpha_i^h + \gamma_t^h + \epsilon_{i,t+h}, \quad (1)$$

for horizons $h = 0, 1, \dots, 10$. The dependent variable, $y_{i,t+h} - y_{i,t-1}$, denotes the cumulative change in outcome y from year $t - 1$ to year $t + h$. The primary regressor of interest, $\text{Disc}_{i,t}$, represents the net present value of large discoveries as a percentage of nominal GDP, whose construction is detailed in Section 2.2.

Our primary dependent variables are the price indices—specifically the logarithms of core CPI, PPI, and the tradable and nontradable price indices. In addition, we examine the logarithms of real GDP, consumption, investment, exports, and imports (all measured in local currency units), alongside the current-account-to-GDP ratio. For variables entered in logarithms, we multiply the series by 100 so that the coefficient of interest, β^h , represents the cumulative percentage change at horizon h in response to a discovery news shock with a net present value (NPV) equal to 1% of GDP. In particular, for the price indices, β^h corresponds to the cumulative change in the price level—or equivalently, cumulative inflation—over the horizon. For share variables, such as the current-account-to-GDP ratio, β^h measures the cumulative percentage-point response.

The local projection method is well suited to our setting for two main reasons. First, it estimates the cumulative impulse response directly at each horizon, rather than deriving

it from the dynamic panel distributed-lag model of [Arezki et al. \(2017\)](#), in which responses and their confidence intervals must be iterated forward using the delta method. This maps directly into our object of interest—the total response of prices, quantities, and external balances over the years following a discovery—and smooths out short-run fluctuations. The local projection method also avoids compounding delta-method approximation errors when projecting responses over a decade-long horizon. Second, lag-augmented local projections deliver more reliable inference in settings with persistent or potentially non-stationary variables, where standard autoregressive inference can perform poorly, especially at longer horizons ([Montiel Olea and Plagborg-Møller, 2021](#)).

Our identification relies on the plausibly exogenous timing and size of discoveries, conditional on country and year fixed effects. As emphasized by [Arezki et al. \(2017\)](#), major commercial discoveries are highly uncertain and primarily governed by geological serendipity and drilling outcomes rather than contemporaneous macroeconomic conditions, so discoveries are unlikely to respond to current inflation, output, or external accounts. Yet, exogenous timing by itself does not make an event a news shock. To ensure that discoveries operate strictly as news about future output rather than as a contemporaneous production windfall, the development lag between initial discovery and first production must be sufficiently protracted. Accordingly, we use large discoveries with realized production lags of at least five years as our baseline measure of news shocks, the details of which we describe in Section 2.2. As a result, the cumulative responses β^h at horizons $h = 0, \dots, 4$ capture the anticipation effect of the oil discovery news. From $h = 5$ onward, oil production has begun for at least some fields in the sample, so β^h captures the combined effect of anticipation and the realization of physical oil output.

The specification (1) includes country and year fixed effects, which address distinct potential confounding factors. Country fixed effects, α_i^h , absorb time-invariant determinants of resource discovery and macroeconomic performance, including baseline geological endowments, institutional quality, and geographic traits. Horizon-specific year fixed effects, γ_t^h , absorb global macroeconomic shocks, including shifts in international oil prices, world interest rates, and global liquidity conditions. We also include five lags of the discovery shocks and five lags of the outcome’s first differences to absorb delayed responses to ear-

lier discoveries and control for domestic business-cycle persistence. We intentionally omit country-specific linear time trends from the baseline specification, as large discoveries can permanently shift the long-run growth trajectory of an economy, and filtering out linear trends risks attenuating the true anticipation response. Standard errors are computed using the [Driscoll and Kraay \(1998\)](#) correction, which is robust to heteroskedasticity, serial correlation, and cross-sectional dependence in the error terms.

We address three potential threats to our identifying assumption. First, if discoveries are anticipated by markets through observable exploration efforts, the timing of the news may be misdated; in [Section 3.2](#), we control directly for exploration expenditures. Second, because our benchmark construction of the NPV of oil discoveries uses the world oil prices prevailing in the discovery year, unobserved commodity price movements could comove with domestic price dynamics; we show that our results are robust to valuing discoveries using backward-looking three-year average prices. Third, to ensure that country-specific exposure to global financial cycles does not bias the estimates, we interact U.S. monetary policy shocks with country-level capital account openness. The estimates are essentially unchanged, as shown in [Section 3.2](#).

2.2 World Oil and Gas Discovery Data

We obtain global discovery data from the upstream database of Rystad Energy ([Rystad Energy, 2025](#)), a leading independent energy research and business intelligence consultancy. Rystad provides comprehensive field-level coverage of the global petroleum sector from 1970 to 2024, encompassing both conventional and unconventional (shale and tight oil) assets worldwide. For each field, it records granular administrative and technical details, including operator, equity ownership structure, geographic coordinates, and geological basin. Crucially, Rystad tracks field life-cycle milestones: the discovery confirmation date, the final investment decision, the year of initial commercial production, and decommissioning dates, along with annual historical production by hydrocarbon stream. It also documents field-level economics within a standardized discounted cash-flow framework that decomposes gross asset revenues into capital expenditures (Capex), exploration expenditures, operating expenditures (Opex), government take (royalties, corporate taxes, and production sharing),

and residual free cash flow to equity holders. These data have been recently adopted in empirical industrial organization studies and macroeconomics (e.g., [Asker et al., 2019](#); [Bornstein et al., 2023](#)).

We focus on conventional oil and gas discoveries spanning 1980 to 2019. We exclude pre-1980 data due to the limited availability of reliable macroeconomic data across many developing economies, and we conclude the sample in 2019 to avoid confounding macroeconomic dynamics associated with the Covid-19 pandemic. In addition, following [Arezki et al. \(2017\)](#), we exclude unconventional resources such as shale formations and oil sands. The geological presence of these reserves has been known for decades; their commercial exploitation has been driven by gradual technological innovations in horizontal drilling and hydraulic fracturing rather than sudden, discrete discovery events, making them distinct from the discovery news shocks we aim to capture.

Shock Construction and Valuation. Whereas [Arezki et al. \(2017\)](#) restrict attention to giant discoveries (defined as ultimate recoverable resources exceeding 500 million barrels of oil equivalent [MMboe]), Rystad’s coverage allows us to study large discoveries, defined as fields with at least 100 MMboe. This lower threshold substantially broadens the sample while still excluding minor discoveries whose aggregate macroeconomic consequences are negligible. Furthermore, when oil fields record subsequent resource reassessments after the initial commercial find, we retain only the initial discovery; later field additions largely capture predictable appraisal drilling rather than unanticipated news shocks.

Adopting the methodology of [Arezki et al. \(2017\)](#), we define the discovery news shock $Disc_{i,t}$ as the net present value (NPV) of large oil discoveries for country i in year t , expressed as a percentage of the country’s GDP:

$$Disc_{i,t} \equiv NPV_{i,t} = \frac{\sum_{d \in D_t} \sum_{j=5}^J \frac{q_{i,t+j}^d \cdot p_t^{\text{oil}}}{(1+r_i)^j}}{GDP_{i,t}} \times 100, \quad (2)$$

where $q_{i,t+j}^d$ represents the projected oil production profile for year $t+j$ associated with a specific large discovery d ; p_t^{oil} denotes the oil price prevailing at the time of discovery; r_i is the country-specific discount rate; and J marks the exhaustion horizon of the oil resource,

which we set to 20 years. The outer summation aggregates the discounted revenues over the set of discoveries D_t occurring in year t , accounting for the fact that a country may experience multiple discoveries in a single year. Finally, following [Arezki et al. \(2017\)](#), we incorporate risk-adjusted discount rates to account for variations in perceived political risk. The discount rate r_i is computed as the sum of a 5% risk-free rate and a country-specific risk premium.

Although Rystad reports realized field-level production, we do not use realized output paths to compute equation (2). Realized extraction profiles are endogenous equilibrium outcomes influenced by subsequent oil price swings, political disruptions, and capacity constraints unknown at the time of the find. Instead, we project $q_{i,t+j}^d$ assuming an initial development lag, an extraction plateau, and an exponential decline rate governed by the field's estimated recoverable reserves.⁵

Expected versus Realized Production Lags. A critical distinction in our framework lies between the *expected* development lag assumed by forward-looking agents when valuing discoveries ex ante, and the *realized* development lag observed ex post. In the valuation formula (2), we follow [Arezki et al. \(2017\)](#) by imposing an expected production lag of five years ($j = 5$). This maintains comparability with the earlier literature and ensures that the timing of future cash flows is treated symmetrically across countries. In Section 3.2, we show that calibrating this expected lag to seven years—the sample median of realized lags—yields near-identical normalized local projections, as altering the discounting horizon primarily rescales the shock (Figure B.5).⁶

However, to guarantee that oil discoveries genuinely operate as an *news* shock—rather than a contemporaneous revenue windfall—the realized development lag must be sufficiently long. If a field is commercialized almost immediately, macroeconomic aggregates will respond to actual extraction revenues, export proceeds, and fiscal royalties rather than expectations. A striking example is the Quesqui onshore field in Tabasco, Mexico, discovered in late 2019 with estimated reserves of 575 MMboe. Utilizing early production systems

⁵See Appendix B of [Arezki et al. \(2017\)](#) for parametric specifications of the projected production profile.

⁶Because the local projection in equation (1) is linear in $Disc_{i,t}$, a uniform change in the assumed expected lag simply rescales the estimated β^h without altering impulse response dynamics, provided that discounting factors are the same across countries.

and existing pipeline networks, commercial extraction began within months and peaked in 2022.

Rystad documents both discovery years and initial production years, so we can identify and purge these rapidly developed fields. In the full sample of large discoveries over 1980–2019 that report production-start year, the median realized production lag is 7 years (with an interquartile range of 3 to 13 years; see Figure A.1).⁷ We exclude 336 discoveries whose realized lag is strictly less than five years (20.4% of the sample).⁸ This leaves 1,309 large discoveries across 79 countries that unambiguously represent pre-extraction news shocks. The inability to observe realized production lags in earlier databases (such as Arezki et al. (2017)) meant that fast-tracked extraction could not be filtered out; our ability to cleanly isolate the anticipation phase constitutes a central empirical advantage.

A caveat is that this restriction conditions on a realized outcome: whether a field’s development lag falls below five years is observed only ex post, and realized lags partly reflect delays unforeseen at the time of discovery. The cutoff may therefore select on ex-post shocks correlated with domestic macroeconomics. Section 3.2 shows that the baseline results are robust to alternative thresholds of four and six years.

Table 1 documents the temporal and geographic distribution of these 1,309 discoveries. Overall, discoveries peaked in the 1980s (387 occurrences) and averaged roughly 307 per decade thereafter, but with marked regional shifts. Discoveries fell sharply across mature basins in Europe and Russia, remained steady in the Middle East and North America, and expanded rapidly across new exploration frontiers: discovery counts in Africa doubled from 41 in the 1980s to 81 in the 2010s, while South American discoveries more than tripled over the same period.

Extreme Outliers and Sample Trimming. Figure 1 presents the histogram of $\log NPV_{i,t}$ for large oil discoveries worldwide, expressed as a percentage of the respective country’s

⁷These realized production lags are slightly longer than the 4–6 year average reported by Arezki et al. (2017) for giant oil discoveries, but the difference is not qualitatively important given the substantial heterogeneity across fields. Moreover, realized lags partly reflect unexpected delays, such as financing constraints, permitting and regulatory delays, political regime changes, and wars, which may not have been foreseeable at the time of discovery.

⁸Of the 1,645 large discoveries, 1,038 report an observed production-start year. The remaining fields have not commenced commercial extraction as of our database vintage, implying an effective realized lag exceeding five years (given discovery dates up to 2019). These unstarted fields are retained in the sample.

GDP, from 1980 to 2019. As depicted, the distribution reveals substantial heterogeneity: in several small economies, discovery values completely dwarf the domestic economy, with NPV exceeding 1,000 percent of GDP ($\ln NPV > 6.91$) and the most extreme case surpassing 300 times GDP ($\ln NPV \approx 10.31$).

We exclude these hyper-windfalls on both conceptual and econometric grounds. Conceptually, windfalls of this scale transform economies into enclave petro-states that operate well outside the scope of standard macroeconomic frameworks. Econometrically, these high-leverage outliers would skew estimates toward the idiosyncratic trajectories of a few hyper-endowed economies rather than broad macroeconomic regularities. We therefore restrict the baseline sample to discoveries where the NPV is below 10 times initial GDP. This threshold removes five small economies—Equatorial Guinea, Guyana, Mozambique, Timor-Leste, and Turkmenistan—yielding a baseline sample of 74 countries with at least one large discovery during 1980–2019. Within this refined sample, the median discovery equals approximately 5 percent of GDP ($\ln NPV = 1.61$), with an interquartile range of 18.49 percent of GDP.

2.3 Macroeconomic Variables and Price Indices

We assemble an annual cross-country panel of aggregate and sectoral price indices and macroeconomic aggregates spanning 1980 to 2019. Table 2 lists the definitions and sources of all variables, and Table 3 presents the descriptive statistics for the key variables used in the empirical analysis.

Aggregate Price Indices. Consumer and producer price dynamics are measured using the World Bank’s [Global Database of Inflation](#) (GDI; Ha et al., 2023, April 2025 release). In our baseline specifications, we evaluate annual core CPI inflation rather than headline CPI. Headline inflation is heavily influenced by volatile global food and energy prices, which represent a large expenditure share in emerging economies and can obscure underlying domestic demand pressures. We complement core CPI with the Producer Price Index (PPI), which assigns greater weight to tradable intermediate and capital goods, thereby helping to differentiate consumer-side from producer-side price adjustments.

Because our local projection framework evaluates cumulative log price-level changes,

we recursively chain the annual inflation rates into continuous price index levels. To ensure our estimates are not distorted by currency collapses or hyperinflationary episodes, we exclude observations with annual inflation rates outside the range of -50% to $+50\%$. Finally, we exclude countries with fewer than six consecutive annual observations for core CPI or PPI, as such limited coverage likely reflects poor data quality and provides insufficient time-series variation for the panel local projection regression.⁹ After these steps, the final sample contains 102 countries, of which 87 have at least six consecutive core CPI observations and 89 have at least six consecutive PPI observations. To maintain a consistent sample across outcomes, we restrict the other macroeconomic variables to the same set of 102 countries. We merge this macro dataset with the refined sample of large oil discoveries to form the baseline estimation sample. Thus, this sample also includes countries without large discoveries, which serve as the control group. In the robustness check, we also restrict our estimation sample to countries with at least one discovery. The online appendix [A.1](#) provides full details on index chaining, break-point resolution, and outlier filtering.

Sectoral Prices: Tradables vs. Nontradables. A primary limitation of the GDI is the absence of sector-specific price indices. To construct price indices for the tradable and nontradable sectors, we obtain sectoral value-added data at the one-digit ISIC Rev. 3 level—in both current and constant national currency units—from the UNNAS. Following [Harding et al. \(2020\)](#), we calculate the price index $P_{i,t}^S$ for sector S in country i at time t as the ratio of sectoral value-added in current prices to its value in constant prices. The sectors denoted by S comprise agriculture, construction, manufacturing, and services.

In our baseline analysis, we categorize manufacturing as the tradable sector and services as the nontradable sector, capturing differences in cross-border trade intensity. We apply the same -50% to $+50\%$ inflation threshold to purge extreme volatility. Because these exclusions can create gaps in the data, we apply the similar procedure used for the CPI and PPI series in the online appendix [A.1](#). In Section [3.2](#), we confirm that our results are robust to alternative sectoral classifications (e.g., adding agriculture to tradables or construction to nontradables).

⁹Six consecutive observations for a given country are the minimum requirement for local projection regression at horizon $h = 0$ with five lagged variables in Equation [\(1\)](#).

Macro variables of real activities. Standard macroeconomic aggregates—real GDP, private consumption, gross fixed capital formation, exports, and imports (in constant local currency units), as well as the current account balance as a share of GDP—are drawn from the World Bank’s World Development Indicators (WDI). To mitigate the impact of outliers, the current-account-to-GDP ratio is winsorized directly at the 1st and 99th percentiles, whereas the real aggregates measured in constant local currency units are flagged as outliers when their share of real GDP falls outside the 1st–99th percentile interval and are then rescaled by the factor required to bring that share back to the breached threshold. This bounds implausible observations while preserving the internal accounting consistency of the national accounts. Online Appendix [A.1](#) details the procedure. Nevertheless, our empirical regression results remain robust if we do not trim these macro variables.

2.4 Bilateral Real Exchange Rate

Rather than evaluating the trade-weighted real effective exchange rate (REER), we investigate real exchange rate dynamics within a bilateral framework. Because the REER is a trade-weighted geometric average across multiple partner economies, its variation conflates domestic discovery news with discoveries and macroeconomic adjustments occurring across trading partners, alongside time-varying trade weights. Moving to a dyadic specification isolates bilateral relative price adjustments and eliminates multilateral confounding factors.

Following [Harding et al. \(2020\)](#), we construct bilateral real exchange rates using sectoral value-added price deflators from the UNNAS. First, we aggregate sectoral current- and constant-price value added across the four major sectors (agriculture, manufacturing, construction, and services) to obtain an aggregate country-level price deflator, $P_{i,t}$. We then convert these price indices into a common currency (US dollars) using annual nominal exchange rates from the IMF. The bilateral real exchange rate between countries i and j at time t is defined as:

$$RER_{ij,t} = \frac{P_{j,t}^{\text{USD}}}{P_{i,t}^{\text{USD}}}, \quad (3)$$

where $P_{i,t}^{\text{USD}}$ and $P_{j,t}^{\text{USD}}$ denote the respective dollar-denominated price indices. Under this definition, a decrease in $RER_{ij,t}$ signifies a real appreciation of country i ’s currency relative

to country j 's currency.

Let $n_{i,t} \equiv NPV_{i,t}/100$ denote the discovery net present value scaled as a decimal share of GDP. For each country pair (i, j) , we construct the relative bilateral discovery shock as

$$Disc_{ij,t}^{\text{rel}} = \log \left(\frac{1 + n_{i,t}}{1 + n_{j,t}} \right) = \log(1 + n_{i,t}) - \log(1 + n_{j,t}), \quad (4)$$

where the $1 + n$ normalization naturally accommodates country-year observations without large discoveries ($n = 0$). Because the mapping is logarithmic, $Disc_{ij,t}^{\text{rel}}$ is not the difference between the two country-level shocks $Disc_{i,t}$ and $Disc_{j,t}$, but rather the relative size of discovery news in country i against country j .

We trace the dynamic response of the bilateral real exchange rate by estimating the following dyadic local projection:

$$\log RER_{ij,t+h} - \log RER_{ij,t-1} = \theta^h Disc_{ij,t}^{\text{rel}} + \sum_{m=1}^5 \delta_m^h Disc_{ij,t-m}^{\text{rel}} + \sum_{k=1}^5 \zeta_k^h \Delta \log RER_{ij,t-k} + \alpha_{ij}^h + \gamma_t^h + \epsilon_{ij,t+h}, \quad (5)$$

for horizons $h = 0, 1, \dots, 10$. The dyad fixed effects, α_{ij}^h , absorb all time-invariant pair characteristics, such as bilateral distance, contiguity, language overlap, and colonial ties. The year fixed effects, γ_t^h , absorb common global macroeconomic shocks and worldwide commodity price cycles. We adjust standard errors using the [Driscoll and Kraay \(1998\)](#) method to account for heteroskedasticity, residual autocorrelation induced by the overlapping local projection horizons, and cross-dyad spatial dependence.

The estimated sequence $\{\theta^h\}_{h=0}^{10}$ maps out the cumulative response of the bilateral real exchange rate. In terms of magnitude, we consider the following example. For a country i that experiences a discovery worth 1 percent of GDP while partner j experiences none ($NPV_{i,t} = 1$, so $n_{i,t} = 0.01$ and $n_{j,t} = 0$), $Disc_{ij,t}^{\text{rel}} = \log(1.01) \approx 0.01$. The cumulative change in the log bilateral real exchange rate over horizon h is approximately $0.01 \theta^h$, meaning that θ^h approximately represents the cumulative percentage appreciation ($\theta^h < 0$) or depreciation ($\theta^h > 0$) of the home currency following a discovery news shock equal to one percentage point of GDP.

3 Empirical results

3.1 Benchmark results

We now present the benchmark empirical results characterizing the dynamic macroeconomic effects of a large oil discovery. The figures below plot the cumulative impulse responses to an oil-discovery news shock with NPV equal to 1% of GDP, estimated via panel local projections. Throughout, the shaded regions denote 90 percent confidence bands computed using [Driscoll and Kraay \(1998\)](#) standard errors. Table [A.1](#) in the online appendix reports the estimated cumulative impulse response coefficients and country coverage.¹⁰ Unless stated otherwise, we report magnitudes for a median-sized discovery, whose NPV equals 5 percent of GDP. Because the local projection is linear in the discovery shock, the impulse responses in Figures [2](#), [3](#), [4](#), [5](#), and [7](#)—estimated for a shock normalized to 1 percent of GDP—are simply rescaled by a factor of 5.

Figure [2](#) documents a striking dichotomy in aggregate price dynamics following the discovery announcement. The estimates reveal that the core Consumer Price Index (CPI) rises significantly and persistently, whereas the Producer Price Index (PPI) exhibits no statistically significant response at any horizon. This result suggests that the inflationary pressures triggered by the news shock are not uniformly distributed across the economy, but are instead highly dependent on the compositional weighting of the respective price indices. Quantitatively, a median-sized discovery—with an NPV equal to 5 percent of initial GDP—raises the core CPI cumulatively by about 0.57 percent by the end of horizon $h = 4$. Because our baseline sample includes only discoveries with realized production lags of at least five years, no field in the sample has begun commercial production by horizon $h = 4$. This response therefore isolates the anticipation effect of the news shock. Over the subsequent decade after discovery, the cumulative increase reaches approximately 1.13 percent, indicating that the inflationary impact persists and continues to build beyond the anticipation window.

To unpack this aggregate divergence, Figure [3](#) examines the underlying sectoral price indices. The results confirm that the inflationary pressure is overwhelmingly concentrated in the nontradable sector, where prices rise sharply in response to the news. In contrast, tradable prices remain broadly stable. This sectoral asymmetry reflects the mechanics of domestic market clearing following a positive wealth shock. The discovery announcement stimulates an immediate expansion in domestic absorption. While the increased demand for tradables is readily accommodated by an influx of imports, the demand boom for nontradables must clear through domestic price adjustments, forcing nontradable prices upward. Because the core CPI basket places a substantial weight on domestically

¹⁰As the country coverage varies with the horizon when using panel local projection, we report the number of countries or country pairs for the horizon $h = 0$.

consumed nontradable goods and services, it captures these demand-driven inflationary pressures. Conversely, the PPI is heavily weighted toward tradable goods, leaving it largely insulated from the domestic demand boom. Quantitatively, a median-sized discovery raises the nontradable price index cumulatively by about 0.29 percent within four years of the discovery. Over the subsequent decade after discovery, the cumulative increase reaches approximately 0.49 percent, implying that roughly 60 percent of the total adjustment in the nontradable price index occurs before extraction.¹¹

Figure 4 traces the responses of real activity to oil-discovery news. The evidence further supports the interpretation that a discovery initially operates as a demand shock. Real GDP, consumption, and investment all rise before newly discovered fields typically begin production, indicating that the announcement affects the economy in advance of any realized increase in oil output. Quantitatively, a median-sized discovery raises consumption and investment cumulatively by about 0.80 percent and 0.37 percent, respectively, within four years of the discovery, while real GDP rises by about 0.14 percent.

The consumption response is consistent with permanent-income behavior. A discovery raises expected future national income when the news arrives, even though the associated revenues materialize only after a development lag. Forward-looking households therefore smooth part of the anticipated windfall into current spending. Consequently, consumption responds upon announcement and expands throughout the first four years after discovery. Beyond horizon $h = 5$, the point estimates remain positive but edge downward slightly and become statistically insignificant as the confidence intervals widen. Cross-country consumption data are notoriously noisy, and measurement error accumulates over horizons, which can inflate the estimated standard errors. In addition, the effective sample for local projections shrinks at longer horizons, further eroding precision.

Investment dynamics display a similarly pronounced anticipation effect. After the announcement of large discoveries, firms anticipate stronger future demand and higher returns to capital, both in the resource sector and other sectors. In the presence of adjustment costs, the desired expansion in productive capacity is optimally spread over time. Investment therefore rises before new oil production begins and reaches its peak at horizon $h = 4$. As new oil production starts, investment demand starts to decline so the cumulative investment response falls slightly after horizon $h = 6$.

Real GDP begins to expand immediately during the pre-production window and continues to

¹¹The magnitudes of the core CPI and nontradable price index responses should not be compared directly as the two indices come from different sources and differ in coverage and construction. The core CPI from the GDI is usually a survey-based retail price index, whereas the nontradable price index is a value-added deflator constructed from the UNNAS. Aggregate price changes in CPI and PPI from GDI are therefore not a weighted sum of sectoral price changes from UNNAS.

rise over the medium run, peaking around eight years after the announcement at approximately 0.31 percent for a median-sized discovery. It then remains elevated through the end of the ten-year horizon. The contrasting dynamics across aggregates are particularly informative: while domestic demand components—consumption and investment—reach their peak at horizon $h = 4$, aggregate output keeps rising at longer horizons as production from newly discovered fields begins to come online after the development lag. This pattern—an immediate, expectation-driven expansion in domestic demand followed by a further rise in output as physical production materializes—provides a clear empirical signature of a macroeconomic news shock.

Figure 5 reports the external-sector responses to oil-discovery news. Total exports do not respond significantly in the first few years after a discovery. The response becomes positive and statistically significant only around the fourth year, after which exports rise sharply and remain elevated over the medium run, peaking at roughly 0.84 percent at horizon $h = 8$ for a median-sized discovery. This delayed adjustment is consistent with the long development lag of newly discovered oil fields: export capacity expands when discoveries begin to translate into production and foreign sales.¹²

Imports, by contrast, expand immediately following a discovery announcement. This surge aligns directly with the anticipation-driven domestic demand boom documented above: higher expected permanent income stimulates household demand for tradable consumer goods, while firms import capital equipment and intermediate inputs in preparation for future expansion. The import response peaks at horizon $h = 3$ before gradually moderating, slightly leading the peaks in consumption and investment at $h = 4$. This lead likely reflects procurement and delivery lags, as firms must order and import foreign capital goods prior to domestic installation and expenditure realization. Beyond horizon $h = 4$, the point estimates remain positive but become statistically insignificant.

The current account reflects the net effect of these trade dynamics. It deteriorates during the anticipation phase, reaching a trough of roughly 0.13 percentage points of GDP in the first year following a median-sized discovery, consistent with front-loaded consumption and investment before any new oil output arrives. From around year four onward, however, the current account improves as exports expand and the import response recedes, turning into a surplus that peaks at about 0.4 percentage points of GDP around year seven. Overall, the evidence suggests that oil-discovery news first raises domestic absorption and import demand, and subsequently strengthens exports and the current account as production capacity comes online. This dynamic pattern is consistent with the

¹²Total exports rise noticeably at $h = 4$, before the onset of official commercial production. This likely reflects pre-commercial flow testing: in major oil projects, test oil is often extracted and exported several months before the official production start date is recorded.

intertemporal approach to the current account (Obstfeld and Rogoff, 1995b): countries borrow more and save less when they expect high future output, and save more once that income is realized.

Figure 6 reports the estimated cumulative response of the bilateral real exchange rate, based on the estimating equation (5). Driven by the anticipated wealth effect and the ensuing demand boom, the real exchange rate appreciates persistently. Quantitatively, a median-sized discovery equal to 5 percent of GDP in country i , relative to a foreign country j with no discovery, generates a cumulative bilateral real exchange rate appreciation of about 0.51 percent within four years of the discovery and about 0.77 percent over the full decade.¹³ Crucially, this implies that roughly two-thirds of the decade-long real appreciation materializes before new oil extraction begins. This front-loaded adjustment strongly supports the forward-looking asset-pricing approach to exchange rates, which emphasizes that exchange rate movements respond to shifts in expectations about future fundamentals (e.g., Engel and West, 2005; Chen et al., 2010). Moreover, the response of the bilateral real exchange rate aligns closely with the sectoral price dynamics documented above. As shown in Figure 3, non-tradable prices rise persistently while tradable prices remain broadly flat, driving an increase in the internal relative price of nontradables. This relative price shift represents a central transmission channel for real exchange rate movements, which we formalize in greater detail in the model. In addition, although both our mechanism and the classic Balassa–Samuelson framework emphasize the role of internal relative prices in governing the real exchange rate (Balassa, 1964; Samuelson, 1964), the standard Balassa–Samuelson effect operates through sectoral productivity differentials. In contrast, in our setting, the increase in the relative price of nontradables is driven by the domestic demand expansion triggered by discovery news.

Our finding of a real exchange rate appreciation following a positive news shock aligns closely with the existing studies. For example, Harding et al. (2020) establish that a giant resource discovery equivalent to 10% of GDP causes the bilateral real exchange rate to appreciate by roughly 1.5% over the subsequent ten years. Crucially, this appreciation begins immediately upon discovery, driven by an increase in the prices of non-tradable goods, well before actual production commences. This anticipation effect mirrors the dynamics documented by Nam and Wang (2015), who find that favorable news about future productivity estimated using a VAR induces a strong, immediate real exchange rate appreciation. In their estimates, this appreciation peaks at roughly 2.1% around four quarters after the shock and remains highly persistent, lasting for up to 16 quarters.

¹³For a median-sized discovery in country i , the bilateral shock against a country with no discovery is $Disc_{ij,t}^{rel} = \ln(1 + 0.05) \approx 0.049$. From Figure 6, $\theta^{(4)} = -0.105$ and $\theta^{(10)} = -0.157$, implying cumulative appreciations of $0.0051 \approx 0.105 \times \ln(1.05)$ within four years and $0.0077 \approx 0.157 \times \ln(1.05)$ over the decade.

The real exchange rate appreciation following oil discoveries can be accommodated through two distinct nominal margins: nominal exchange rate appreciation and domestic price inflation. Figure 7 examines these margins by estimating inflation responses separately across two different exchange-rate regimes, using the classification in Ilzetzi et al. (2019). We aggregate their coarse regime codes into two groups: inflexible regimes, which include pegs and narrow bands of at most ± 2 percent (Codes 1 and 2), and flexible regimes, which include wider bands and floating arrangements (Codes 3 and 4). As the exchange-rate regime code for a given country may change over time during 1980–2019, we classify a country as having an inflexible (flexible) regime for the whole sample period if its regime code is 1 or 2 (3 or 4) for more than half of the years during the sample period.¹⁴ This classification separates economies in which nominal exchange rate movements are tightly constrained from those in which the exchange rate can adjust more freely to macroeconomic shocks. In our sample, exchange-rate regime data is available for 100 countries; after applying our grouping criteria, 60 countries are classified as inflexible-regime countries and 22 as flexible-regime countries. Eighteen countries are unclassified and thus not included in either group.

Because the exchange rate regimes do change over time for some countries, we deliberately assign each country to a single regime for the entire sample rather than adopting a time-varying, year-by-year classification. The key reason is that exchange-rate regime transitions are endogenous to macroeconomic outcomes: countries typically abandon pegs and move toward more flexible arrangements during currency crises, balance-of-payments stress, or inflationary spirals. Because inflation crises can both trigger devaluations and mechanically raise measured inflation, such a time-varying split would build a positive correlation between the “flexible regime” group and the inflation outcome directly into the empirical design. By contrast, classifying countries according to their predominant regime over the full sample yields a fixed country-level characteristic. This makes the regime grouping constant with respect to any individual discovery event and keeps it orthogonal to the post-discovery responses we estimate.

This distinction is central to the transmission of the anticipated resource boom. In a standard open-economy framework, a large oil discovery raises expected future income, which generates pressure for real appreciation. When the nominal exchange rate is constrained, this adjustment must occur through domestic prices; when the exchange rate is flexible, nominal appreciation can absorb part of the adjustment. Figure 7 shows that the data conform closely to this prediction.

¹⁴Ilzetzi et al. (2019) also include a “freely falling” category (Code 5) to denote severe macroeconomic instability or currency crises. We do not classify this group as either a flexible or an inflexible exchange rate regime, as it does not represent a deliberate exchange-rate policy.

Under inflexible exchange-rate regimes, shown in the left panels, the nominal exchange rate cannot appreciate sufficiently in response to the discovery news. The real appreciation is therefore achieved primarily through higher domestic prices. Both core CPI and nontradable prices rise significantly and persistently, indicating that domestic inflation is the main margin of adjustment when the nominal exchange rate is constrained.

By contrast, under flexible exchange-rate regimes, shown in the right panels, the nominal exchange rate can appreciate following the discovery. This forward-looking nominal adjustment absorbs much of the pressure generated by the anticipated resource boom. As a result, the cumulative responses of core CPI and nontradable prices are small and statistically indistinguishable from zero. These findings underscore the exchange-rate regime as a key determinant of how anticipated wealth shocks are transmitted into domestic inflation.

3.2 Robustness check

To assess the robustness of our baseline findings, we consider alternative sample constructions, variable definitions, and econometric specifications. For each check, the online appendix [B](#) reports the impulse responses for related macro variables. In all cases, the estimated responses remain quantitatively similar to the baseline, confirming that our main conclusions are not driven by any single specification choice.

Alternative sample. A potential concern is that unobserved, time-invariant structural differences between oil-endowed nations and countries that never discover oil might confound the estimates. To mitigate this, we limit our estimation sample exclusively to countries that experienced at least one large oil discovery during the 1980–2019 period. As depicted in Figure [B.1](#), the impulse responses of key variables including aggregate and sectoral price indices, real activity, external adjustment, and real exchange rates remain similar to the baseline results. This confirms the generalizability of the transmission mechanism and ensures the findings are not artifacts of structural heterogeneity between discovering and non-discovering economies.

Alternative sample selection based on ex-post production lags. Our baseline excludes discoveries with realized production lags shorter than five years to separate news from contemporaneous production. Since this restriction uses ex-post information, we check whether the results depend on the cutoff by re-estimating the model with alternative thresholds of four and six years. The former readmits faster-developing fields, while the latter retains only discoveries with longer realized lags.

Figures B.2 and B.3 show that the impulse responses of prices, real activity, and the external accounts remain close to the baseline. Thus, our findings are not driven by the specific five-year cutoff.

Alternative oil price in the construction of NPV. In the baseline, the NPV of discoveries is constructed using the prevailing spot oil price at the time of discovery. However, short-term price volatility might distort the measured size of the news shock, as current prices may not perfectly reflect the expected future value of the extracted oil. To address this, we reconstruct the NPV using the average of global oil prices in the discovery year and the two preceding years.¹⁵ Figure B.4 demonstrates that this alternative valuation yields nearly identical macroeconomic dynamics, indicating that our results are robust to the assumptions underlying the valuation of the resource windfall.

Alternative production lag used in the construction of NPV. The baseline analysis follows the established literature by assuming a standard five-year delay. However, the median realized production lag in our dataset is approximately seven years. To ensure our temporal assumptions do not artificially drive the results, we reconstruct the discovery NPVs assuming a seven-year expected production lag, which matches the median realized lag in the data. Figure B.5 demonstrates that the impulse responses of both nominal and real variables are virtually unchanged, indicating that the macroeconomic adjustments are not sensitive to minor variations in the assumed development horizon.

Alternative definitions of tradable and nontradable sectors. To verify that the observed sectoral inflation divergence is not strictly an artifact of our baseline classification (which defines manufacturing as tradable and services as nontradable), we adopt a broader definition. We redefine the tradable sector to encompass both manufacturing and agriculture, and the nontradable sector to combine services and construction. Figure B.6 illustrates the impulse responses under these alternative groupings. The fundamental dichotomy persists: the nontradable composite experiences significant inflationary pressure, whereas the tradable composite remains insulated.

Alternative dependent-variable definitions. In our baseline analysis, the dependent variables for prices are expressed as cumulative log changes in price indices. Alternatively, we re-estimate using annual inflation rates as the dependent variables. Figure B.7 displays the impulse responses of inflation rates to oil discovery news. The core CPI inflation rate and the nontradable inflation rate

¹⁵We obtain similar results when we exclude the discovery year and value discoveries at the average oil price in the three years before the discovery.

continue to exhibit significant positive responses, while the PPI inflation rate and tradable inflation rate show no significant reaction. This reaffirms the core inflationary findings across alternative definitions of the dependent variables.

Controlling for exploration capex and U.S. monetary policy. To ensure our discovery shocks are not confounded by pre-existing domestic exploration or global financial cycles, we augment our baseline specification with two additional controls. First, because extensive exploration activities could independently stimulate demand or make discoveries partially anticipated, we control for log exploration capital expenditure (capex) in the current period and the previous five periods. Second, while year fixed effects absorb uniform global shocks, countries' heterogeneous responses to global financial conditions could potentially bias the results. To account for this, we introduce an interaction term between the country's lagged capital account openness-to-GDP ratio (measured as total foreign assets and liabilities over GDP) and U.S. monetary policy shocks, measured by the federal funds rate. Figure B.8 demonstrates that simultaneously including these controls does not materially alter the estimated impulse responses, confirming that our baseline dynamics are robust to both pre-discovery investments and U.S. monetary policy spillovers.

Lag selection using BIC. The baseline specification uniformly incorporates five lags of the dependent and independent variables to capture macroeconomic persistence. To avoid arbitrary lag selection and ensure an optimal balance between model fit and parsimony, we utilize the Bayesian Information Criterion (BIC) to determine the optimal lag structure. Figure B.9 demonstrates that optimizing the lag length does not alter the fundamental macroeconomic dynamics.

Nickell bias correction in panel local projections. Recently Mei et al. (2026) has pointed out the presence of the Nickell bias in the panel fixed-effect estimator of the panel local projection due to inherent dynamic structures of the predictive specifications, even if the regressors have no lagged dependent variables. We therefore re-estimate the model using the Split-Panel Jackknife (SPJ) bias correction proposed by Mei et al. (2026), combined with the BIC-selected lag structure from the previous exercise. Figure B.10 shows that the resulting responses for key macro variables closely track the baseline estimates, confirming that our findings are not driven by dynamic panel estimation bias.

4 Model

In this section, we develop a three-sector New Keynesian model of a small open economy with oil, nontradables, and tradables. The model is designed to account for the dynamic responses documented in the previous section, including the effects of oil-discovery news on inflation, real activity, and the real exchange rate. Moreover, we use the model to illustrate the key channels through which news about future resource income affects inflation. The framework is closely related to the small open economy environment of [Galí and Monacelli \(2005\)](#), and the oil sector builds on [Arezki et al. \(2017\)](#).

The home economy is populated by a representative household, a set of firms operating in three production sectors, and a monetary authority. The production side consists of an oil sector, a nontradable sector, and a home tradable sector. Oil is exported in full and sold at an exogenous world price. Nontradable and home tradable firms are monopolistically competitive and face Rotemberg price adjustment costs. Final tradable goods are produced by combining domestically produced tradables with imported foreign tradables, while households consume and invest in final expenditure baskets that aggregate tradable and nontradable goods. This structure allows the model to distinguish between inflationary pressures arising from domestic demand for nontradables and the disinflationary effects of exchange-rate appreciation on imported tradable inputs. For simplicity, we assume the Foreign country produces only a tradable good, and the Home economy is small relative to the Foreign, so Home exogenous shocks do not affect the Foreign country's aggregate price level.

The central disturbance is an oil-discovery news shock at Home. Upon announcement, agents learn that the economy's productive oil reserves will increase in the future. Current oil production is unaffected, however, because newly discovered reserves become productive only after a time-to-build, or time-to-connect, delay. Specifically, a discovery announced in period t enters the stock of producing reserves only in period $t + j$, where j denotes the expected development delay that agents internalize at the time of the announcement. The shock therefore changes expectations about future oil output, export revenues, and national income before any additional oil is extracted. This timing distinction is essential: it allows the model to isolate the macroeconomic effects of anticipated future resource income from the contemporaneous supply effects of realized oil production.

4.1 Household

The household chooses consumption, labor supply, domestic and foreign bond holdings, and sector-specific investment. International financial markets are incomplete: the household can trade one-period domestic nominal bonds and one-period foreign-currency bonds, but not a complete set of state-contingent claims. Preferences are given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{L_t^{1+\gamma}}{1+\gamma} \right], \quad (6)$$

where C_t is aggregate consumption, L_t is aggregate hours worked, $\beta \in (0, 1)$ is the subjective discount factor, $\sigma > 0$ is the coefficient of relative risk aversion, $\gamma > 0$ is the inverse Frisch elasticity, and $\chi > 0$ governs the disutility of labor. Labor is perfectly mobile across sectors and is allocated to the nontradable, tradable, and oil sectors:

$$L_t = L_{N,t} + L_{T,t} + L_{O,t}. \quad (7)$$

The household owns the sector-specific capital stocks $K_{h,t-1}$, $h \in \{N, T, O\}$, and rents them to firms at nominal rental rates $R_{h,t}$. Capital is predetermined at the beginning of period t and is sector-specific in the short run. This assumption captures the limited short-run mobility of installed capital across sectors, which means that once capital has been installed in a given sector, it cannot be immediately reallocated to another sector. For example, capital used in oil extraction is not readily transformed into capital used in nontradable production. Sectoral rental rates may therefore differ in the short run. New capital is accumulated through sector-specific investment $I_{h,t}$. The household also receives nominal wages W_t , firm profits Φ_t , and returns on domestic and foreign bonds. Its period budget constraint is

$$\begin{aligned} P_{C,t}C_t + P_{I,t} \sum_{h \in \{N,T,O\}} I_{h,t} + B_t + \mathcal{E}_t B_t^* = W_t L_t + \sum_{h \in \{N,T,O\}} R_{h,t} K_{h,t-1} + \Phi_t \\ + (1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^*)\mathcal{E}_t B_{t-1}^*, \end{aligned} \quad (8)$$

where B_t denotes one-period nominal domestic bond holdings, B_t^* denotes foreign-currency bond holdings, i_t is the domestic nominal interest rate, and i_t^* is the interest rate paid on foreign bonds. The nominal exchange rate $\mathcal{E}_t$ is expressed as units of domestic currency per unit of foreign currency, so that $\mathcal{E}_t B_t^*$ is the domestic-currency value of foreign bond holdings. The household's budget

constraint therefore incorporates both the standard intertemporal saving decision and the choice between domestic and foreign bond holdings, as in canonical small open economy models with incomplete international asset markets.

The aggregate consumption good is a CES composite of final nontradable and final tradable goods:

$$C_t = \left[\omega_C^{1/\xi} C_{N,t}^{\frac{\xi-1}{\xi}} + (1 - \omega_C)^{1/\xi} C_{T,t}^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}}, \quad (9)$$

where $\omega_C \in (0, 1)$ is the weight on nontradables in consumption and $\xi > 0$ is the elasticity of substitution between tradables and nontradables. This aggregator allows the model to distinguish movements in the CPI from movements in producer prices, since the CPI places direct weight on nontradable consumption goods while final tradable prices are disciplined in part by world prices and the exchange rate. The corresponding consumer price index is

$$P_{C,t} = \left[\omega_C P_{N,t}^{1-\xi} + (1 - \omega_C) P_{T,t}^{1-\xi} \right]^{\frac{1}{1-\xi}}. \quad (10)$$

Cost minimization implies the sectoral consumption demands

$$C_{N,t} = \omega_C \left(\frac{P_{N,t}}{P_{C,t}} \right)^{-\xi} C_t, \quad C_{T,t} = (1 - \omega_C) \left(\frac{P_{T,t}}{P_{C,t}} \right)^{-\xi} C_t. \quad (11)$$

Thus, relative prices determine the composition of consumption expenditure, while $P_{C,t}$ is the unit price of the aggregate consumption bundle.

The household purchases an aggregate investment good, also produced from nontradable and final tradable goods, and allocates it across sector-specific capital stocks. The aggregate investment good is

$$I_t = \left[\omega_I^{1/\xi} J_{N,t}^{\frac{\xi-1}{\xi}} + (1 - \omega_I)^{1/\xi} J_{T,t}^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}}, \quad (12)$$

where $\omega_I \in (0, 1)$ is the nontradable share in investment. The investment price index is

$$P_{I,t} = \left[\omega_I P_{N,t}^{1-\xi} + (1 - \omega_I) P_{T,t}^{1-\xi} \right]^{\frac{1}{1-\xi}}, \quad (13)$$

and the associated demands for two goods are

$$J_{N,t} = \omega_I \left(\frac{P_{N,t}}{P_{I,t}} \right)^{-\xi} I_t, \quad J_{T,t} = (1 - \omega_I) \left(\frac{P_{T,t}}{P_{I,t}} \right)^{-\xi} I_t. \quad (14)$$

Total investment is allocated across sectors according to

$$I_t = I_{N,t} + I_{T,t} + I_{O,t}. \quad (15)$$

Capital accumulation in each sector is subject to investment adjustment costs:

$$K_{h,t} = (1 - \delta)K_{h,t-1} + I_{h,t} \left[1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 \right], \quad h \in \{N, T, O\}, \quad (16)$$

where δ is the depreciation rate and $\psi_h \geq 0$ controls the size of sector-specific investment adjustment costs. These costs are specified in terms of changes in the investment flow. They prevent unrealistically sharp reallocations of investment across sectors in response to the oil discovery news shock. They also imply that the shadow value of installed capital can differ across sectors after a shock, even though households own all capital stocks.

Optimality conditions.¹⁶ Let λ_t denote the multiplier on the household budget constraint after normalizing nominal revenues and expenditures by the consumption price index $P_{C,t}$. Equivalently, λ_t is the shadow value of one additional unit of resources measured in units of the consumption good. The first-order condition for consumption is

$$\lambda_t = C_t^{-\sigma}. \quad (17)$$

The intratemporal labor-supply condition is

$$\chi L_t^\gamma = \lambda_t \frac{W_t}{P_{C,t}}. \quad (18)$$

This condition determines aggregate hours worked as a function of the real wage and the marginal utility of consumption. Because labor is perfectly mobile across sectors, all sectors face the same nominal wage. Sectoral labor allocations are then determined by firms' labor demand conditions together with aggregate labor supply.

The Euler equation for domestic bonds is

$$1 = \beta(1 + i_t) \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \right]. \quad (19)$$

This equation equates the utility cost of giving up one unit of current consumption to the expected

¹⁶See the detailed derivation in Appendix C.1.

discounted utility value of the nominal payoff next period, deflated by CPI inflation. It is the standard intertemporal optimality condition for a nominal risk-free asset.

The Euler equation for foreign bonds is

$$1 = \beta(1 + i_t^*)\mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{\mathcal{E}_{t+1}}{\mathcal{E}_t} \frac{P_{C,t}}{P_{C,t+1}} \right]. \quad (20)$$

Relative to the domestic bond Euler equation, the payoff on the foreign bond is adjusted by the change in the nominal exchange rate, $\mathcal{E}_{t+1}/\mathcal{E}_t$. Combining the two Euler equations yields the usual uncovered-interest-parity condition. Later we will introduce the debt-elastic risk premium to ensure stationarity of the equilibrium of a small open economy, following [Schmitt-Grohé and Uribe \(2003\)](#).

Let $Q_{h,t}$ denote the real shadow value of installed capital in sector h (the real Tobin's Q), measured in units of the consumption good. The asset-pricing equation for installed capital in sector h is

$$Q_{h,t} = \beta\mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \left(\frac{R_{h,t+1}}{P_{C,t+1}} + (1 - \delta)Q_{h,t+1} \right) \right]. \quad (21)$$

The current value of installed capital equals the expected discounted value of its future real rental income plus the continuation value of the undepreciated capital stock. Because capital is sector specific, $Q_{h,t}$ may vary across h , reflecting sector-specific expected returns and adjustment costs.

The optimal investment condition is

$$\begin{aligned} \frac{P_{I,t}}{P_{C,t}} = Q_{h,t} & \left[1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 - \psi_h \frac{I_{h,t}}{I_{h,t-1}} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right) \right] \\ & + \beta\mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} Q_{h,t+1} \psi_h \left(\frac{I_{h,t+1}}{I_{h,t}} \right)^2 \left(\frac{I_{h,t+1}}{I_{h,t}} - 1 \right) \right], \quad h \in \{N, T, O\}. \end{aligned} \quad (22)$$

The condition equates the relative price of investment to the shadow value of installed capital, adjusted for the marginal effect of current investment on adjustment costs today and tomorrow. At the steady state, the adjustment-cost terms vanish and $P_{I,t}/P_{C,t} = Q_{h,t}$. Away from steady state, adjustment costs dampen the response of sectoral investment and generate gradual capital accumulation.

Lastly, the household's choices are also subject to standard no-Ponzi and transversality conditions for domestic bonds, foreign bonds, and sectoral capital stocks.

4.2 Production

The production side of the economy consists of three sectors. The nontradable sector produces nontradable goods for domestic absorption. The home tradable sector produces differentiated trad-

able goods that are used both domestically and abroad. The oil sector produces a homogeneous export good that is sold on world markets at the exogenous foreign-currency oil price.

4.2.1 Nontradable sector

The nontradable sector consists of a competitive final-good producer and a continuum of monopolistically competitive firms indexed by $i \in [0, 1]$ that supply differentiated varieties and face Rotemberg price adjustment costs. The final good producer combines these varieties into a composite nontradable good through the CES form:

$$Y_{N,t} = \left(\int_0^1 Y_{N,t}(i)^{\frac{\varepsilon_N - 1}{\varepsilon_N}} di \right)^{\frac{\varepsilon_N}{\varepsilon_N - 1}}, \quad (23)$$

where $\varepsilon_N > 1$ denotes the elasticity of substitution across nontradable varieties. Cost minimization implies the nontradable sectoral price index

$$P_{N,t} = \left(\int_0^1 P_{N,t}(i)^{1 - \varepsilon_N} di \right)^{\frac{1}{1 - \varepsilon_N}}, \quad (24)$$

and the demand for variety i is:

$$Y_{N,t}(i) = \left(\frac{P_{N,t}(i)}{P_{N,t}} \right)^{-\varepsilon_N} Y_{N,t}. \quad (25)$$

Each nontradable variety producer produces one differentiated variety through a Cobb–Douglas technology using labor and sector-specific capital:

$$Y_{N,t}(i) = A_{N,t} K_{N,t-1}(i)^{\alpha_N} L_{N,t}(i)^{1 - \alpha_N}, \quad (26)$$

where $A_{N,t}$ is sectoral productivity and $\alpha_N \in (0, 1)$ is the capital share.¹⁷ Given factor prices, cost minimization implies a common real marginal cost across firms producing different varieties. Expressed in units of the final nontradable good, the real marginal cost is

$$MC_{N,t} = \frac{1}{A_{N,t}} \left(\frac{R_{N,t}/P_{N,t}}{\alpha_N} \right)^{\alpha_N} \left(\frac{W_t/P_{N,t}}{1 - \alpha_N} \right)^{1 - \alpha_N}. \quad (27)$$

This expression is the unit cost of producing one additional unit of the nontradable intermediate

¹⁷We slightly abuse the notation of $K_{N,t-1}(i)$ here as it is chosen by each variety producer at period t . However, in the symmetric equilibrium, all atomic firms rent the same amount of capital, equal to aggregate sector-specific capital stock $K_{N,t-1}$, which is predetermined by household at the end of period $t - 1$.

good, normalized by the nontradable price level.

In a symmetric equilibrium, all firms choose the same factor inputs, and set the same price and produce the same quantity. Aggregate nontradable output is therefore

$$Y_{N,t} = A_{N,t} K_{N,t-1}^{\alpha_N} L_{N,t}^{1-\alpha_N}. \quad (28)$$

The associated factor demand conditions are

$$W_t = (1 - \alpha_N) \frac{P_{N,t} MC_{N,t} Y_{N,t}}{L_{N,t}}, \quad R_{N,t} = \alpha_N \frac{P_{N,t} MC_{N,t} Y_{N,t}}{K_{N,t-1}}. \quad (29)$$

These equations equate factor prices to the corresponding marginal revenue products, taking into account that the marginal unit of output is valued at $P_{N,t} MC_{N,t}$.

Nontradable variety producers face nominal price rigidity in the form of quadratic Rotemberg adjustment costs (Rotemberg, 1982).¹⁸ A firm that changes its price from $P_{N,t-1}(i)$ to $P_{N,t}(i)$ incurs the real resource cost in units of the final nontradable good:

$$\frac{\phi_N}{2} \left(\frac{P_{N,t}(i)}{P_{N,t-1}(i)} - 1 \right)^2 Y_{N,t},$$

where $\phi_N \geq 0$ governs the degree of price rigidity.

The firm chooses its price sequence to maximize the expected discounted value of profits, taking its demand curve as given and discounting nominal payoffs using the household's stochastic discount factor. The derivation details are presented in Appendix C.2. In symmetric equilibrium, the first-order condition for optimal price setting yields the nonlinear Phillips curve

$$(\Pi_{N,t} - 1)\Pi_{N,t} = \frac{\varepsilon_N}{\phi_N} \left(MC_{N,t} - \frac{\varepsilon_N - 1}{\varepsilon_N} \right) + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \frac{P_{N,t+1}}{P_{N,t}} \frac{Y_{N,t+1}}{Y_{N,t}} (\Pi_{N,t+1} - 1)\Pi_{N,t+1} \right], \quad (30)$$

where $\Pi_{N,t} = P_{N,t}/P_{N,t-1}$ denotes gross inflation in the nontradable sector.

Log-linearizing around a zero-inflation steady state gives the New Keynesian Phillips curve for the nontradable sector:

$$\pi_{N,t} = \beta \mathbb{E}_t \pi_{N,t+1} + \kappa_N \widehat{MC}_{N,t}, \quad \kappa_N \equiv \frac{\varepsilon_N - 1}{\phi_N}. \quad (31)$$

¹⁸We use Rotemberg pricing rather than Calvo pricing for tractability. In symmetric equilibrium, Rotemberg pricing does not generate price dispersion across firms, which avoids introducing price dispersion as an additional state variable. Around a zero-inflation steady state, the resulting Phillips curve is closely related to that obtained under Calvo pricing.

where $\pi_{N,t} \equiv \log \Pi_{N,t}$ is the log-deviation of nontradable inflation from steady state and $\widehat{MC}_{N,t}$ is the log-deviation of real marginal cost from its flexible-price level. This equation links current nontradable inflation to real marginal cost and expected future inflation. The slope of the Phillips curve, κ_N , is decreasing in the adjustment-cost parameter ϕ_N . Thus, greater nominal price rigidity implies a flatter Phillips curve and a more gradual response of nontradable inflation to movements in real marginal cost.

Nominal profits in the nontradable sector are

$$\Phi_{N,t} = P_{N,t} Y_{N,t} \left[1 - MC_{N,t} - \frac{\phi_N}{2} (\Pi_{N,t} - 1)^2 \right]. \quad (32)$$

These profits are rebated lump-sum to households.

The nontradable goods market clears according to

$$Y_{N,t} = C_{N,t} + J_{N,t} + \frac{\phi_N}{2} (\Pi_{N,t} - 1)^2 Y_{N,t}. \quad (33)$$

Thus, the output of the final nontradable good is absorbed by nontradable consumption demand, nontradable investment demand, and the real resources used in price adjustment.

4.2.2 Tradable sector

The tradable sector has two layers. First, a competitive final-good producer combines a domestically produced tradable input with an imported foreign input. Second, the domestic tradable input is produced by monopolistically competitive firms that supply differentiated varieties and face Rotemberg price adjustment costs. Since the pricing and production problem of home tradable firms is isomorphic to that of nontradable firms, we present the main equilibrium conditions and emphasize the additional open-economy setting.

The final tradable good is a CES composite of home and foreign tradable inputs:

$$Y_{T,t} = \left[\omega_H^{1/\eta} (Y_{H,t}^{dom})^{\frac{\eta-1}{\eta}} + (1 - \omega_H)^{1/\eta} Y_{F,t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad (34)$$

where $\omega_H \in (0, 1)$ measures home bias in the tradable bundle and $\eta > 0$ is the elasticity of substitution between home and foreign tradable goods. The corresponding price index is

$$P_{T,t} = \left[\omega_H P_{H,t}^{1-\eta} + (1 - \omega_H) (\mathcal{E}_t P_{F,t}^*)^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad (35)$$

where $P_{F,t}^*$ is the price of foreign tradable goods in foreign currency and $\mathcal{E}_t$ denotes the nominal exchange rate, measured in domestic currency per unit of foreign currency.

In our model, we define the producer price index as the final tradable price index $P_{T,t}$. This mapping reflects the construction of our PPI data. First, PPIs typically cover goods-producing, tradable sectors and exclude services (International Monetary Fund, 2004). Second, the Global Database of Inflation explicitly notes that its PPI series covers both domestic and imported products (Ha et al., 2023). Moreover, producer prices are exposed to trade policy shocks such as tariff changes through pervasive imported inputs (Brandt et al., 2017; Flaaen and Pierce, 2026; Gilchrist et al., 2026). Thus, the empirical PPI corresponds naturally to the final tradable price index in our model, which combines home-produced and imported tradable inputs. This mapping also captures the two features relevant to our analysis: producer prices are dominated by tradable goods, and they respond to exchange-rate movements through the imported-input channel.

The demand for the home and foreign inputs is

$$Y_{H,t}^{dom} = \omega_H \left(\frac{P_{H,t}}{P_{T,t}} \right)^{-\eta} Y_{T,t}, \quad M_t \equiv Y_{F,t} = (1 - \omega_H) \left(\frac{\mathcal{E}_t P_{F,t}^*}{P_{T,t}} \right)^{-\eta} Y_{T,t}. \quad (36)$$

Thus, an appreciation of the domestic currency (a decline in $\mathcal{E}_t$) shifts domestic tradable demand toward imports, and reduces the overall tradable sector price level $P_{T,t}$.

The market structure of home tradable firms is the same as that of the nontradable sector. First, the home tradable input is itself a CES aggregate of differentiated varieties:

$$Y_{H,t} = \left(\int_0^1 Y_{H,t}(i)^{\frac{\varepsilon_H - 1}{\varepsilon_H}} di \right)^{\frac{\varepsilon_H}{\varepsilon_H - 1}}, \quad (37)$$

with associated price index

$$P_{H,t} = \left(\int_0^1 P_{H,t}(i)^{1 - \varepsilon_H} di \right)^{\frac{1}{1 - \varepsilon_H}}, \quad (38)$$

where $\varepsilon_H > 1$. The demand for variety i is

$$Y_{H,t}(i) = \left(\frac{P_{H,t}(i)}{P_{H,t}} \right)^{-\varepsilon_H} Y_{H,t}. \quad (39)$$

There is a continuum of monopolistically competitive firms that supply differentiated varieties and face quadratic Rotemberg price adjustment costs, by hiring labor and renting sector-specific

capital. In a symmetric equilibrium, aggregate home tradable production is

$$Y_{H,t} = A_{T,t} K_{T,t-1}^{\alpha_T} L_{T,t}^{1-\alpha_T}. \quad (40)$$

Real marginal cost, expressed in units of the home tradable good, is

$$MC_{H,t} = \frac{1}{A_{T,t}} \left(\frac{R_{T,t}/P_{H,t}}{\alpha_T} \right)^{\alpha_T} \left(\frac{W_t/P_{H,t}}{1-\alpha_T} \right)^{1-\alpha_T}. \quad (41)$$

The associated factor demand conditions are

$$W_t = (1 - \alpha_T) \frac{P_{H,t} MC_{H,t} Y_{H,t}}{L_{T,t}}, \quad R_{T,t} = \alpha_T \frac{P_{H,t} MC_{H,t} Y_{H,t}}{K_{T,t-1}}. \quad (42)$$

As in the nontradable sector, home tradable producers set prices subject to Rotemberg adjustment costs with the quadratic parameter ϕ_T , which allows us to capture different price rigidities across sectors. The derivation details are presented in Appendix C.2. In symmetric equilibrium, optimal price setting implies

$$(\Pi_{H,t} - 1)\Pi_{H,t} = \frac{\varepsilon_H}{\phi_T} \left(MC_{H,t} - \frac{\varepsilon_H - 1}{\varepsilon_H} \right) + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \frac{P_{H,t+1}}{P_{H,t}} \frac{Y_{H,t+1}}{Y_{H,t}} (\Pi_{H,t+1} - 1)\Pi_{H,t+1} \right], \quad (43)$$

where $\Pi_{H,t} = P_{H,t}/P_{H,t-1}$ denotes the gross inflation of home tradable inputs. This equation is the home tradable counterpart of the nontradable Phillips curve. The key difference is that demand for home tradables depends not only on domestic absorption but also on import substitution and foreign export demand.

Similarly, we can obtain the log-linearized Phillips curve for home-produced tradable goods:

$$\pi_{H,t} = \beta E_t \pi_{H,t+1} + \kappa_H \widehat{MC}_{H,t}, \quad \kappa_H \equiv \frac{\varepsilon_H - 1}{\phi_T}. \quad (44)$$

Note that in the calibration exercise, we set $\phi_N > \phi_T$ to capture the reality that the price rigidity in the nontradable sector is typically higher than in the tradable sector.

Nominal profits in the home tradable sector are

$$\Phi_{H,t} = P_{H,t} Y_{H,t} \left[1 - MC_{H,t} - \frac{\phi_T}{2} (\Pi_{H,t} - 1)^2 \right]. \quad (45)$$

The home tradable goods market clears according to

$$Y_{H,t} = Y_{H,t}^{dom} + X_{H,t} + \frac{\phi_T}{2}(\Pi_{H,t} - 1)^2 Y_{H,t}, \quad (46)$$

where $X_{H,t}$ denotes exports of home tradables, which will be explained later. Hence, home tradable input production is absorbed by domestic use in the final tradable good, foreign demand, and Rotemberg adjustment costs.

4.2.3 Oil sector

The oil sector is modeled as an extractive sector in which current production depends not only on variable factor inputs, but also on the stock of producing reserves that has already been connected to extraction capacity. The production function is given by:

$$Y_{O,t} = A_{O,t} K_{O,t-1}^{\alpha_k} L_{O,t}^{\alpha_L} Res_{t-1}^{1-\alpha_k-\alpha_L}, \quad (47)$$

where $A_{O,t}$ is oil-sector productivity, $K_{O,t-1}$ is the stock of oil-sector capital predetermined at the end of the period $t - 1$, $L_{O,t}$ is labor employed in oil production, and Res_{t-1} is the stock of producing reserves. The parameters satisfy $\alpha_k > 0$, $\alpha_L > 0$, and $\alpha_k + \alpha_L < 1$. The final restriction implies that producing reserves are an essential factor in extraction. A larger stock of producing reserves raises the productivity of capital and labor, capturing the idea that oil fields with greater available reserves are easier and less costly to operate.

The stock of producing reserves evolves according to

$$Res_t = Res_{t-1} + \varsigma - Y_{O,t} + \varepsilon_{t-j}^{Res}. \quad (48)$$

The constant term ς denotes an exogenous constant reserve inflow, which ensures that the economy admits a non-degenerate steady state with positive oil production and a positive stock of producing reserves. Current extraction $Y_{O,t}$ depletes the stock of producing reserves one-for-one. The term ε_{t-j}^{Res} denotes the oil discovery news shocks, and the time lag captures the time-to-connect nature of oil discoveries: news about an oil discovery arrives at period $t - j$, but the associated reserves become productive only at period t . Thus, the discovery affects agents' expectations and forward-looking decisions immediately upon announcement, while actual oil production capacity expands only after a delay of j periods. Because the model abstracts from uncertainty about the timing of development,

this delay is simultaneously the expected and the realized lag: agents know at announcement when production will begin.

Oil is fully exported and sold at the world price. The domestic-currency oil price is therefore

$$P_{O,t} = \mathcal{E}_t P_{O,t}^*, \quad (49)$$

where $P_{O,t}^*$ is the world oil price in foreign currency and $\mathcal{E}_t$ is the nominal exchange rate, expressed as units of domestic currency per unit of foreign currency. Since the home economy is small in world oil markets, $P_{O,t}^*$ is taken as exogenous.

The oil producer hires labor, rents sector-specific capital from the households, and utilizes the reserve stock to maximize the expected present value of profits, taking as given the oil price and the stochastic discount factor of the household and subject to the production technology and reserve accumulation constraint. The details of the oil producer's dynamic extraction problem are provided in Appendix C.3. Below we discuss the main intuitions and key results.

The oil producer is forward-looking because current extraction reduces the stock of reserves available for future production. Let ζ_t denote the nominal shadow value of one additional unit of producing reserves, i.e., the multiplier on the reserves law of motion. Equivalently, ζ_t is the scarcity value of leaving one unit of oil in the ground. The producer internalizes this opportunity cost when choosing current inputs. As a result, the effective price relevant for current production is not the spot price $P_{O,t}$, but the net price $P_{O,t} - \zeta_t$. Optimal labor and capital demand satisfy

$$W_t = \alpha_L \frac{(P_{O,t} - \zeta_t) Y_{O,t}}{L_{O,t}}, \quad R_{O,t} = \alpha_k \frac{(P_{O,t} - \zeta_t) Y_{O,t}}{K_{O,t-1}}. \quad (50)$$

These conditions equate factor prices to the corresponding marginal revenue products, evaluated at the net-of-depletion price. They differ from static factor demand conditions because an additional unit of output today lowers tomorrow's reserve stock. When the shadow value of reserves is high, the net price $P_{O,t} - \zeta_t$ is low, and the producer has an incentive to restrain current extraction.

The shadow value of reserves satisfies the intertemporal condition

$$\zeta_t = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \left\{ (P_{O,t+1} - \zeta_{t+1})(1 - \alpha_k - \alpha_L) \frac{Y_{O,t+1}}{Res_t} + \zeta_{t+1} \right\} \right]. \quad (51)$$

The stochastic discount factor is inherited from the household because oil firms are owned by households. Equation (51) states that the value of one unit of reserves today equals its expected discounted value tomorrow. This equation is the model's Hotelling-type condition, modified to allow reserves

to enter production directly as a productive state variable.

Oil-sector profits are defined as revenues net of payments to labor and rented capital:

$$\Phi_{O,t} = P_{O,t}Y_{O,t} - W_tL_{O,t} - R_{O,t}K_{O,t-1}. \quad (52)$$

Using the optimal factor demand conditions, these profits can also be interpreted as the sum of an operating surplus and a scarcity-rent component. In particular, substituting (50) into (52) gives

$$\begin{aligned} \Phi_{O,t} &= P_{O,t}Y_{O,t} - \alpha_L(P_{O,t} - \zeta_t)Y_{O,t} - \alpha_k(P_{O,t} - \zeta_t)Y_{O,t} \\ &= [(1 - \alpha_k - \alpha_L)P_{O,t} + (\alpha_k + \alpha_L)\zeta_t] Y_{O,t}. \end{aligned} \quad (53)$$

Equivalently,

$$\Phi_{O,t} = (1 - \alpha_k - \alpha_L)(P_{O,t} - \zeta_t)Y_{O,t} + \zeta_t Y_{O,t}. \quad (54)$$

Thus, oil-sector dividends reflect both the current operating surplus from extraction and the scarcity value associated with ownership of the reserve stock. These profits are rebated lump-sum to households. Total profits received by households are

$$\Phi_t = \Phi_{N,t} + \Phi_{H,t} + \Phi_{O,t}. \quad (55)$$

4.3 Aggregate output and external account

Aggregate output. Aggregate real GDP is measured as value added across the three sectors, evaluated at steady-state relative prices:

$$Y_t = \frac{\bar{P}_N Y_{N,t} + \bar{P}_H Y_{H,t} + \bar{\mathcal{E}} \bar{P}_O^* Y_{O,t}}{\bar{P}_C}. \quad (56)$$

This definition holds relative prices fixed at their steady-state values, isolating movements in real quantities from valuation effects driven by changes in the terms of trade or the real exchange rate.

External account. The open-economy block links the domestic production structure to the rest of the world. The home economy exports both home-produced tradable goods and oil, and imports foreign tradable inputs used in the production of the domestic final tradable composite. Foreign prices and foreign demand are taken as exogenous, consistent with the small open economy assumption.

Foreign demand for the home tradable good is downward sloping:

$$X_{H,t} = \left(\frac{P_{H,t}}{\mathcal{E}_t P_{T,t}^*} \right)^{-\eta^*} Y_t^*, \quad (57)$$

where $P_{T,t}^*$ is the foreign tradable price index, Y_t^* is an exogenous foreign demand shifter, and $\eta^* > 0$ is the price elasticity of foreign demand. The relevant price is the price of the home tradable good relative to the foreign tradable price, both expressed in the same currency. An appreciation of the domestic currency, i.e., a decrease in $\mathcal{E}_t$, reduces the exports of the home tradable good, all else equal.

Oil exports equal oil production:

$$X_{O,t} = Y_{O,t}. \quad (58)$$

This assumption reflects the focus of the model on oil-exporting economies in which oil is sold on world markets and is not used as an intermediate input by domestic firms.

Import demand for foreign tradable input is given by equation (36), reproduced here for convenience: $M_t = (1 - \omega_H) \left(\frac{\mathcal{E}_t P_{F,t}^*}{P_{T,t}} \right)^{-\eta} Y_{T,t}$. Imports rise with aggregate tradable absorption, $Y_{T,t}$, and fall with the relative price of imported inputs. Since the domestic-currency price of imports is $\mathcal{E}_t P_{F,t}^*$, an appreciation lowers this relative price and therefore raises import volumes, other things equal.

The nominal trade balance is

$$TB_t = P_{H,t} X_{H,t} + P_{O,t} Y_{O,t} - \mathcal{E}_t P_{F,t}^* M_t. \quad (59)$$

The first two terms denote non-oil and oil export revenues, respectively, while the last term is import expenditure. Net foreign assets evolve according to

$$\mathcal{E}_t B_t^* = (1 + i_{t-1}^*) \mathcal{E}_t B_{t-1}^* + TB_t. \quad (60)$$

Hence, the current account, defined as the change in the domestic-currency value of net foreign assets, is

$$CA_t \equiv \mathcal{E}_t B_t^* - \mathcal{E}_t B_{t-1}^* = i_{t-1}^* \mathcal{E}_t B_{t-1}^* + TB_t. \quad (61)$$

These identities discipline the model's external adjustment. The impact of oil discovery news on the current account is consistent with the classic intertemporal approach to the current account, i.e., the current account reflects forward-looking saving and investment decisions in response to anticipated changes in future income (Obstfeld and Rogoff, 1995b). An oil-discovery news shock raises the

present value of national income before oil revenues are realized. Households and firms therefore increase consumption and investment on impact, raising imports, while the associated appreciation may reduce non-oil exports. Since oil exports do not yet rise, the trade balance and the current account can deteriorate initially despite higher lifetime wealth.

Once the new reserves become productive, oil export revenues increase, improving the trade balance and rebuilding net foreign assets. The model therefore delivers the current-account dynamics emphasized by [Arezki et al. \(2017\)](#): oil-discovery news can generate a short-run current-account deficit before the associated resource revenues materialize.

4.4 Monetary Policy and Uncovered Interest Parity

To close the model, we specify the monetary policy rule and the conditions that determine the nominal exchange rate. Together, the Taylor rule, the uncovered interest parity (UIP) condition, and the debt-elastic risk premium form a three-equation system that jointly determines the domestic nominal interest rate i_t , the effective foreign interest rate i_t^* , and the nominal exchange rate $\mathcal{E}_t$. In the next subsection, we characterize the forward-looking nature of the nominal exchange rate and the role of exchange-rate flexibility ϕ_e , and we derive the response of the real exchange rate, which is the object examined in the empirical analysis of Section 3.

Monetary policy. The central bank follows a generalized Taylor-type rule with interest-rate smoothing ([Bergin, 2006](#); [Itskhoki and Mukhin, 2025](#)). It adjusts the nominal interest rate in response to CPI inflation, aggregate output, and nominal exchange-rate depreciation:

$$\frac{1+i_t}{1+\bar{i}} = \left(\frac{1+i_{t-1}}{1+\bar{i}} \right)^{\rho_i} \left[\left(\frac{\Pi_{C,t}}{\bar{\Pi}_C} \right)^{\phi_\pi} \left(\frac{Y_t}{\bar{Y}} \right)^{\phi_y} \left(\frac{\mathcal{E}_t}{\mathcal{E}_{t-1}} \right)^{\phi_e} \right]^{1-\rho_i}, \quad (62)$$

where $\Pi_{C,t} = \frac{P_{C,t}}{P_{C,t-1}}$ denotes CPI inflation, $\rho_i \in [0, 1)$ governs interest-rate smoothing, $\phi_\pi > 1$ ensures an active response to inflation, and $\phi_y, \phi_e \geq 0$ are the output and exchange-rate depreciation response coefficients. The variables with bars are steady-state values.

The rule responds to exchange-rate changes rather than to the level because this formulation captures a crawl-like policy in which the central bank resists exchange-rate movements without committing to a fixed parity, consistent with the fear-of-floating evidence that managed regimes stabilize the rate of change of the exchange rate rather than its level ([Itskhoki and Mukhin, 2025](#)). The coefficient ϕ_e governs the degree of exchange-rate flexibility. A higher value of ϕ_e means the central

bank responds more aggressively to exchange-rate movements, corresponding to a tightly managed or inflexible exchange-rate regime. Conversely, a lower value of ϕ_e allows the exchange rate to adjust more freely, corresponding to a more flexible regime. This parameter is central to the paper's analysis of how the inflationary effects of oil-discovery news depend on the exchange-rate regime.

Exchange rate determination. The nominal exchange rate is determined by the uncovered interest parity (UIP) condition. Equating the household's Euler equations for domestic bonds (19) and foreign bonds (20) and dropping the covariance between exchange-rate changes and the household's stochastic discount factor, which is of second order and vanishes in the log-linear approximation, yields

$$1 + i_t = (1 + i_t^*) \mathbb{E}_t \left[\frac{\mathcal{E}_{t+1}}{\mathcal{E}_t} \right]. \quad (63)$$

This condition states that the expected returns on domestic and foreign bonds must be equal in equilibrium. To ensure that the equilibrium dynamics of net foreign assets are stationary, following [Schmitt-Grohé and Uribe \(2003\)](#) we allow the effective foreign interest rate faced by domestic residents to depend on the economy's net foreign asset position:

$$1 + i_t^* = (1 + i_t^w) \exp \left[-\chi_B \left(\frac{\mathcal{E}_t B_t^*}{\bar{P}_{C,t} \bar{Y}_t} - \bar{b} \right) \right], \quad (64)$$

where i_t^w is the exogenous world interest rate, $\chi_B > 0$ controls the sensitivity of the borrowing rate to the external position, and $\bar{b} \equiv \bar{\mathcal{E}} \bar{B}^* / (\bar{P}_C \bar{Y})$ is the steady-state net foreign asset position relative to output. When net foreign assets fall below their steady-state level, the effective foreign rate rises above the world rate, making borrowing more expensive and encouraging a return to the steady state. Combining equations (63) and (64), we have the risk premium term in the traditional UIP condition:

$$1 + i_t = (1 + i_t^w) \mathbb{E}_t \left[\frac{\mathcal{E}_{t+1}}{\mathcal{E}_t} \right] \exp \left[-\chi_B \left(\frac{\mathcal{E}_t B_t^*}{\bar{P}_{C,t} \bar{Y}_t} - \bar{b} \right) \right], \quad (65)$$

To save space, the equilibrium conditions including the first-order conditions and market-clearing conditions are listed in [Appendix C.4](#).

4.5 Exchange Rate Dynamics

Having laid out the model, we now log-linearize several key equations to develop intuition for the impact of an oil-discovery news shock on the exchange rate. Throughout this section, variables topped with a hat, such as $\hat{X}_t$, denote log-deviations from the deterministic steady state, defined as

$\hat{X}_t \equiv \log(X_t / \bar{X})$, except for interest rates, which are defined as $\hat{i}_t \equiv \log[(1 + i_t) / (1 + \bar{i})]$, and similarly for i_t^* and i_t^w .

Log-linearizing the generalized Taylor rule in equation (62) and the risk-adjusted UIP condition in equation (65) around the steady state, after some algebra, yields the forward-looking difference equation for the nominal exchange rate stated in the following lemma.

Lemma 1 (Forward-looking nominal exchange rate). *Let $\hat{\mathcal{E}}_t$ denote the log-deviation of the nominal exchange rate from its steady state, and define $\Gamma \equiv 1 + \chi_B \bar{b} + (1 - \rho_i)\phi_e$, where $\bar{b} \equiv \bar{\mathcal{E}}\bar{B}^* / (\bar{P}_C \bar{Y})$ is the steady-state net foreign asset position relative to output. The equilibrium nominal exchange rate satisfies the forward-looking difference equation*

$$\hat{\mathcal{E}}_t = \frac{1}{\Gamma} \left\{ \mathbb{E}_t[\hat{\mathcal{E}}_{t+1}] + \underbrace{\hat{i}_t^w - \rho_i \hat{i}_{t-1} - (1 - \rho_i)[\phi_\pi \pi_{C,t} + \phi_y \hat{Y}_t]}_{=: D_t} + (1 - \rho_i)\phi_e \hat{\mathcal{E}}_{t-1} + \chi_B \bar{b} (\hat{P}_{C,t} + \hat{Y}_t - \hat{B}_t^*) \right\}, \quad (66)$$

Lemma 1 makes explicit that the nominal exchange rate is a forward-looking asset price: the current exchange rate loads with weight $1/\Gamma$ on its expected future value and on the driving variables D_t , rather than on current fundamentals alone. Iterating (66) forward, the exchange rate equals a present value of the entire expected future path of the driving variables, geometrically discounted by $1/\Gamma$, as shown in Appendix C.6.¹⁹ The signs of the underlying effects are standard. An increase in the world interest rate $\hat{i}_{t+k}^w$ depreciates the home currency, as capital flows out toward higher foreign yields. An increase in CPI inflation $\pi_{C,t+k}$ or output $\hat{Y}_{t+k}$ appreciates the home currency, because both trigger monetary policy tightening through the Taylor rule; higher output also lowers the risk premium by raising GDP in the denominator of the net foreign asset ratio. An increase in net foreign assets $\hat{B}_{t+k}^*$ appreciates the currency by lowering the risk premium and making domestic bonds relatively more attractive.

This forward-looking structure is central for understanding the effects of oil-discovery news. An oil discovery changes the expected future path of the driving variables even though current oil production is unchanged: agents anticipate stronger domestic demand and thus higher future inflation and output, and they anticipate an improvement in the external position once oil exports begin. Because the exchange rate loads on the entire expected future path of these fundamentals, it appreciates on impact—at the time of the announcement—well before the newly discovered oil enters production.

The coefficient ϕ_e plays a central role in shaping the exchange-rate response to news, through

¹⁹In our calibration, $\chi_B = 0.0001$ and $\bar{b}$ is small, so $\chi_B \bar{b}$ is close to zero, and with $\phi_e > 0$, we have $\Gamma > 1$.

both the discount factor $\Gamma = 1 + \chi_B \bar{b} + (1 - \rho_i)\phi_e$ and the persistence term $(1 - \rho_i)\phi_e \hat{\mathcal{E}}_{t-1}$ embedded in D_t . A higher ϕ_e —corresponding to a more tightly managed or inflexible exchange-rate regime—raises Γ , thereby reducing the weight the exchange rate places on expected future fundamentals and anchoring it more firmly to its past level. Conversely, a lower ϕ_e —corresponding to a more flexible regime—implies a smaller Γ , so the exchange rate responds more aggressively to news about future inflation, output, and the external position. This mechanism is essential for understanding why oil-discovery news shocks are more inflationary under inflexible exchange-rate regimes: when ϕ_e is large, the exchange rate appreciates less on impact, so the disinflationary import-price channel is weaker and the domestic demand channel dominates.

Next, we turn to the real exchange rate. We define the real exchange rate as

$$e_t = \frac{\mathcal{E}_t P_{F,t}^*}{P_{C,t}}, \quad (67)$$

where a decline in e_t denotes a real appreciation. We have the following proposition:

Proposition 2 (Real exchange rate decomposition). *Let $\hat{e}_t$ denote the log-deviation of the real exchange rate from its steady state. Assuming $\hat{P}_{F,t}^* = 0$, the real exchange rate can be decomposed as*

$$\hat{e}_t = -s_C \hat{p}_t^{N/T} - s_H \hat{p}_t^{H/F}, \quad (68)$$

where s_C is the steady-state expenditure share of nontradables in consumption, s_H is the steady-state expenditure share of home-produced goods in the tradable bundle, $\hat{p}_t^{N/T} \equiv \hat{P}_{N,t} - \hat{P}_{T,t}$ is the internal relative price of nontradables, and $\hat{p}_t^{H/F} \equiv \hat{P}_{H,t} - \hat{\mathcal{E}}_t$ is the relative price of home tradables against the foreign tradable good.

The functional forms of s_C and s_H , together with the derivation, are provided in Appendix C.7.

Proposition 2 shows that the real exchange rate moves through two channels. The first is the internal relative-price channel: a rise in the relative price of nontradables, $P_{N,t}/P_{T,t}$, appreciates the real exchange rate with elasticity s_C . The second is the home-tradable price channel: an increase in the price of home-produced tradables relative to the foreign tradable input, $P_{H,t}/(\mathcal{E}_t P_{F,t}^*)$, appreciates the real exchange rate with elasticity s_H .

Following an oil-discovery news shock, both channels contribute to a real appreciation. The anticipated future resource windfall raises domestic absorption and increases demand for nontradable goods, driving up $\hat{p}_t^{N/T}$. At the same time, stronger domestic demand and tighter factor markets raise home tradable marginal costs and therefore $\hat{P}_{H,t}$, while the nominal exchange rate appreciates on impact through the forward-looking mechanism in Lemma 1, raising $\hat{p}_t^{H/F}$. Both forces push $\hat{e}_t$

downward, generating a real appreciation following the oil discovery news shock that is consistent with the empirical response of the bilateral real exchange rate documented in Section 3.

The decomposition also clarifies how the exchange-rate regime shapes the real appreciation, and thus the inflationary consequences of the news. Under a flexible regime, the appreciation operates primarily through the nominal exchange rate: the currency appreciates on impact, which raises $\hat{p}_t^{H/F}$ and lowers the domestic-currency price of imported goods, dampening tradable inflation. Under an inflexible regime, nominal appreciation is limited by the high value of ϕ_e , so the real appreciation is achieved instead through the internal relative-price channel: the demand boom raises nontradable prices, increasing $\hat{p}_t^{N/T}$ and therefore CPI inflation. Hence, the same real appreciation has very different inflationary consequences across regimes—it operates through nominal adjustment with muted inflation under flexibility, but through higher nontradable inflation under an inflexible regime.

4.6 News Shocks and Inflation Dynamics

Now we turn to the key question of the model: how do oil discovery news shocks affect aggregate inflation dynamics? Note that the model distinguishes among several price indices because oil-discovery news moves sectoral relative prices through different channels. Log-linearizing the CPI price index (10) around the steady state gives

$$\pi_{C,t} = s_C \pi_{N,t} + (1 - s_C) \pi_{T,t}, \quad (69)$$

where s_C denotes the steady-state expenditure share of nontradable goods in the consumption basket, $\pi_{N,t} \equiv \log(P_{N,t}/P_{N,t-1})$ is nontradable inflation, and $\pi_{T,t} \equiv \log(P_{T,t}/P_{T,t-1})$ is inflation in the final tradable composite.

The distinction between the final tradable price $P_{T,t}$ and the price of home-produced tradable goods $P_{H,t}$ is important. The final tradable composite includes both domestically produced tradable goods and imported goods, whereas $P_{H,t}$ refers only to home-produced tradable varieties. Log-linearizing the tradable price index (35) yields

$$\pi_{T,t} = s_H \pi_{H,t} + (1 - s_H) (\Delta \epsilon_t + \pi_{F,t}^*), \quad (70)$$

where s_H is the steady-state share of home-produced goods in the tradable bundle, $\pi_{H,t} \equiv \log(P_{H,t}/P_{H,t-1})$ denotes inflation in home-produced tradable goods, and $\Delta \epsilon_t \equiv \log \mathcal{E}_t - \log \mathcal{E}_{t-1}$ denotes nominal exchange-rate depreciation. Imported goods are priced in foreign currency at $P_{F,t}^*$, so their domestic-

currency inflation rate is $\Delta\epsilon_t + \pi_{F,t}^*$. Assuming that foreign-currency import prices are constant, $\pi_{F,t}^* = 0$, equation (70) simplifies to

$$\pi_{T,t} = s_H \pi_{H,t} + (1 - s_H) \Delta\epsilon_t. \quad (71)$$

The following proposition summarizes the CPI inflation decomposition.

Proposition 3 (CPI inflation decomposition). *Assume that foreign-currency import prices are constant, $\pi_{F,t}^* = 0$. CPI inflation can be decomposed as*

$$\pi_{C,t} = s_C \pi_{N,t} + (1 - s_C) [s_H \pi_{H,t} + (1 - s_H) \Delta\epsilon_t], \quad (72)$$

where s_C is the steady-state expenditure share of nontradables in the consumption basket and s_H is the steady-state expenditure share of home-produced goods in the tradable bundle. Thus, CPI inflation can be decomposed into three components: nontradable inflation, inflation in home-produced tradable goods, and domestic-currency import-price inflation. Substituting the sectoral New Keynesian Phillips curves (31) and (44) yields

$$\pi_{C,t} = s_C \left(\beta E_t \pi_{N,t+1} + \kappa_N \widehat{MC}_{N,t} \right) + (1 - s_C) \left[s_H \left(\beta E_t \pi_{H,t+1} + \kappa_H \widehat{MC}_{H,t} \right) + (1 - s_H) \Delta\epsilon_t \right]. \quad (73)$$

Transmission of oil discovery news shocks. Equation (73) decomposes CPI inflation into two opposing channels. The first is a domestic demand channel that is inflationary. News of a future oil windfall generates a positive wealth effect. Households increase consumption and firms increase investment before the newly discovered reserves become productive. The resulting increase in domestic absorption raises demand for nontradable goods, which must be produced domestically. Higher nontradable demand increases labor demand and raises real marginal cost $\widehat{MC}_{N,t}$. Through the nontradable Phillips curve, this raises nontradable inflation $\pi_{N,t}$, which feeds into CPI inflation with weight s_C . Stronger domestic absorption and tighter factor markets may also raise marginal cost in the home tradable sector, $\widehat{MC}_{H,t}$, creating additional upward pressure on home-produced tradable inflation $\pi_{H,t}$.

The second is a disinflationary exchange-rate channel. As discussed in Lemma 1, we show that the nominal exchange rate tends to appreciate following oil discoveries, implying $\Delta\epsilon_t < 0$. This reduces the domestic-currency price of imported goods and dampens inflation in the final tradable composite $\pi_{T,t}$. Because tradable goods enter the CPI with weight $1 - s_C$, the exchange-rate channel offsets part of the inflationary pressure generated by stronger domestic demand.

Whether CPI inflation rises or falls on net depends on the relative strength of these two channels.

CPI inflation is more likely to rise when the nontradable share s_C is large and monetary policy limits the nominal exchange-rate appreciation, thereby weakening the disinflationary import-price channel. It also responds more strongly when κ_N and κ_H are larger: these coefficients fall as price-adjustment costs rise, so greater price flexibility lets sectoral firms pass increases in marginal costs through to prices more rapidly, amplifying the domestic-demand channel.

Equation (73) also highlights the role of inflation expectations. The terms $E_t\pi_{N,t+1}$ and $E_t\pi_{H,t+1}$ imply that current inflation depends on expected future inflation in each sector. Oil-discovery news raises expected future nontradable inflation because agents anticipate sustained demand pressure on the nontradable sector. Through the forward-looking NKPC, these higher inflation expectations feed directly into current nontradable price-setting, amplifying the demand channel even before current marginal costs fully adjust. The inflationary effect of news shocks thus operates not only through current cost pressures but also through the expectations-driven propagation built into the Phillips curve.

Equation (71) and (72) also explain the divergent CPI and PPI responses. In the model, PPI corresponds to tradable inflation $\pi_{T,t}$. The oil discovery raises home tradable marginal cost $\widehat{MC}_{H,t}$, putting upward pressure on $\pi_{H,t}$, but the associated appreciation ($\Delta\epsilon_t < 0$) lowers import prices and dampens $\pi_{T,t}$. Because PPI excludes the nontradable component, it is less exposed to the inflationary demand channel and more exposed to the disinflationary exchange-rate channel. The model thus generates a positive CPI response alongside a muted PPI response following oil discovery news shocks.

Exchange-rate regime and the inflation response. We examine how the exchange-rate regime shapes the inflation response to oil discoveries. Recall that the coefficient ϕ_e in the Taylor rule (62) governs the degree of exchange-rate management: a higher ϕ_e corresponds to a more tightly managed, or less flexible, exchange-rate regime, whereas a lower ϕ_e corresponds to a more flexible regime. This parameter affects CPI inflation through both terms in equation (73). When ϕ_e is high, the central bank leans against nominal appreciation, weakening the disinflationary effect of the $\Delta\epsilon_t$ term. Resisting appreciation also requires a lower nominal interest rate than would otherwise prevail, even as output and inflation rise. This monetary accommodation further stimulates domestic demand and raises nontradable inflation through the Phillips curve. Both forces amplify the CPI inflation response.

When ϕ_e is low, the opposite occurs. The exchange rate appreciates more sharply on impact, strengthening the disinflationary contribution of the $\Delta\epsilon_t$ term. At the same time, a central bank

that puts less weight on exchange-rate stabilization can raise the nominal interest rate in response to the demand-driven increase in output and inflation, which further supports appreciation through uncovered interest parity. The exchange-rate channel therefore offsets a larger share of the demand-driven pressure, leaving CPI inflation only weakly responsive to the same news shock. In summary, our model implies that oil-discovery news shocks are more likely to generate a stronger CPI inflation response under inflexible exchange-rate regimes than under flexible exchange-rate regimes.

5 Calibration and simulation

Next, we calibrate the model to simulate the dynamic effects of oil-discovery news shocks on inflation and other macro variables. The calibration does not aim to match the empirical cumulative responses in Section 3 quantitatively, as the underlying model parameters may vary substantially across countries. We instead take parameters from the literature and ask whether the model reproduces the empirical patterns qualitatively.

The oil discovery shock is normalized so that the present discounted value of the future increase in oil production equals one percent of initial GDP. The defining feature of the experiment is that agents receive the news before oil production rises. In the benchmark calibration, newly discovered oil becomes part of producing reserves after $j = 5$ years, matching the expected production lag assumed in the empirical NPV construction.

5.1 Calibration

The model is calibrated at annual frequency. Parameter values are chosen from standard small open economy DSGE models, from the oil-discovery framework of [Arezki et al. \(2017\)](#), and from steady-state targets. Table 4 reports the benchmark calibration.

Most preference and technology parameters are standard. The discount factor is set to $\beta = 0.943$, implying a 6 percent annual steady-state real interest rate, which is relatively high as our sample covers many developing economies with relatively high returns to capital. The exogenous world interest rate i_t^w is calibrated to an annualized rate of six percent, sharing the same steady-state value as the domestic interest rate $\bar{i} = 1/\beta - 1$. Preferences are logarithmic in consumption, $\sigma = 1$. The inverse Frisch elasticity is set to $\gamma = 1$, and the labor disutility parameter is calibrated to $\chi = 5.78$ so that steady-state labor supply is $\bar{L} = 1/3$.

The annual depreciation rate is $\delta = 0.10$. The capital share is $\alpha_T = 0.40$ in the home tradable

sector and $\alpha_N = 0.33$ in the nontradable sector. The oil-sector production parameters follow [Arezki et al. \(2017\)](#): the capital share is $\alpha_k = 0.49$, the labor share is $\alpha_L = 0.13$, and the remaining share, $1 - \alpha_k - \alpha_L = 0.38$, is assigned to producing reserves. The steady-state reserve inflow is set to $\zeta = 0.03$, which helps match an oil-output-to-GDP ratio of six percent.

The expenditure shares and elasticities govern our mechanism. Nontradables account for 70 percent of both consumption and investment ($\omega_C = \omega_I = 0.70$). Home goods account for 60 percent of the tradable bundle ($\omega_H = 0.60$); the 40 percent import share gives exchange-rate appreciation substantial scope to dampen tradable inflation. The open-economy block features imperfect substitution between domestic and foreign goods. The elasticity between tradables and nontradables is $\zeta = 0.44$, as in [Stockman and Tesar \(1995\)](#); the home-foreign tradable elasticity η and the foreign demand elasticity η^* are both 1.5. Foreign prices and the world oil price are normalized to one: $P_F^* = P_T^* = P_O^* = 1$. The foreign demand shifter is set to $Y^* = 0.044$, which helps match a steady-state trade-balance-to-GDP ratio of four percent. The steady-state nominal exchange rate is normalized to $\bar{\mathcal{E}} = 1$.

Nominal rigidities are asymmetric, and the asymmetry matters. Nontradable price adjustment is costlier than tradable ($\phi_N = 10$ versus $\phi_T = 5$), consistent with evidence that services prices are stickier than goods prices. The elasticity of substitution across varieties is set to $\varepsilon_N = \varepsilon_H = 6$, implying a steady-state markup of 1.2 in both sectors. The investment adjustment cost parameter is $\psi_h = 1.4$ for each sector, which prevents implausibly sharp reallocation of investment across sectors after the news.

For the Taylor rule, the interest-rate smoothing parameter is $\rho_i = 0.8^4$, the annual counterpart of the quarterly value of 0.8 estimated by [Smets and Wouters \(2007\)](#). The responses to CPI inflation and output are $\phi_\pi = 1.50$ and $\phi_y = 0.125$, as in [Schmitt-Grohé and Uribe \(2022\)](#). For the exchange-rate response coefficient, [Itskhoki and Mukhin \(2025\)](#) set $\phi_e = 0$ under a fully floating exchange-rate regime and $\phi_e = 5$ under a pegged regime in their small open economy New Keynesian model (online appendix, Table A4). Their rule responds to the deviation of the exchange rate from its steady-state level, whereas our rule responds to exchange-rate movements and therefore admits greater exchange-rate volatility; we accordingly treat $\phi_e = 5$ as an upper bound. Because our empirical panel pools countries under different exchange-rate regimes, the benchmark calibration uses an intermediate $\phi_e = 2.4$. To characterize the role of exchange-rate flexibility, we compare a high value of $\phi_e = 4$, corresponding to an inflexible regime, against a low value of $\phi_e = 0.5$, corresponding to a flexible regime. Finally, the debt-elastic premium $\chi_B = 0.0001$ follows [Schmitt-Grohé and Uribe](#)

(2003) and serves only to pin down a stationary net foreign asset position.

5.2 Simulation results

To be consistent with the empirical impulse responses, the simulated responses are cumulative and correspond to a single normalized oil-discovery news shock.

5.2.1 Inflation and Real Exchange Rate

Proposition 3 predicts that oil-discovery news raises CPI inflation through nontradable inflation, while the exchange-rate channel damps tradable inflation. Figure 8 confirms this prediction. Panel (a) shows the cumulative CPI response: the model generates a positive response on impact, the cumulative increase rises during the first four years following the oil discovery, and afterwards the cumulative effects start to decline gradually as domestic supply catches up. The increase is driven by the nontradable sector: panel (c) shows nontradable inflation rising sharply after the news shock, while panel (d) shows tradable inflation nearly flat and very limited throughout the simulation horizon. Panel (b) shows the cumulative impulse response of the real exchange rate, suggesting persistent appreciation, which partially offsets the inflation pressure.

The mechanism is the demand channel described in Proposition 3. The news shock raises expected lifetime wealth and stimulates domestic absorption before oil production begins. Since nontradable goods must be produced domestically and their supply is constrained by installed capital and nominal rigidities, the increase in demand pushes up nontradable marginal costs and nontradable prices through the nontradable Phillips curve (31).

Tradable inflation remains muted for two reasons, both visible in equation (71). First, the tradable price index $P_{T,t}$ is a composite of the price of domestically produced tradable goods $P_{H,t}$ and the domestic-currency price of imported foreign goods $\mathcal{E}_t P_{F,t}^*$. Even though stronger domestic demand puts some upward pressure on home tradable prices, this pressure is largely offset by the decline in the import price of foreign inputs as the home currency appreciates. Second, the weight of home-produced goods in the tradable composite, s_H , is less than one, so the pass-through of home tradable marginal cost increases to final tradable prices is incomplete.

The appreciation of the real exchange rate in Panel (b) follows Lemma 1 and Proposition 2. The anticipated future resource windfall raises demand for nontradables, driving up the relative price of nontradables, $\hat{p}_t^{N/T}$, while stronger demand and the nominal exchange rate appreciation raise the relative price of home tradable inputs to the foreign tradable input, driving up $\hat{p}_t^{H/F}$. Both channels

push the real exchange rate downward—an appreciation, given the convention in equation (67).

The simulated aggregate and sectoral price responses reproduce the empirical patterns in Figures 2 and 3.²⁰ Nontradable prices rise while tradable prices stay flat, and CPI rises while PPI—which corresponds to the tradable price index $P_{T,t}$ in the model—does not respond. The model thus rationalizes why the inflationary effect of oil discovery news is concentrated in consumer prices, which place substantial weight on nontradables, rather than in producer prices, which are dominated by tradable goods.

Production lags and the persistence of the inflation response. The empirical and model-implied cumulative CPI responses in Figures 2 and 8 move together in the first few years after an oil-discovery news shock but diverge at longer horizons. In the baseline simulation, the expected development lag is fixed at five years, so the inflation response peaks early and declines as the anticipated expansion in oil production approaches and begins. In the data, by contrast, the estimated response averages across discoveries whose realized production lags vary substantially, as shown in Figure A.1. Realized lags are observed ex post and partly reflect delays that were not foreseeable at the time of discovery, so they proxy imperfectly for the dispersion of expected lags. Longer development delays sustain news about future income and postpone the arrival of new oil production, keeping demand-driven inflationary pressure alive at horizons where the baseline simulation has already reverted. To illustrate this mechanism, Figure 9 plots model simulations of the cumulative impulse response of inflation following a large oil discovery with alternative expected lags of 5, 7, 10, and 13 years. Across all specifications, inflation increases after the discovery, confirming the robustness of the inflationary response. At the same time, longer lags dampen the initial increase but make the response more persistent, which suggests one reason the empirical cumulative response is flatter and more sustained than the five-year benchmark simulation.

5.2.2 Real activity and trade flows

Panels (a)–(c) of Figure 10 report output, consumption, and investment, all of which increase following the oil discovery news shock. Consumption rises immediately on impact because households anticipate higher future income from oil production and exports. Investment also expands, reflecting both the wealth effect and the incentive to accumulate capital in anticipation of stronger

²⁰Because the model defines inflation as the log difference in price level between two periods, the cumulative inflation response from period 0 to h equals the cumulative change in the log price level from period -1 to h —the object estimated by the local projections. The simulated cumulative impulse responses of inflation in Figure 8 are therefore directly comparable to the empirical cumulative changes in log price in Figures 2 and 3.

future production. Output rises gradually during the anticipation phase and accelerates once the newly discovered oil becomes productive at $t = 5$. Because activity expands before oil production begins, oil-discovery news operates like an aggregate demand shock.

Panels (d)–(f) of Figure 10 report the cumulative responses of exports, imports, and the current account-to-GDP ratio. These patterns are largely consistent with the empirical responses in Figure 5. In the short run, imports increase because the news shock raises expected lifetime wealth and stimulates domestic absorption before oil exports expand, and the real appreciation makes foreign goods relatively cheaper. By contrast, non-oil exports decline slightly at first. This decline reflects a Dutch disease mechanism: the real appreciation raises the relative price of domestically produced tradable goods in foreign markets, weakening the competitiveness of the home tradable sector. As a result, the current account deteriorates before oil production begins.

However, once the new reserves become productive, oil output and export revenues increase. Total exports then rise sharply, and the current account turns positive, consistent with the data. The model thus captures the intertemporal pattern of the current account emphasized by [Arezki et al. \(2017\)](#): oil discoveries initially worsen the current account because the economy borrows more and saves less before production from the new fields begins, but the external position improves once oil exports commence.

5.2.3 Exchange-rate regime and inflation

Our model implies that oil-discovery news shocks are more likely to generate a stronger CPI inflation response under inflexible exchange-rate regimes than under flexible exchange-rate ones. Figure 11 tests this implication by comparing an inflexible exchange-rate regime with a high $\phi_e = 4$ against a flexible regime with a low $\phi_e = 0.5$. The results confirm the prediction: the cumulative CPI response is substantially larger under the inflexible regime and remains positive throughout the simulation horizon. When monetary policy places a larger weight on exchange-rate stabilization, nominal exchange-rate adjustment is limited, and the central bank leans against the appreciation rather than responding fully to the rise in inflation and output. The economy therefore cannot absorb the oil-discovery news shock through nominal appreciation and lower imported goods prices. The wealth effect instead translates more directly into stronger domestic absorption and higher demand for nontradable goods, pushing up nontradable prices and aggregate CPI inflation. By contrast, under the flexible exchange-rate regime, the nominal exchange rate adjusts freely and appreciates on impact. The resulting decline in the domestic-currency price of imported goods dampens tradable

inflation and offsets part of the increase in nontradable inflation, so the aggregate CPI response is weaker and eventually turns slightly negative.

This simulated ordering reproduces the empirical split documented in the regime analysis: the response of core CPI is significantly larger under inflexible exchange-rate regimes and small and statistically insignificant under flexible regimes (Figure 7).

Moreover, Figure 11 also presents the cumulative CPI responses to an oil-discovery news shock under two additional calibrations of inflexible exchange-rate regimes, with $\phi_e = 3$ and $\phi_e = 5$. A higher value of ϕ_e corresponds to a more tightly managed nominal exchange rate. The figure shows that the qualitative pattern of the CPI response is similar across $\phi_e \in \{3, 4, 5\}$. However, the magnitude of the response increases with ϕ_e : cumulative CPI inflation becomes larger and more persistent as the exchange rate is managed more tightly. Intuitively, stronger intervention limits nominal appreciation and weakens the disinflationary import-price channel, allowing demand-driven pressures in the nontradable sector to play a larger role in aggregate consumer prices. This sensitivity exercise therefore reinforces the paper’s main mechanism: exchange-rate management amplifies the inflationary effects of oil-discovery news.

5.2.4 Summary

In summary, our calibrated model reproduces the main qualitative patterns of the empirical impulse responses, although our model abstracts from many additional frictions or institutional features, such as sticky wages, financial frictions, information frictions, and fiscal policies. Oil-discovery news shocks raise CPI inflation through the domestic demand channel operating on the nontradable sector, while tradable and PPI inflation remain muted because exchange-rate appreciation offsets the increase in home tradable costs. Real activity expands during the anticipation phase, and the current account follows an intertemporal path of initial deterioration followed by improvement once oil production begins. The inflationary effect is amplified under less flexible exchange-rate regimes, consistent with the empirical evidence.

6 Conclusion

Whether news shocks are inflationary is a central question not only for canonical New Keynesian models but also for central banks seeking to formulate appropriate monetary policy. This paper shows that news about future output can be inflationary in open economies. Large oil discoveries

with a realized production lag of at least five years provide a useful setting because they reveal information about higher future output while leaving current productive capacity essentially unchanged.

The evidence indicates that oil-discovery news sets in motion a broad macroeconomic adjustment before new oil is produced. Domestic absorption rises, the current account initially weakens, and the real exchange rate appreciates on impact. The price response, however, is highly uneven across sectors and exchange-rate regimes. Consumer prices rise because nontradable prices increase, whereas producer prices and tradable prices respond little. Moreover, the inflationary effect is concentrated in economies with inflexible exchange-rate regimes, where nominal appreciation is limited and the required real appreciation is instead achieved through higher domestic prices.

These findings contribute to the literature on news shocks by emphasizing that the inflationary consequences of news depend on both the nature of the news and the openness of the economy. In closed-economy settings, news about future productivity is often disinflationary because it signals lower future marginal costs. By contrast, oil-discovery news raises expected wealth without expanding current supply, generating an immediate increase in demand. In an open economy, the exchange rate then determines how this pressure is allocated between nominal appreciation and domestic inflation. For policymakers in resource-rich economies, the exchange-rate regime is therefore not a background feature but a central determinant of how anticipated resource windfalls pass through to consumer inflation.

Finally, the canonical shock analyzed in the news-driven business-cycle literature is a shock to future aggregate TFP. Oil-discovery news differs from TFP news along two margins. The first is the nature of the shock: productivity news raises future supply and lowers expected future marginal costs, whereas an oil discovery raises expected future income without expanding current supply, so its inflationary impulse operates through demand. The second is that oil-discovery news is a sectoral shock. It announces higher future output in a specific, export-oriented resource sector rather than an economy-wide productivity gain, and the resulting wealth effect falls disproportionately on nontradable goods, whose supply cannot be expanded through imports and whose prices are sticky. The implication for the TFP-news literature is that the inflation response to news is not a property of news per se, but depends on what the news is about and on the sectoral and open-economy structure through which expectations reach prices.

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Table 1: The Spatial and Temporal Distribution of Large Discoveries (1980–2019)

Regions	Decade				Total
	1980s	1990s	2000s	2010s	
Africa	41	60	69	81	251
Asia	67	81	58	56	262
Europe	45	20	10	12	87
Middle East	71	46	58	56	231
North America	25	14	20	19	78
Oceania	9	11	22	15	57
Russia	110	32	43	32	217
South America	19	10	37	60	126
World	387	274	317	331	1,309

Notes: This table aggregates the number of large oil and gas discoveries with a realized production lag of at least five years by regions and decade. The data covers the period from 1980 to 2019. Regions are classified by Rystad. One country can have multiple large discoveries for a given year. Discoveries with a production lag of less than five years are excluded from the table (336 of the 1,645 large discoveries).

Table 2: Data Definition and Sources

Variable	Definition	Source
Core CPI	Core consumer price index (first year in data = 100)	Global database of inflation (2025)
PPI	Producer price index (first year in data = 100)	Global database of inflation (2025)
Tradable price	Price index in tradable sector (2015 = 100), local currency unit	UN Stats (2025)
Nontradable price	Price index in nontradable sector (2015 = 100), local currency unit	UN Stats (2025)
Real GDP	GDP in constant prices, local currency unit	World Bank (2025)
Real final consumption	Final consumption expenditure in constant local currency unit	World Bank (2025)
Real gross capital formation	Gross capital formation in constant local currency unit	World Bank (2025)
Real imports	Imports in constant local currency unit	World Bank (2025)
Real exports	Exports in constant local currency unit	World Bank (2025)
Current account/GDP	—	World Bank (2025)

Table 3: Summary Statistics of Macro Variables

Variable	Countries	Obs.	Mean	Std. Dev.	Min	Max
Log core CPI	87	2,020	527.130	57.631	452.453	744.481
Log PPI	89	2,170	534.651	77.381	387.324	912.921
Log nontradable price	97	3,099	580.432	108.011	362.897	1182.689
Log tradable price	97	3,022	555.327	96.428	403.913	1037.169
Log real GDP	101	3,772	2778.752	308.249	2111.892	3723.383
Log real final consumption	96	3,451	2468.717	192.547	1966.330	3049.034
Log real gross capital formation	99	3,542	2345.136	203.595	1613.249	2945.173
Log real imports	96	3,470	2383.324	184.702	1853.932	2877.243
Log real exports	98	3,540	2372.994	199.438	1344.919	2860.770
Current account/GDP	96	3,216	-1.784	7.302	-32.096	25.758
Log real exchange rate	88	181,204	0.043	0.506	-10.660	11.513

Notes: The sample covers the period from 1980 to 2019. As detailed in Table 2, all real variables are expressed in constant local currency units, with the exception of the real exchange rate. Variables prefixed with Log are constructed by taking the natural logarithm of the original series and multiplying by 100; however, the logarithmic transformation of the real exchange rate is not scaled by 100 owing to its different specification. Finally, the current account-to-GDP ratio is not log-transformed and is expressed directly as a percentage.

Table 4: Baseline Calibration

Parameter	Value	Description	Source / Rationale
<i>Preferences</i>			
β	0.943	Subjective discount factor	Arezki et al. (2017)
σ	1	Relative risk aversion	Log utility
γ	1	Inverse Frisch elasticity	Arezki et al. (2017)
χ	5.78	Labor disutility scale	Match $\bar{L} = 1/3$
<i>Technology and production</i>			
δ	0.10	Annual depreciation rate	Arezki et al. (2017)
α_T	0.40	Capital share in home tradables	Benchmark calibration
α_N	0.33	Capital share in nontradables	Benchmark calibration
α_L	0.13	Labor share in oil production	Arezki et al. (2017)
α_k	0.49	Capital share in oil production	Arezki et al. (2017)
ς	0.03	Steady-state reserve inflow	Match $\bar{Y}_O/\bar{Y} = 0.06$
<i>Open economy and trade</i>			
ω_H	0.60	Home share in tradable bundle	Galí and Monacelli (2005)
ω_C	0.70	Nontradable share in consumption	Benchmark calibration
ω_I	0.70	Nontradable share in investment	Benchmark calibration
ζ	0.44	Elasticity between tradables and nontradables	Stockman and Tesar (1995)
η	1.50	Elasticity between home and foreign tradables	Schmitt-Grohé and Uribe (2022)
η^*	1.50	Foreign demand elasticity	Benchmark calibration
P_F^*	1	Foreign input price	Normalization
P_T^*	1	Foreign tradable price index	Normalization
P_O^*	1	World oil price	Normalization
i_t^w	0.06	World interest rate	Normalization
Y^*	0.044	Foreign demand shifter	Match $\bar{T}\bar{B}/\bar{Y} = 0.04$
$\bar{\mathcal{E}}$	1	Steady-state exchange rate	Normalization
χ_B	0.0001	Debt-elastic premium	Schmitt-Grohé and Uribe (2003)
<i>Nominal and investment rigidities</i>			
ϕ_N	10	Nontradable price adjustment cost	Moderate price stickiness
ϕ_T	5	Home tradable price adjustment cost	Moderate price stickiness
ψ_h	1.4	Investment adjustment cost	Christiano et al. (2005)
$\varepsilon_N, \varepsilon_H$	6	Elasticity across differentiated varieties	Markup of 1.2
<i>Monetary policy</i>			
ρ_i	0.8 ⁴	Interest-rate smoothing	Smets and Wouters (2007)
ϕ_π	1.50	Inflation response	Schmitt-Grohé and Uribe (2022)
ϕ_y	0.125	Output response	Schmitt-Grohé and Uribe (2022)
ϕ_e	2.4	Exchange-rate response	Itskhoki and Mukhin (2025)

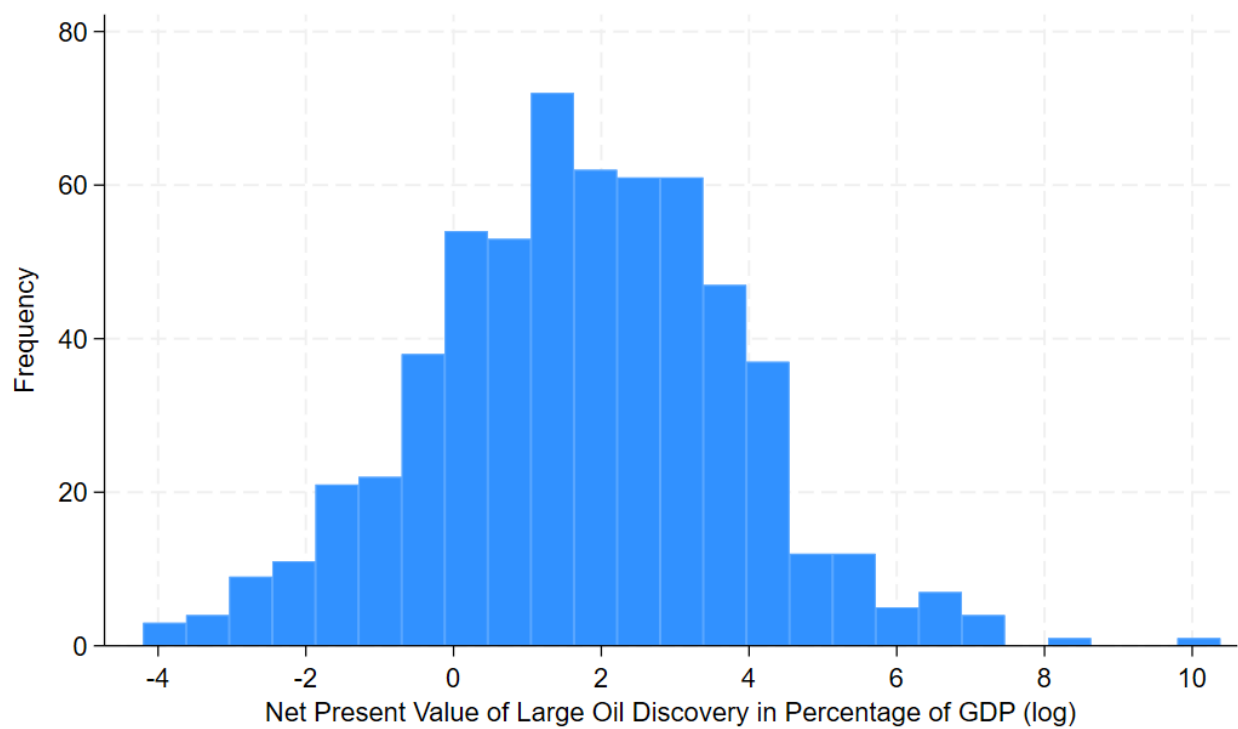
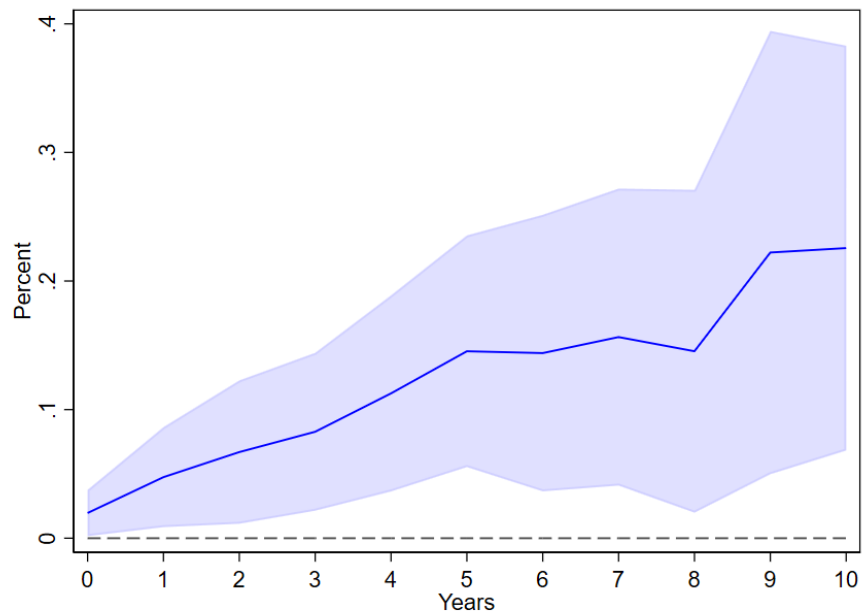
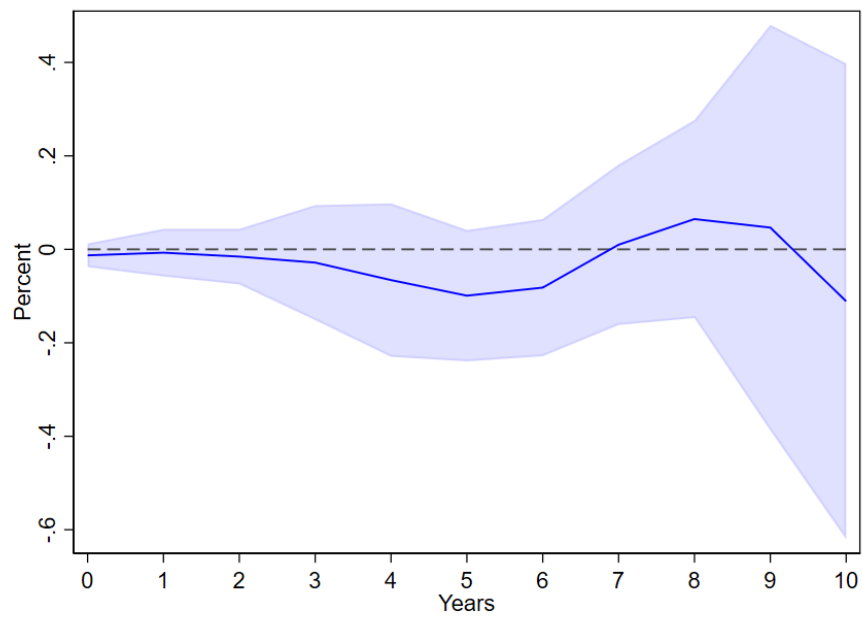


Figure 1: Distribution of the NPV of Large Oil Discoveries, 1980–2019

Notes: The figure presents the distribution of the log NPVs of large oil discoveries listed in Table 1. As in that table, the sample covers only discoveries with a realized production lag of at least five years. While the large discoveries in Table 1 are reported at the field level, their NPVs are aggregated to the country-year level, as shown in equation (2).



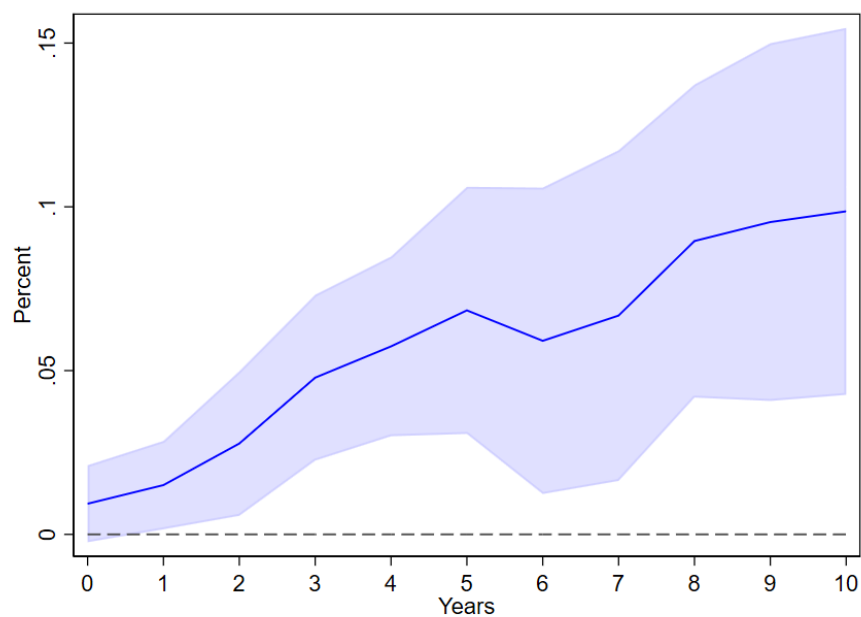
(a) Core CPI



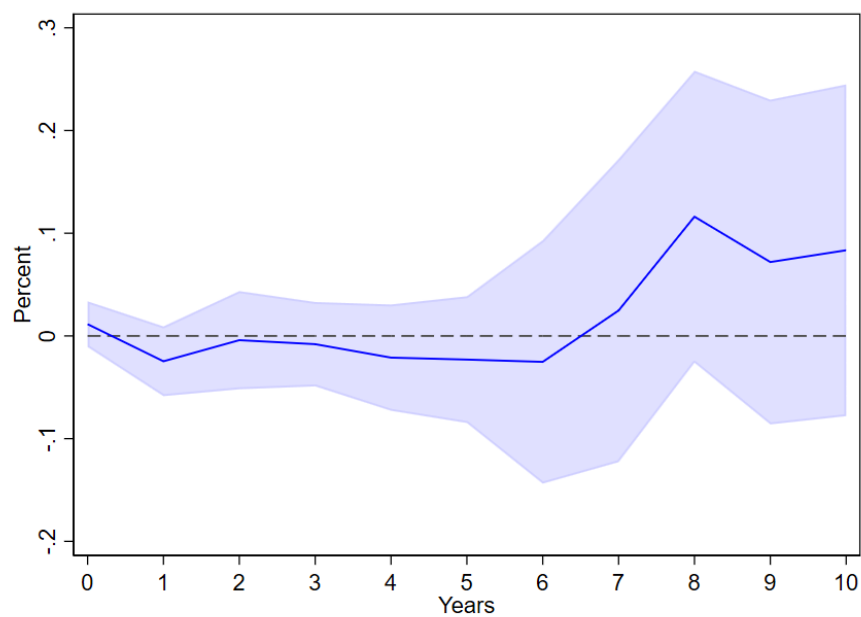
(b) PPI

Figure 2: Impact of Large Oil Discoveries on Core CPI and PPI

Notes: The figure presents the cumulative impulse responses of Core CPI and PPI to large oil discoveries with an NPV equal to 1% of GDP. The line indicates point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis shows the percentage change.



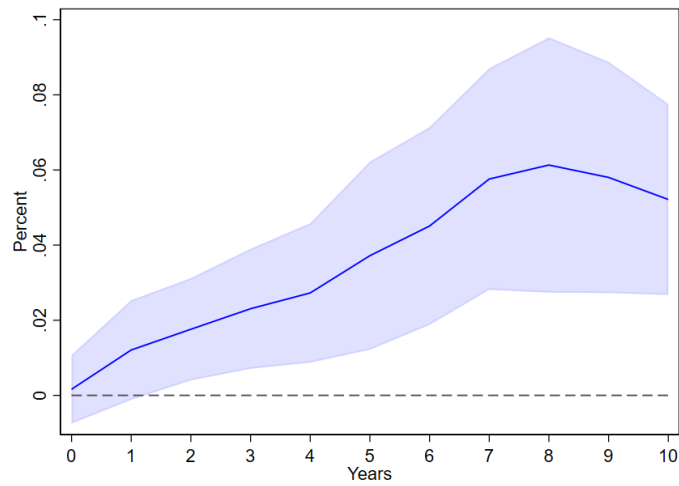
(a) Nontradable Price Index



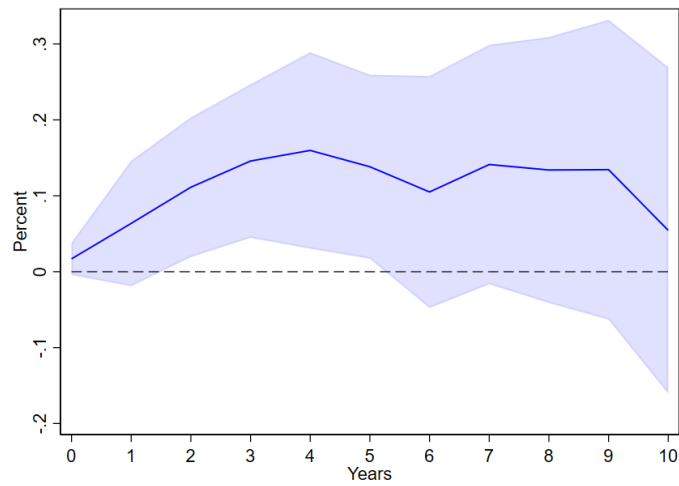
(b) Tradable Price Index

Figure 3: Impact of Large Oil Discoveries on Tradable and Nontradable Price Indices

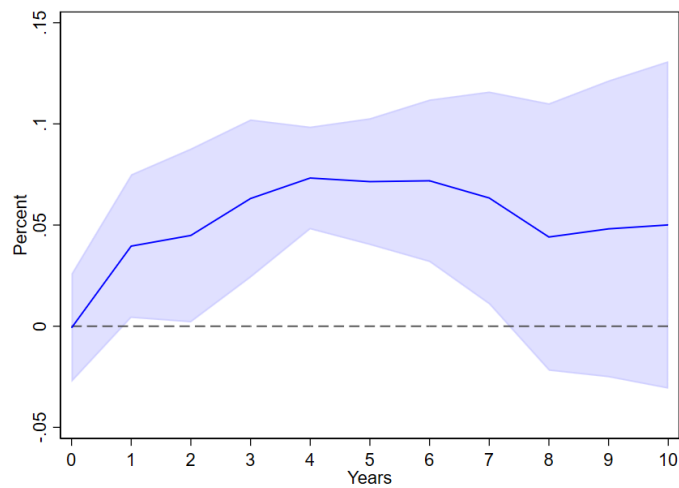
Notes: The figure presents the cumulative impulse responses of tradable and nontradable sector price indices to large oil discoveries with an NPV equal to 1% of GDP. The tradable and nontradable sectors refer to the manufacturing and services sectors, respectively. The line indicates point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis shows the percentage change.



(a) GDP



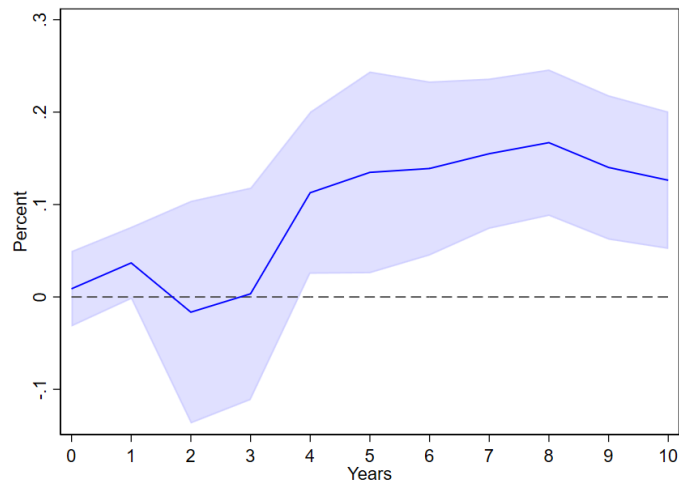
(b) Consumption



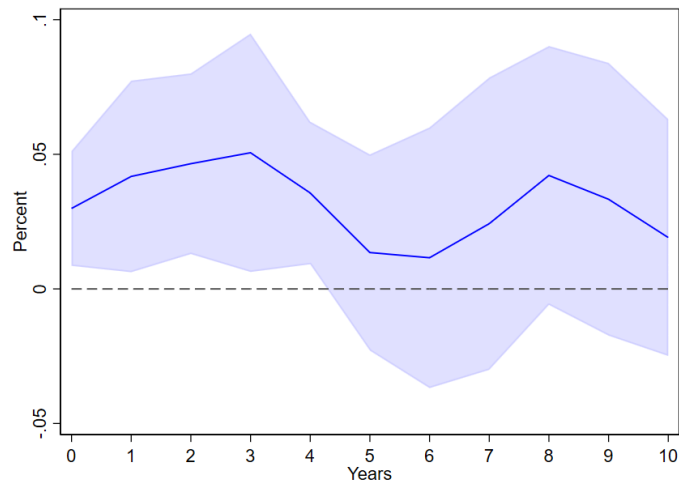
(c) Investment

Figure 4: Impact of Large Oil Discoveries on GDP, Consumption, and Investment

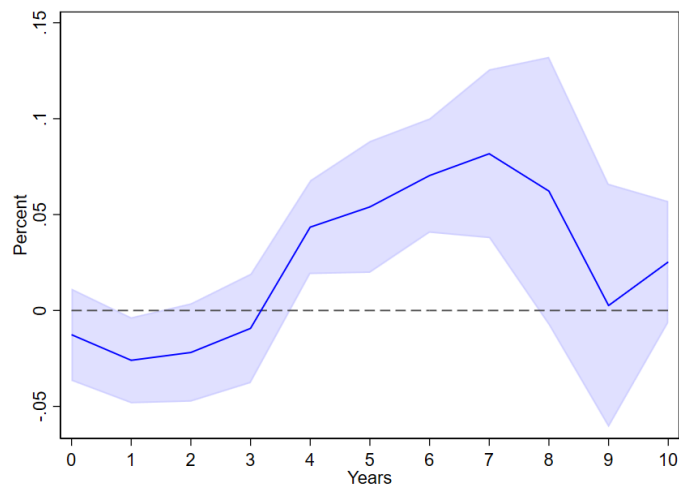
Notes: The figure presents the cumulative impulse responses of GDP, consumption, and investment to large oil discoveries with an NPV equal to 1% of GDP. The line indicates point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis shows the percentage change.



(a) Exports



(b) Imports



(c) Current Account/GDP

Figure 5: Impact of Large Oil Discoveries on Exports, Imports, and Current Account

Notes: The figure presents the cumulative impulse responses of real exports, real imports, and the current account/GDP ratio to large oil discoveries with an NPV equal to 1% of GDP. The line indicates point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis shows the percentage change.

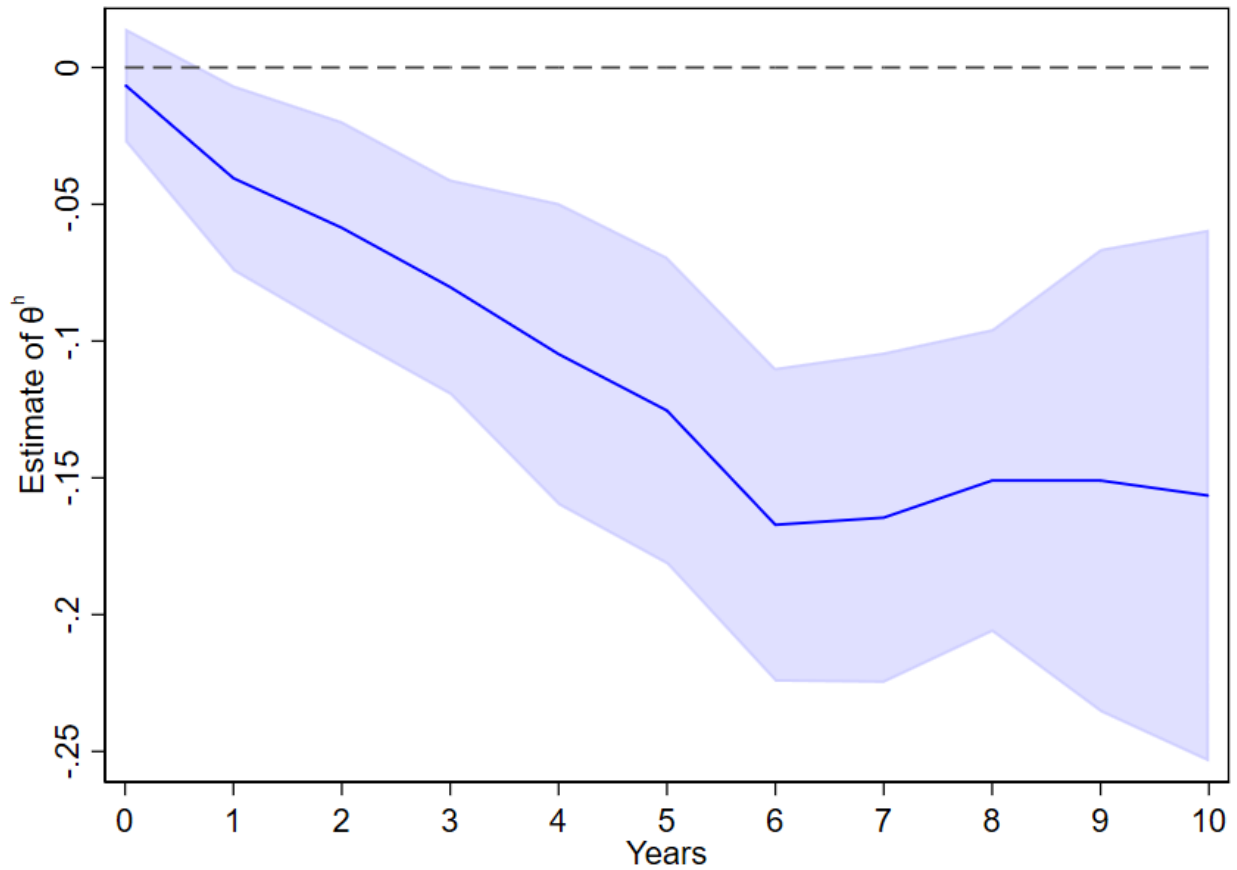


Figure 6: Impact of Large Oil Discoveries on the Bilateral Real Exchange Rate

Notes: The figure presents the cumulative response of the bilateral real exchange rate between country i and country j to a relative oil-discovery news shock, estimated from equation (5) on the 1980–2019 panel of country pairs. Intuitively, the vertical axis can be approximately read as the cumulative percentage change in the bilateral real exchange rate in response to a discovery in country i with an NPV equal to 1 percent of GDP and no discovery in country j . The solid line denotes point estimates and the shaded band the 90 percent confidence interval. Negative values indicate a real appreciation of country i 's currency relative to country j .

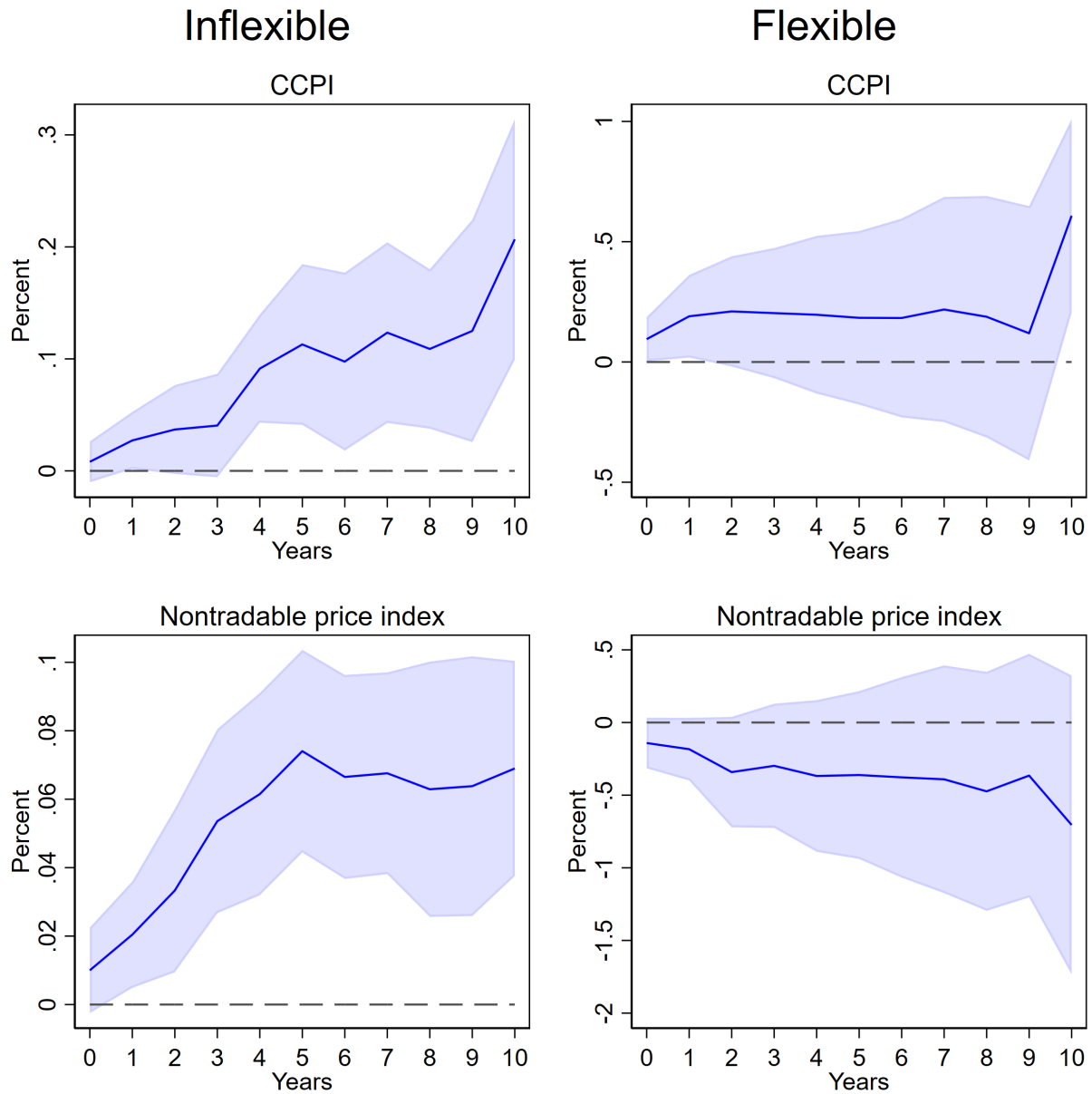


Figure 7: Cumulative Responses of Inflation under Different Exchange Rate Regimes

Notes: The figure presents the cumulative impulse responses of core CPI and the nontradable price index to a large oil discovery with an NPV equal to 1% of GDP, partitioned by exchange rate regime (inflexible versus flexible) based on [Ilizetzi et al. \(2019\)](#)'s classification. The line indicates point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis shows the percentage change.

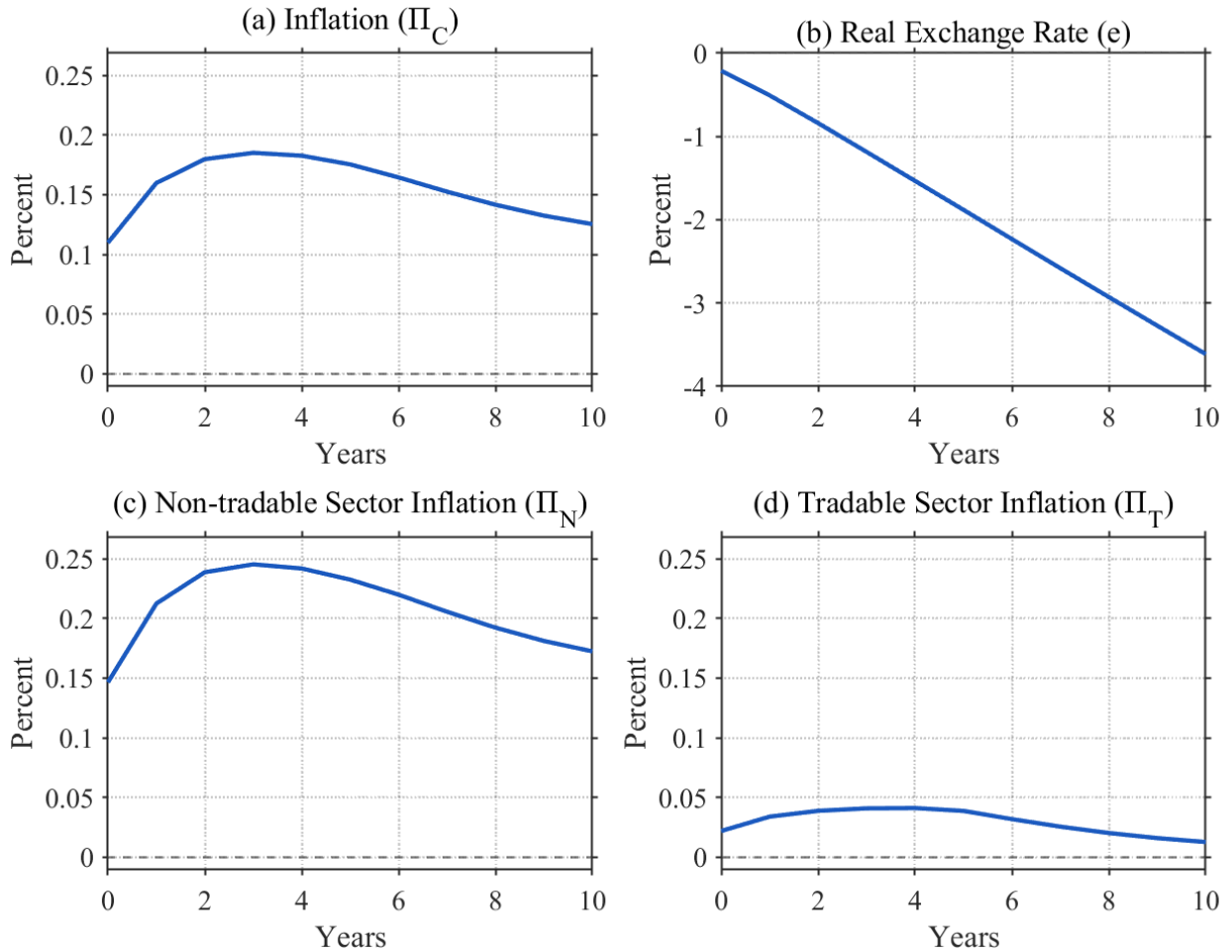


Figure 8: Cumulative Impulse Responses of Inflation and the Real Exchange Rate to an Oil-Discovery News Shock

Notes: The figure reports cumulative impulse responses to an oil-discovery news shock whose present discounted value equals one percent of initial GDP. All responses are cumulative: the value plotted at horizon h equals the sum of the responses from period 0 through h . The panels show (a) CPI inflation, (b) the real exchange rate, (c) nontradable inflation, and (d) tradable inflation. The oil discovery is announced in year 0, and the newly discovered oil becomes productive after $j = 5$ years. The vertical axis is in percent.

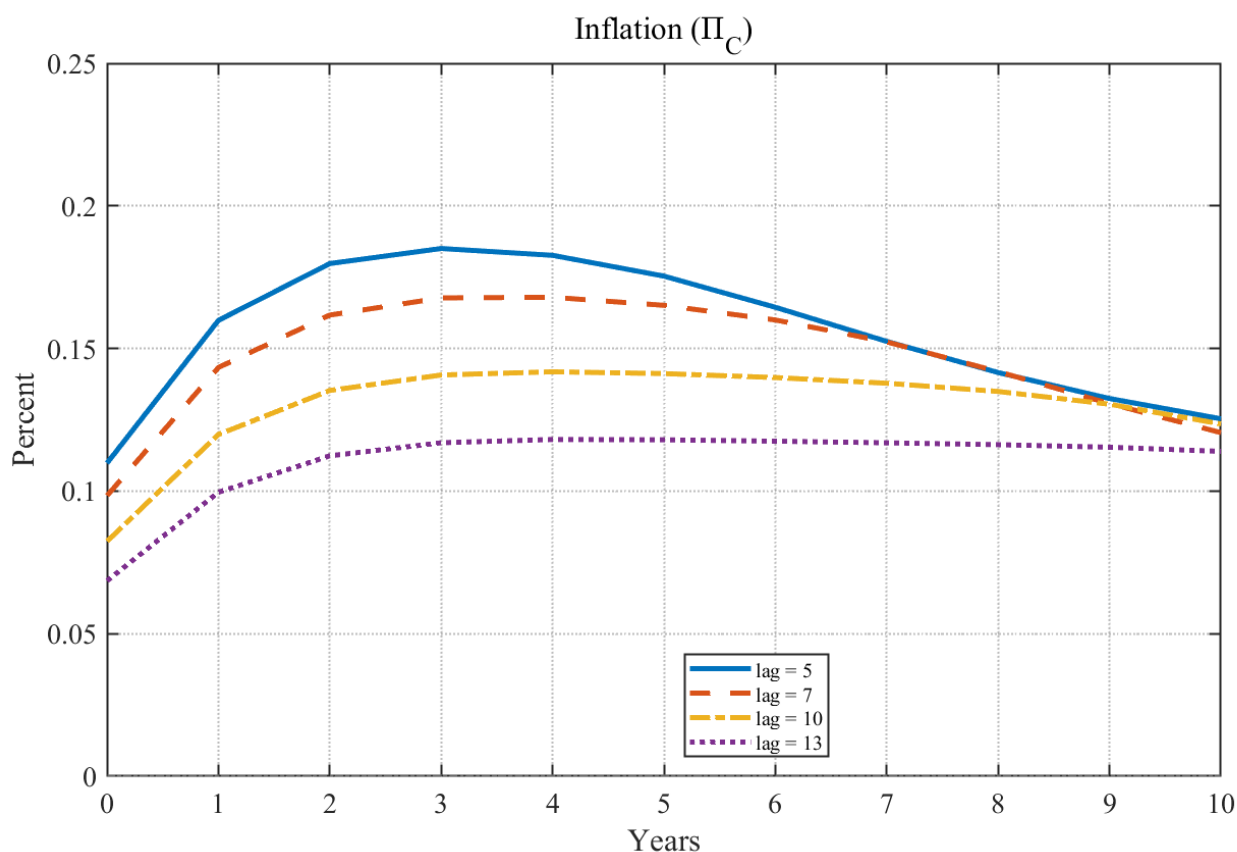


Figure 9: Cumulative Impulse Responses of Inflation to an Oil-Discovery News Shock with Different Expected Production Lags

Notes: The figure reports cumulative impulse responses of CPI inflation to an oil-discovery news shock whose present discounted value equals one percent of initial GDP. All responses are cumulative: the value plotted at horizon h equals the sum of the responses from period 0 through h . The oil discovery is announced in year 0, and the newly discovered oil becomes productive after j years, where j denotes the expected production lag. The lines correspond to $j = 5$ (solid blue), $j = 7$ (dashed red), $j = 10$ (dash-dotted yellow), and $j = 13$ (dotted purple). The vertical axis is in percent.

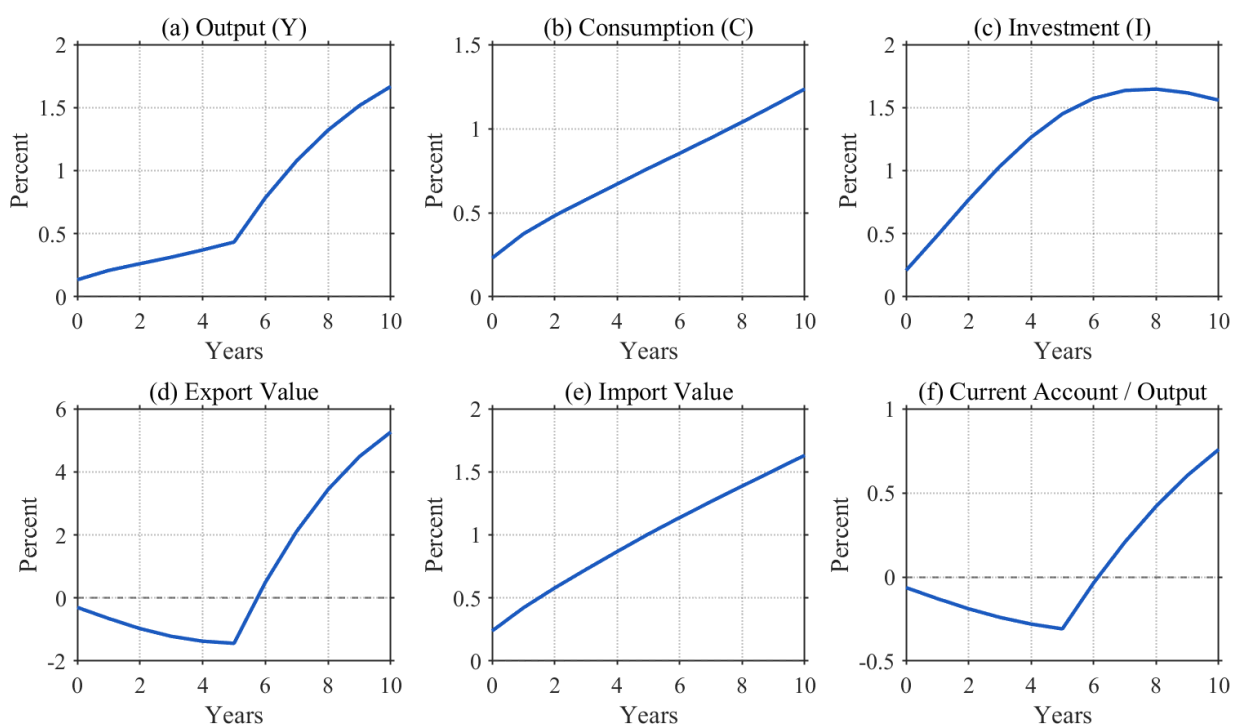


Figure 10: Cumulative Impulse Responses of Macroeconomic Variables to an Oil-Discovery News Shock

Notes: The figure reports cumulative impulse responses to an oil-discovery news shock whose present discounted value equals one percent of initial GDP. The panels show (a) GDP, (b) consumption, (c) investment, (d) exports, (e) imports, and (f) the current account-to-GDP ratio. The shock is announced in year 0, and the newly discovered oil becomes productive after $j = 5$ years. The vertical axis is in percent.

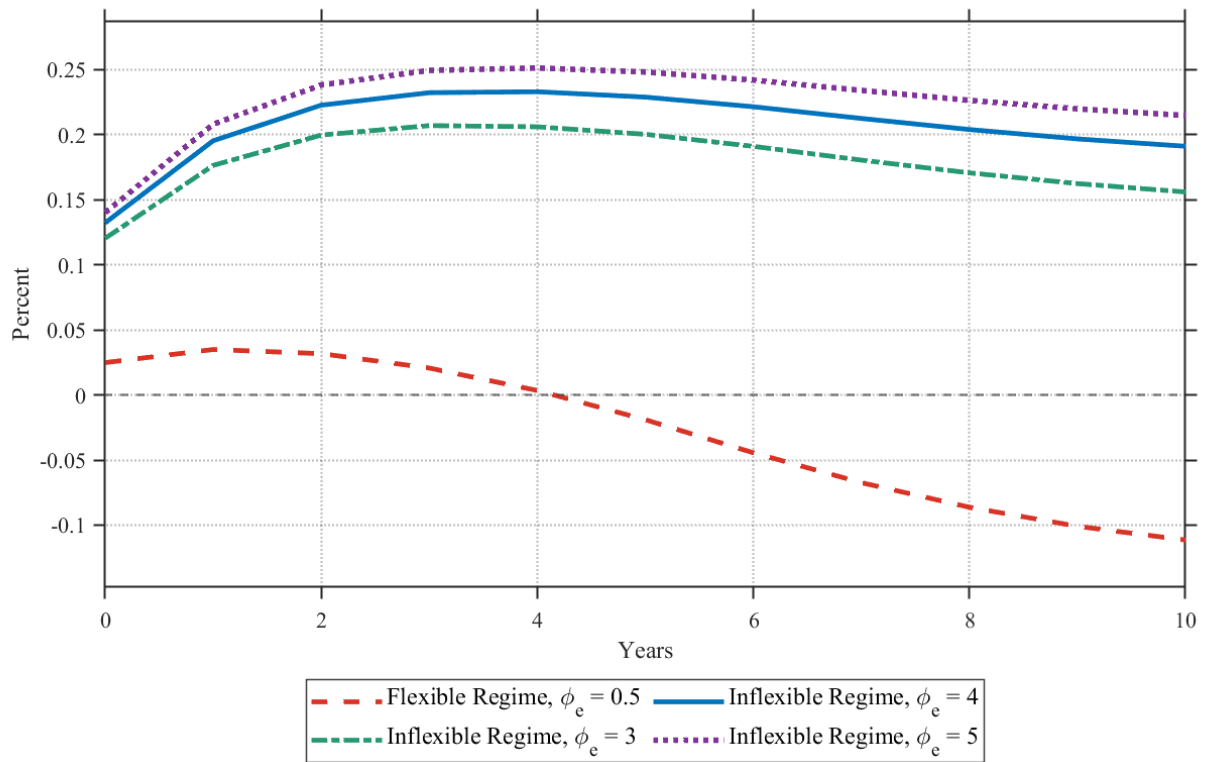


Figure 11: Cumulative Response of CPI Inflation under Different Exchange Rate Regimes

Notes: The figure reports cumulative impulse responses of CPI inflation to an oil-discovery news shock under different exchange-rate regimes. The dashed red line denotes the flexible exchange-rate regime ($\phi_e = 0.5$). The solid blue, dash-dotted green, and dotted purple lines denote inflexible exchange-rate regimes with $\phi_e = 4$, $\phi_e = 3$, and $\phi_e = 5$, respectively.

Online Appendix for “News Shocks and Inflation: Theory and Evidence from Large Oil Discoveries”

Bohou Cai, Liugang Sheng, and Hongyan Zhao

Not for Publication

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A Additional Empirical Analysis

A.1 Macroeconomic Data Construction and Cleaning

This subsection details the step-by-step procedures used to clean the macroeconomic series, construct continuous price indices, and treat extreme observations.

A.1.1 Price Index Construction and Break-Point Protocol

The Global Database of Inflation (GDI; [Ha et al., 2023](#)) provides annual inflation rates for headline CPI, core CPI, food CPI, energy CPI, and PPI. Because our local projection specification requires cumulative log price differences, we construct chained price indices $P_{i,t}^c$ via the recursive relation:

$$P_{i,t}^c = P_{i,t-1}^c \times \left(1 + \frac{\pi_{i,t}}{100}\right), \quad (74)$$

where $\pi_{i,t}$ is the annual percentage inflation rate. To ensure that the resulting series are statistically consistent across time, we implement the following four-step protocol:

1. **Continuous Segment Isolation:** Several developing and transition economies exhibit breaks in their historical reporting. For countries with missing intermediate years, we isolate a single, continuous time series. We default to retaining the most recent continuous episode, as modern statistical data collection adheres to more harmonized standards. However, if the latest segment contains fewer than six consecutive observations, we select the longest available historical block to preserve time-series degrees of freedom. This adjustment is applied to 10 countries for core CPI and 14 countries for PPI.
2. **Hyperinflation and Extreme Noise Truncation:** To prevent episodes of hyperinflation, extreme stagflation, or post-conflict currency resets from dominating the ordinary least squares (OLS) criterion, we trim country-year observations where annual inflation exceeds +50% or falls below −50%. If this truncation introduces an internal gap in a country’s series, we reapply the continuous-segment selection criteria described in Step 1.
3. **Base-Year Normalization:** For each country, the first observation of the final continuous sequence is normalized to $P_{i,t_0}^c = 100$. We use country-specific base years because no single year has non-missing observations for all countries. Because the dependent variable in our local projection is the cumulative change $\ln P_{i,t+h}^c - \ln P_{i,t-1}^c$, any choice of base year functions as an

additive constant in logs and cancels out mechanically.

4. **Sample of countries:** We exclude countries with fewer than six consecutive annual observations for core CPI or PPI, as such limited coverage likely reflects poor data quality and provides insufficient time-series variation for the empirical analysis. After these steps, the final sample contains 102 countries, of which 87 have at least six consecutive core CPI observations and 89 have at least six consecutive PPI observations.

For the sectoral price indices, only the truncation and continuous-segment isolation rules apply. Because the UNNAS sectoral value-added deflators are already level series, we do not chain them from inflation rates and do not re-base them; they remain expressed in the reference year reported in Table 2 (2015 = 100).

A.1.2 Winsorization of Macroeconomic Aggregates

Macroeconomic aggregates from the World Development Indicators (WDI) occasionally contain reporting anomalies, redenomination errors, or severe measurement disruptions. We treat outliers using a scale-preserving winsorization procedure:

- **Ratios to GDP:** For the current account balance, which enters directly as a share of nominal GDP, we winsorize the ratio at the 1st and 99th percentiles across the full pooled panel.
- **Constant LCU Levels:** Real aggregates (private consumption, investment, exports, and imports) are denominated in constant local currency units. Comparing raw levels across countries is uninformative due to vast scale differences and trend growth. We therefore evaluate outliers relative to domestic economic size: for each aggregate $X_{i,t}$, we compute its contemporaneous share of real GDP, $s_{i,t}^X = X_{i,t}/GDP_{i,t}$. If $s_{i,t}^X$ falls outside the 1st–99th percentile interval $[\underline{s}^X, \bar{s}^X]$, we calculate the scaling factor $\lambda_{i,t} = s_{\text{bound}}^X / s_{i,t}^X$ (where s_{bound}^X is the breached threshold) and rescale the level as $\tilde{X}_{i,t} = \lambda_{i,t} X_{i,t}$. This bounds implausible shares while preserving the internal national accounts scaling.

A.2 Distribution of Realized Production Lags

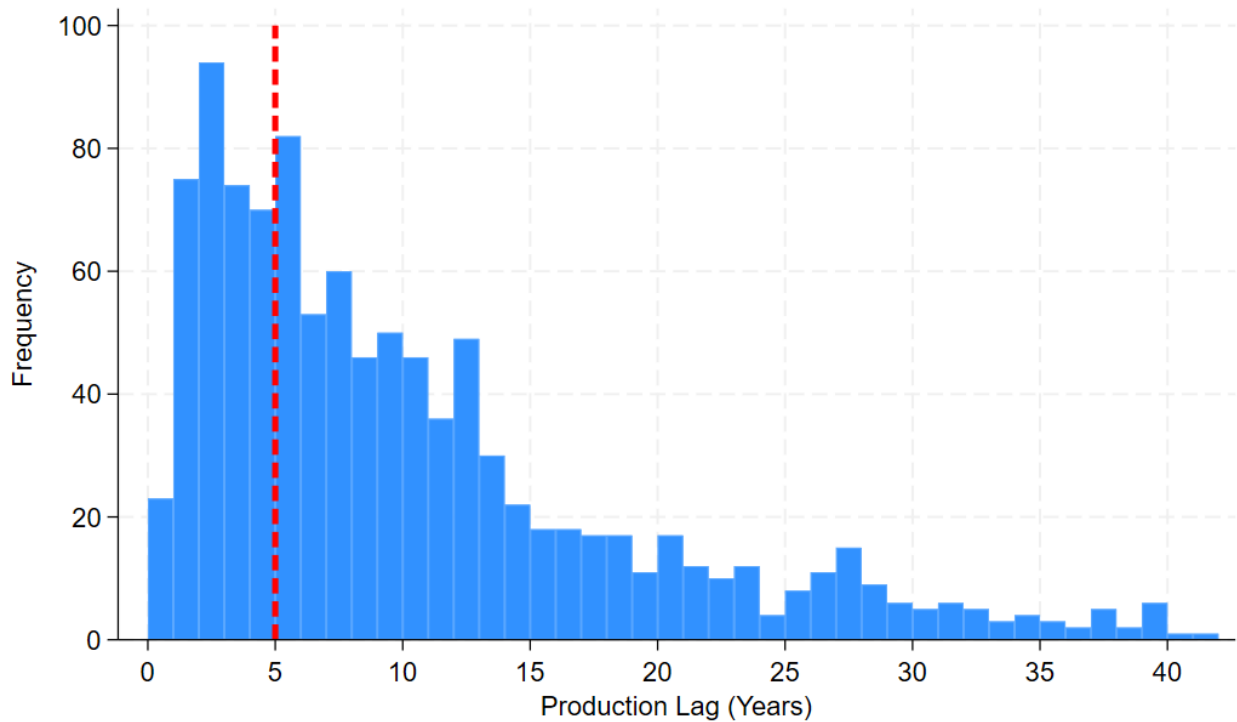


Figure A.1: Distribution of Realized Production Lags for Large Oil Discoveries (1980–2019)

Notes: This figure displays the frequency distribution of realized production lags—defined as the number of years between the initial discovery and the start of commercial production—for large oil discoveries made between 1980 and 2019. The plotted distribution consists of the 1,038 discoveries that have a recorded startup year prior to 2025. While this figure illustrates this broader set of discoveries, our baseline analysis sample further restricts this group by excluding discoveries with a realized production lag of less than five years.

A.3 Cumulative Impulse Response Estimates and Country Coverage

Table A.1: Cumulative Impulse Response Coefficients and Country Coverage

Horizon h	Price indices					Real activity			External sector		
	Core CPI (1)	PPI (2)	Nontradable (3)	Tradable (4)	GDP (5)	Consumption (6)	Investment (7)	Exports (8)	Imports (9)	CA/GDP (10)	Bilateral RER (11)
0	0.020	-0.013	0.009	0.011	0.002	0.017	-0.001	0.009	0.030	-0.013	-0.006
1	0.047	-0.007	0.015	-0.025	0.012	0.064	0.040	0.037	0.042	-0.026	-0.040
2	0.067	-0.016	0.028	-0.004	0.018	0.111	0.045	-0.016	0.047	-0.022	-0.059
3	0.083	-0.028	0.048	-0.008	0.023	0.146	0.063	0.003	0.051	-0.009	-0.080
4	0.113	-0.066	0.057	-0.021	0.027	0.160	0.073	0.113	0.036	0.043	-0.105
5	0.145	-0.099	0.068	-0.023	0.037	0.138	0.071	0.135	0.014	0.054	-0.125
6	0.144	-0.082	0.059	-0.025	0.045	0.105	0.072	0.139	0.012	0.070	-0.167
7	0.156	0.010	0.067	0.025	0.058	0.141	0.063	0.155	0.024	0.082	-0.165
8	0.145	0.065	0.090	0.116	0.061	0.134	0.044	0.167	0.042	0.062	-0.151
9	0.222	0.047	0.095	0.072	0.058	0.134	0.048	0.140	0.033	0.003	-0.151
10	0.226	-0.111	0.099	0.083	0.052	0.054	0.050	0.126	0.019	0.025	-0.157
Countries	85	87	96	96	101	96	99	98	96	96	5270
Country pairs											

Notes: Columns (1)-(10) report the cumulative impulse response coefficients β^h from the panel local projection in equation (1), and column (11) reports the coefficients θ^h from the dyadic specification in equation (5), each estimated at horizons $h = 0, \dots, 10$ years after the discovery. All coefficients refer to a discovery with an NPV equal to 1 percent of GDP. Columns (1)-(9) are expressed in percent, and column (10) is expressed in percentage points of GDP. In column (11), θ^h approximates the cumulative percentage appreciation ($\theta^h < 0$) or depreciation ($\theta^h > 0$) of country i 's currency against country j following a discovery news shock equal to 1 percent of GDP in country i , assuming partner j experiences no such shock. Standard errors are computed using the Driscoll and Kraay (1998) method but are omitted to conserve space; the corresponding 90 percent confidence intervals are plotted in the figures. The penultimate row reports the number of countries entering the horizon-0 regression for each outcome in columns (1)-(10), and the last row reports the number of country pairs in the regression of the bilateral real exchange rate at horizon-0.

B Robustness Check

B.1 Alternative sample

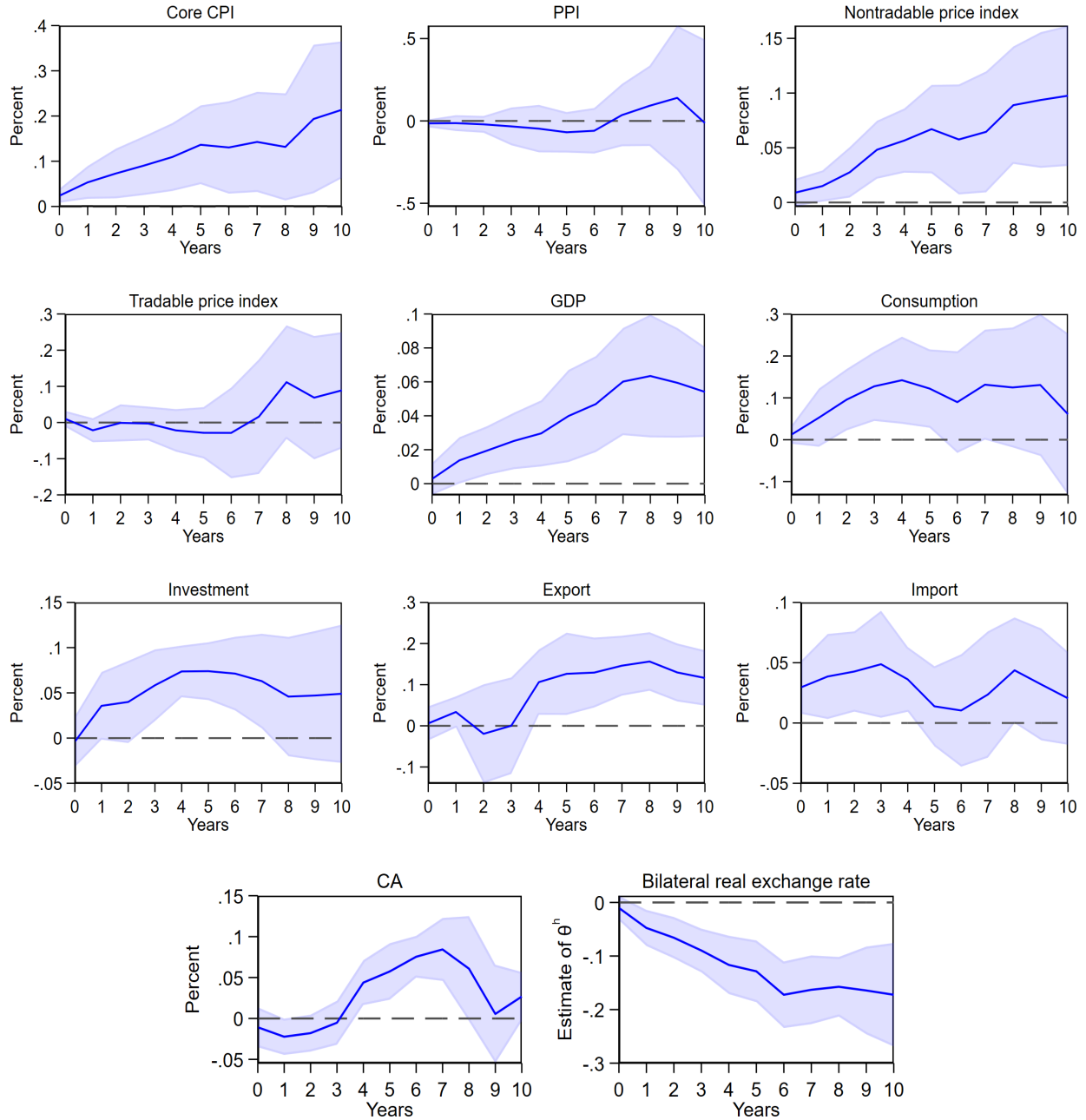


Figure B.1: Alternative sample: countries with at least one large oil discovery.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. Unlike the baseline specification, this robustness check restricts the sample exclusively to countries that have experienced at least one large oil discovery during the sample period. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.2 Excluding discoveries with realized production lags below four years

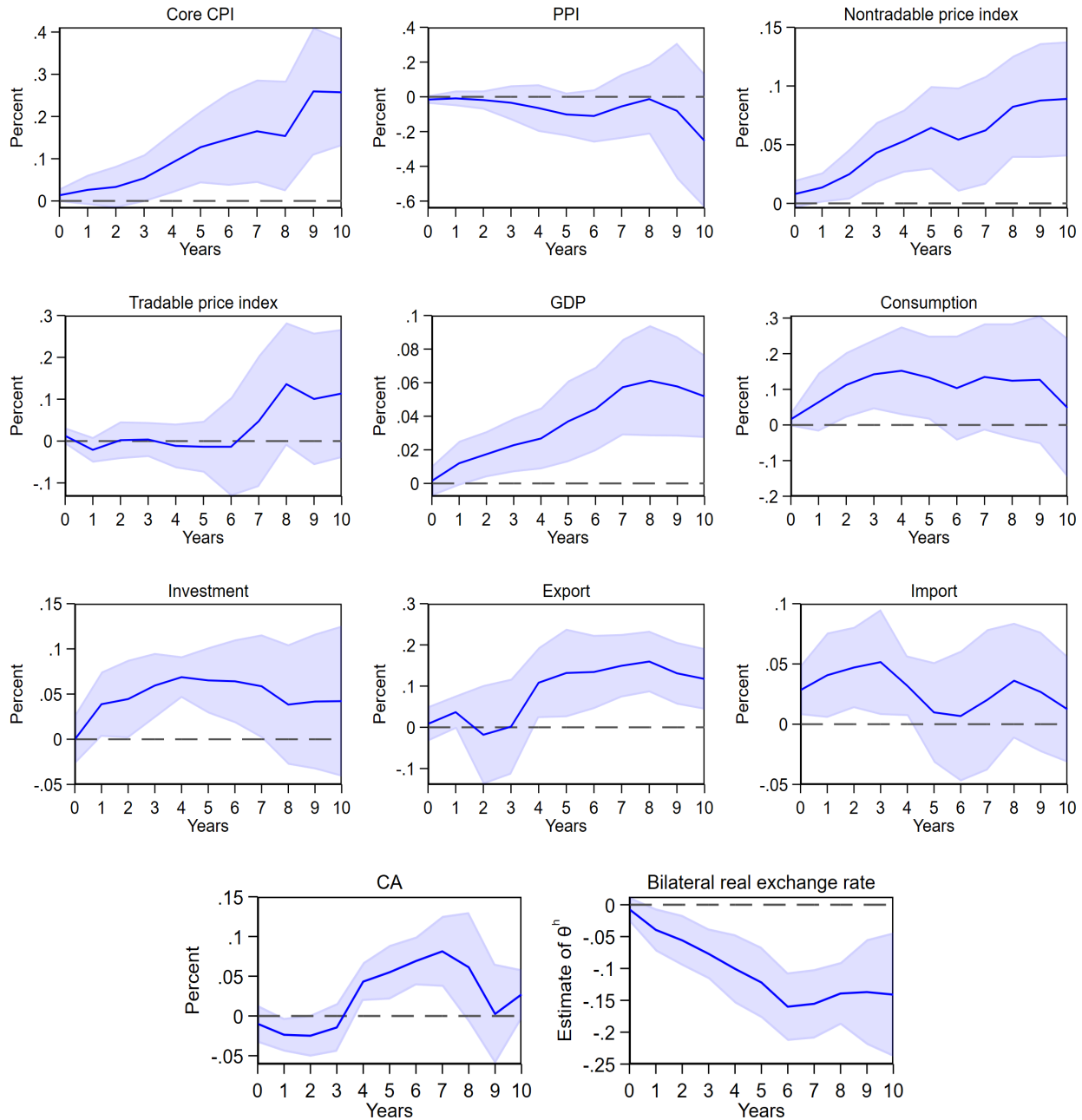


Figure B.2: Alternative sample: excluding large oil discoveries with realized production lags less than four years.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. Unlike the baseline specification which excludes discoveries with realized production lags of less than five years, this robustness check restricts the sample by excluding discoveries with realized production lags of less than four years. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.3 Excluding discoveries with realized production lags below six years

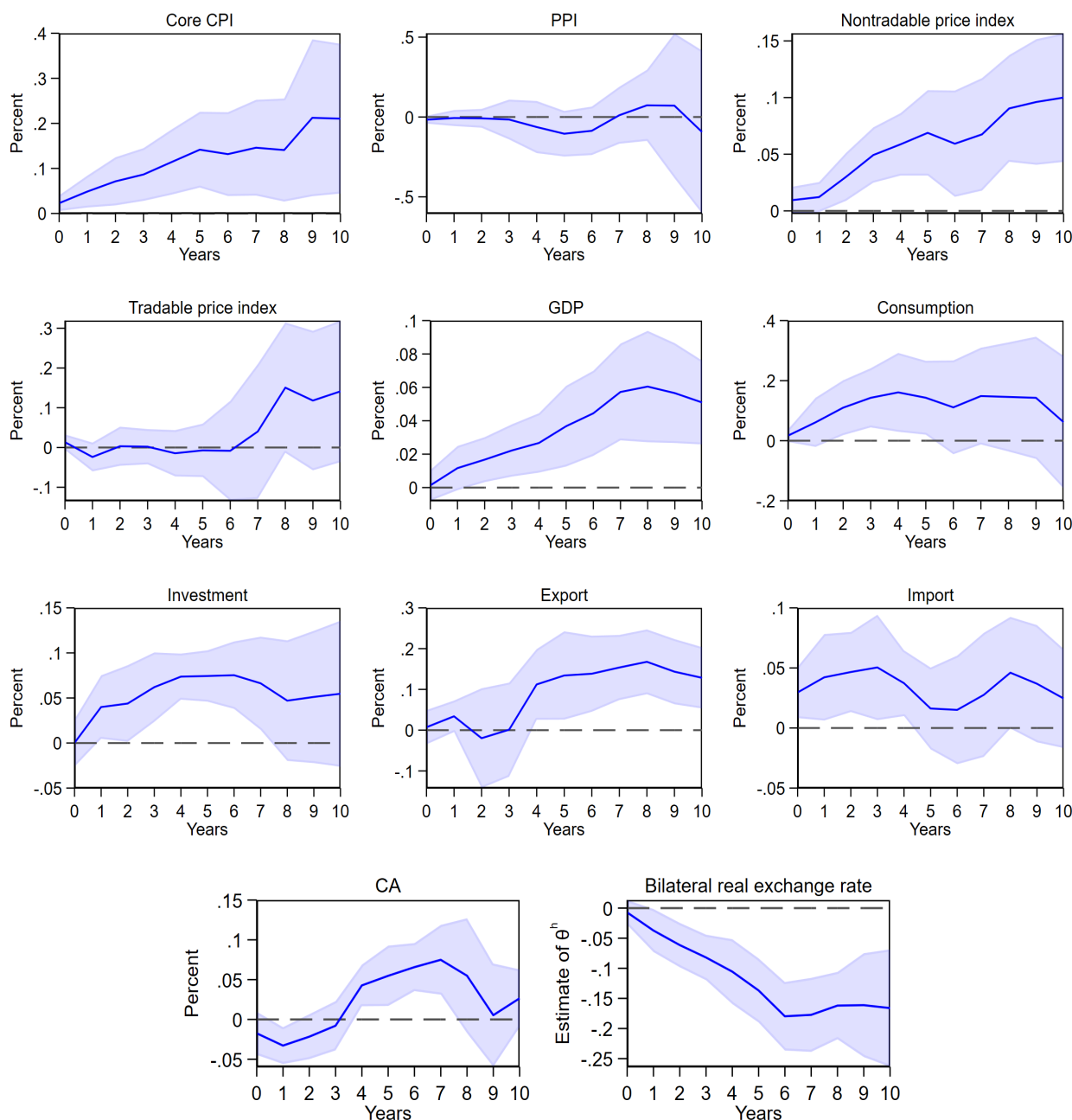


Figure B.3: Alternative sample: excluding large oil discoveries with realized production lags less than six years.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. Unlike the baseline specification which excludes discoveries with realized production lags of less than five years, this robustness check restricts the sample by excluding discoveries with realized production lags of less than six years. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.4 Alternative oil price in the construction of NPV

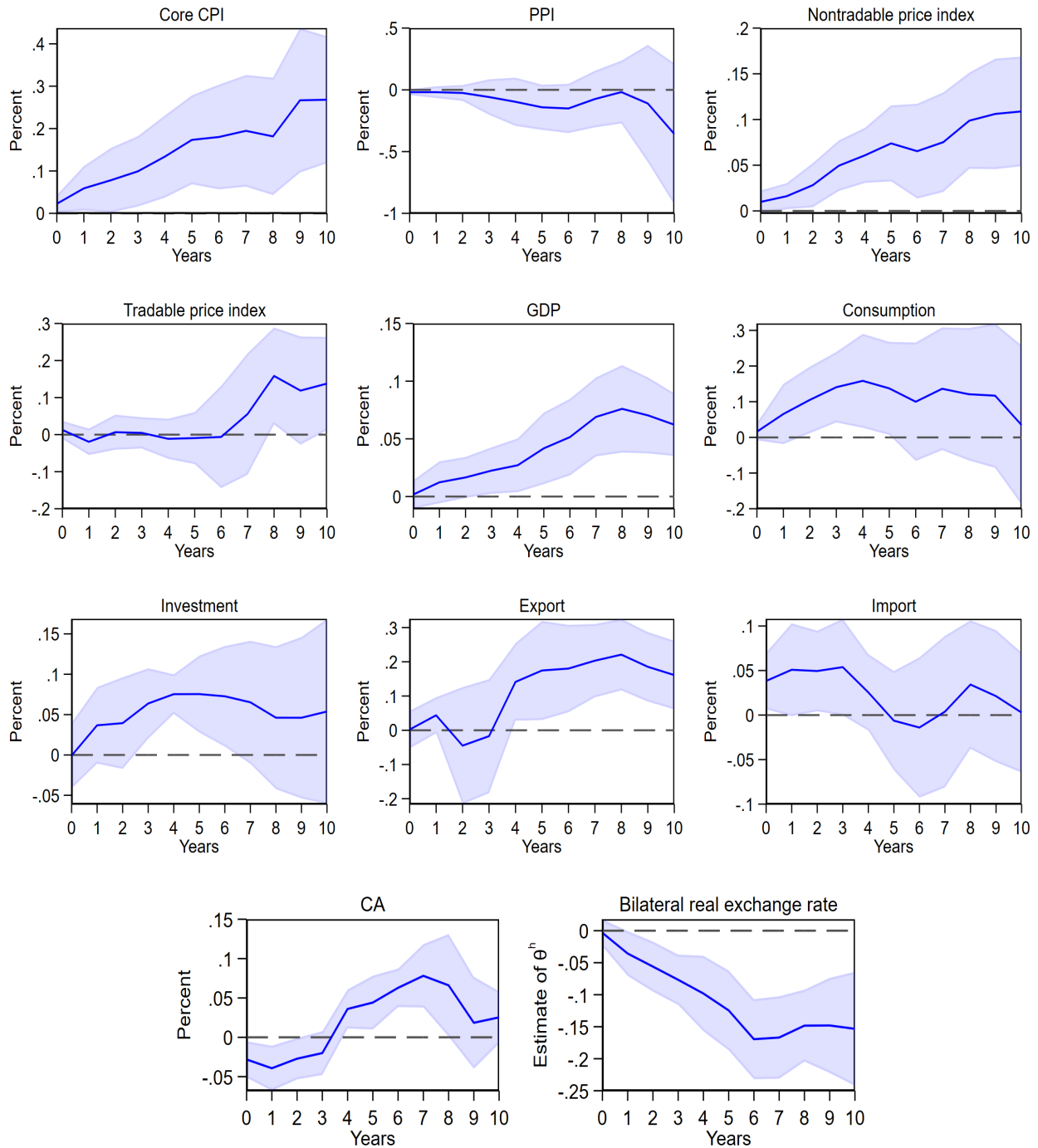


Figure B.4: Construction of NPV using the three-year average of world oil prices before the discovery.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. Unlike the baseline specification, which uses the current oil price to value the discoveries, this robustness check reconstructs the NPV using the average of global oil prices in the discovery year and the two preceding years. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.5 Alternative production lag used in the construction of NPV

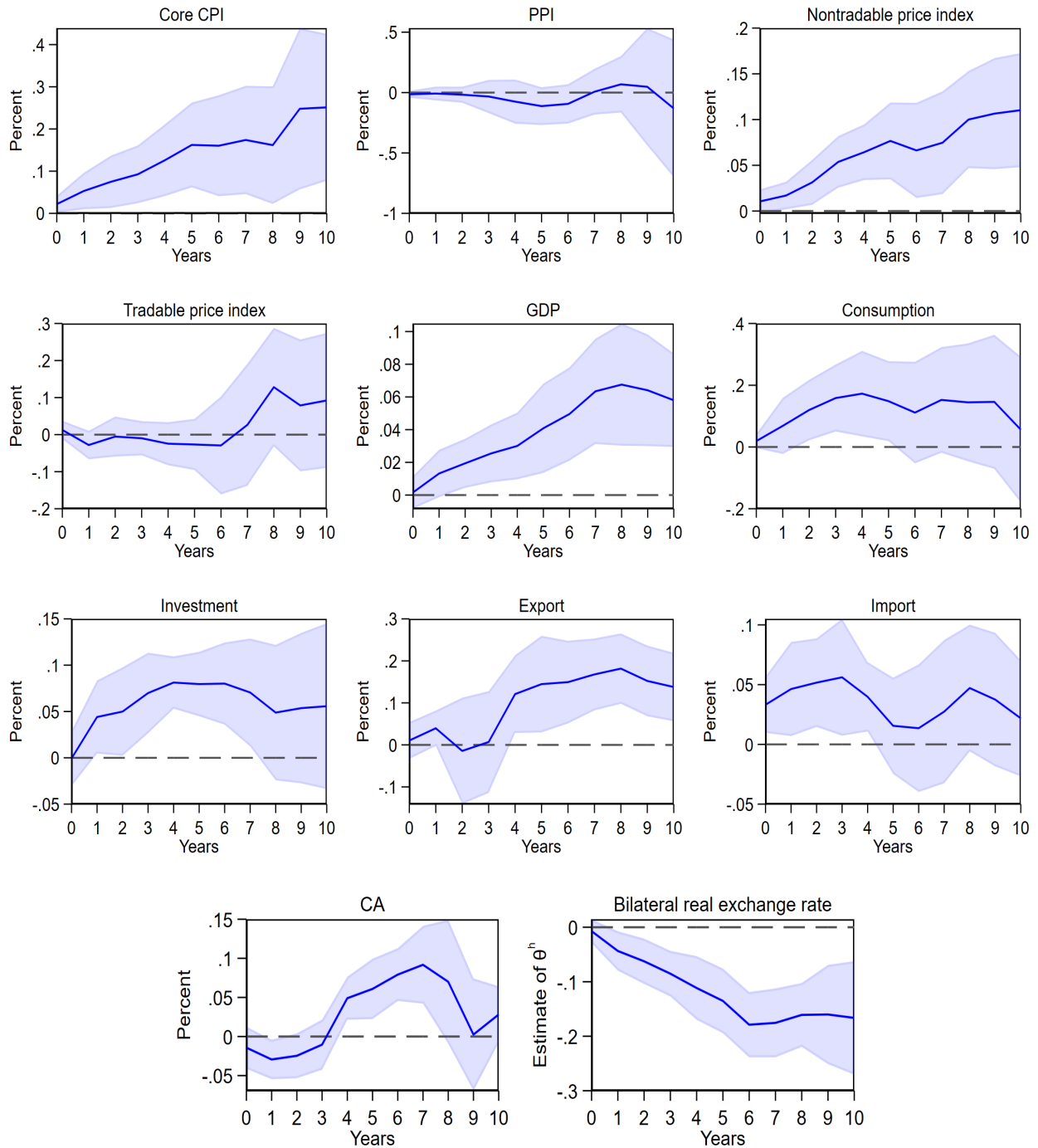


Figure B.5: Construction of NPV using 7-year production lags.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. In contrast to the baseline model, which assumes a standard five-year delay between discovery and actual production, this exercise utilizes a seven-year production lag to construct the discovery's NPV. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.6 Alternative definitions of tradable and nontradable sectors

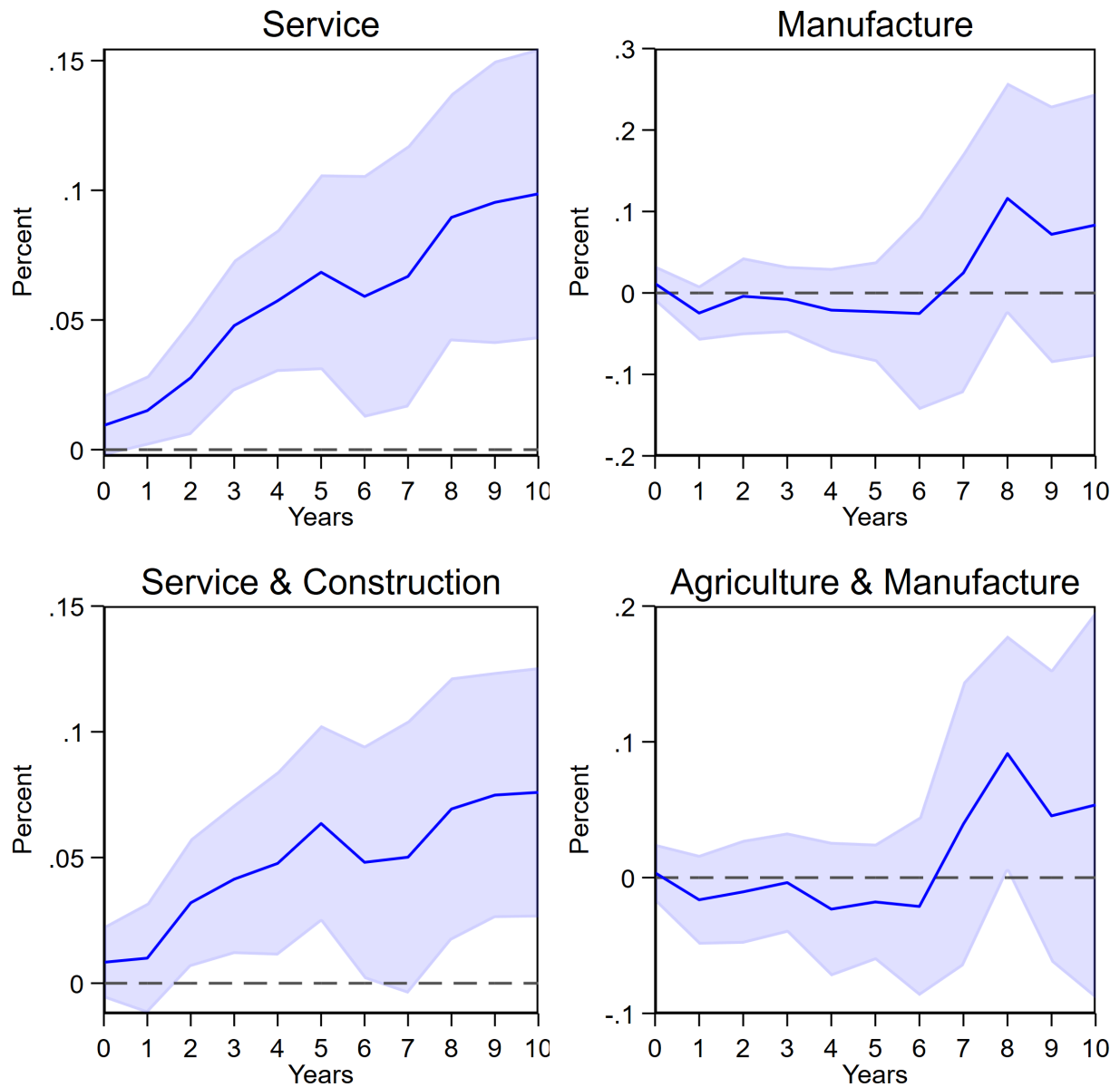


Figure B.6: Alternative definitions of tradable and nontradable sectors.

Notes: The figure presents the cumulative impulse responses of prices in specific sectors following a large oil discovery with an NPV equal to 1% of GDP. This robustness check explores alternative definitions of tradable and nontradable sectors compared to the baseline analysis. Specifically, it examines the responses of the services, manufacturing, combined services & construction (proxying alternative nontradables), and combined agriculture & manufacturing (proxying alternative tradables) price indices. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.7 Alternative dependent-variable definitions

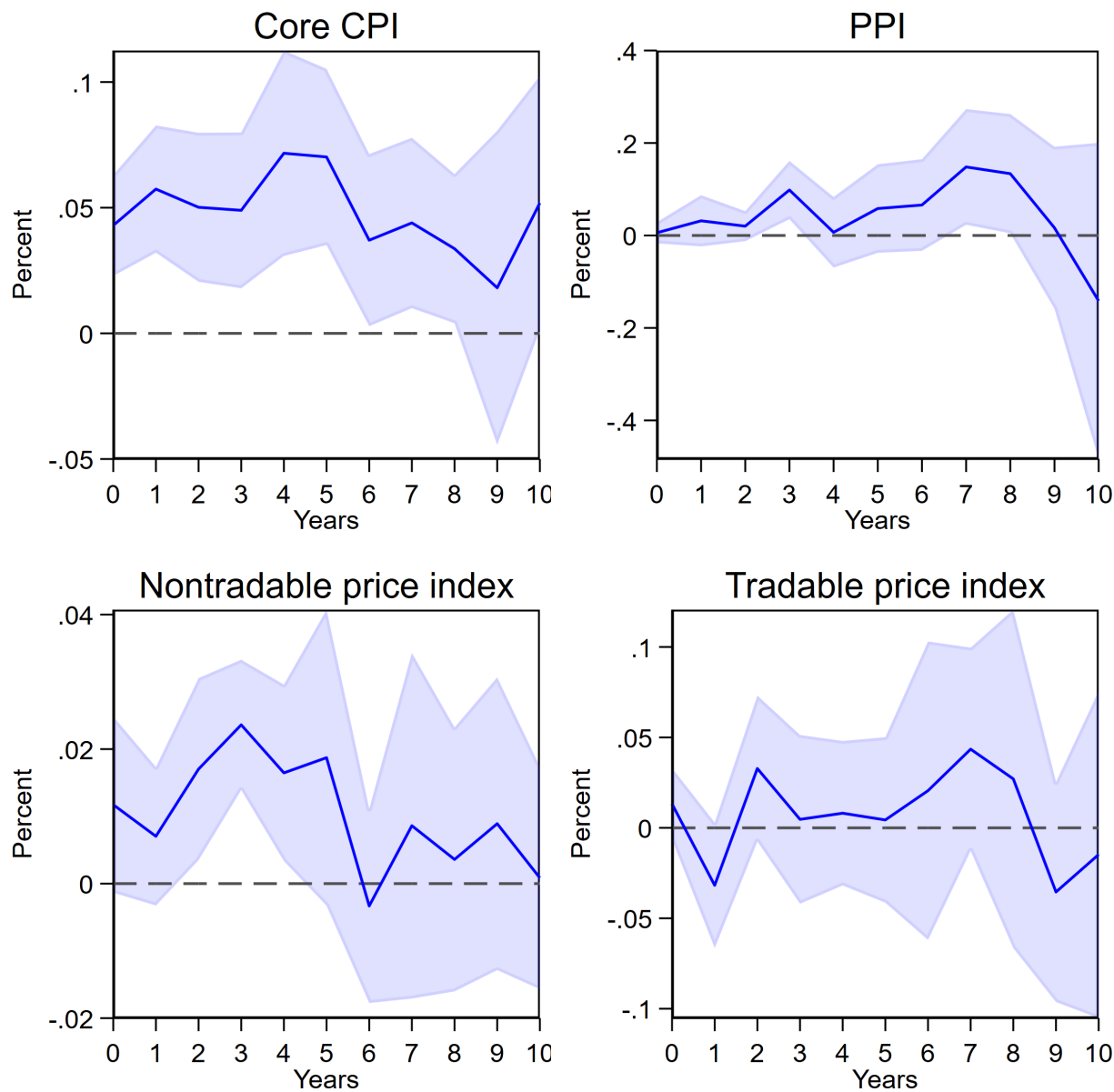


Figure B.7: Alternative dependent-variable definitions: inflation rate instead of price index.

Notes: The figure presents the impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices) following a large oil discovery with an NPV equal to 1% of GDP. Departing from the baseline approach, which evaluates prices as cumulative log deviations of base-year indices, this alternative specification directly employs annual inflation rates as the dependent variables. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the percentage change.

B.8 Controlling for exploration capex and U.S. monetary policy

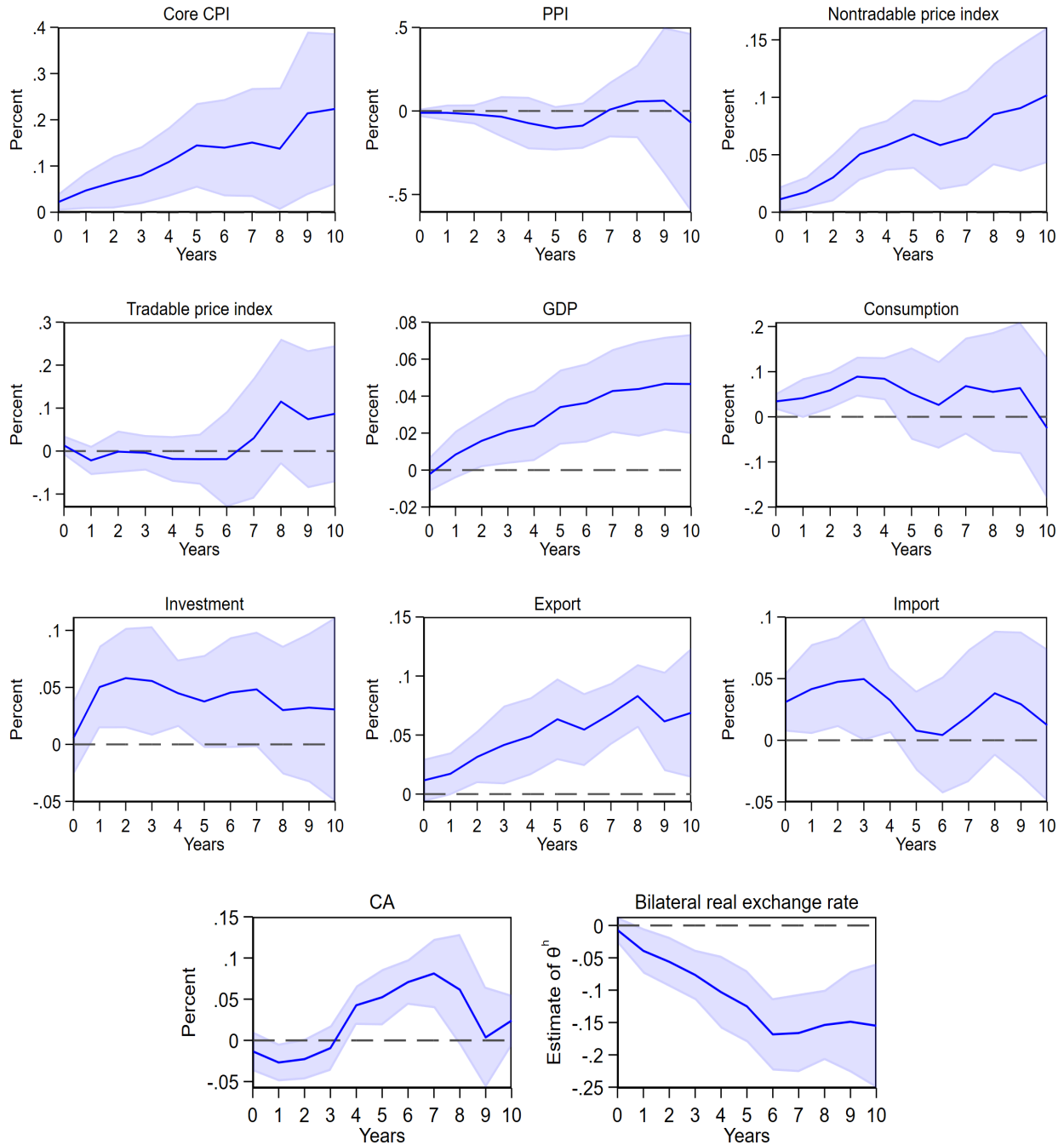


Figure B.8: Controlling for exploration capex and U.S. monetary policy.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. This specification augments the baseline regression with two additional controls: exploration capital expenditures (capex) in the current and previous five periods and an interaction term between the country's lagged capital account openness-to-GDP ratio and U.S. monetary policy shocks measured by the federal funds rate. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.9 Lag selection using BIC

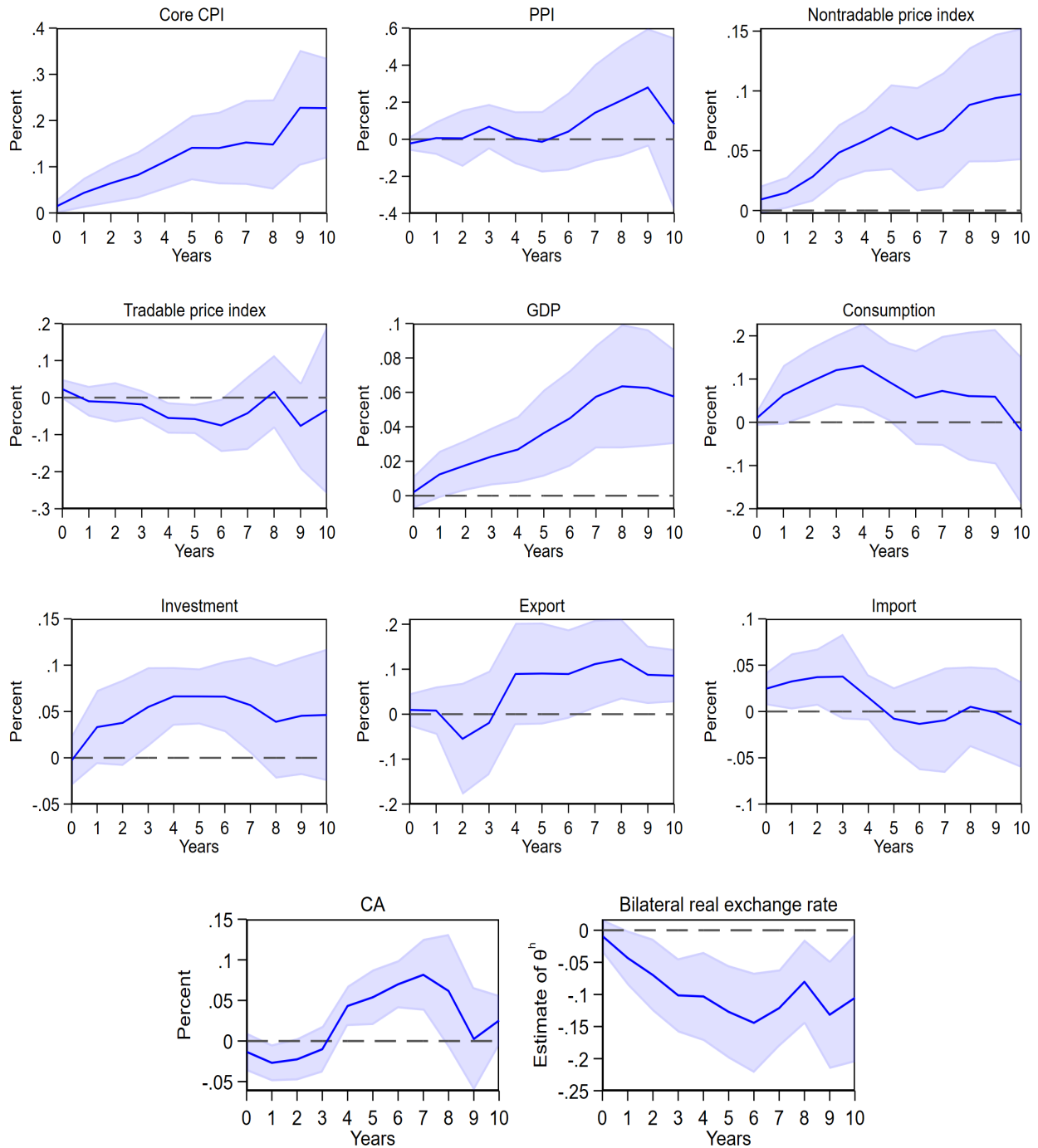


Figure B.9: Lag selection using BIC.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. Rather than imposing a uniform five-lag structure as in the baseline analysis, this robustness check utilizes the Bayesian Information Criterion (BIC) to select the optimal lag length for the model. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

B.10 Nickell bias correction in panel local projections

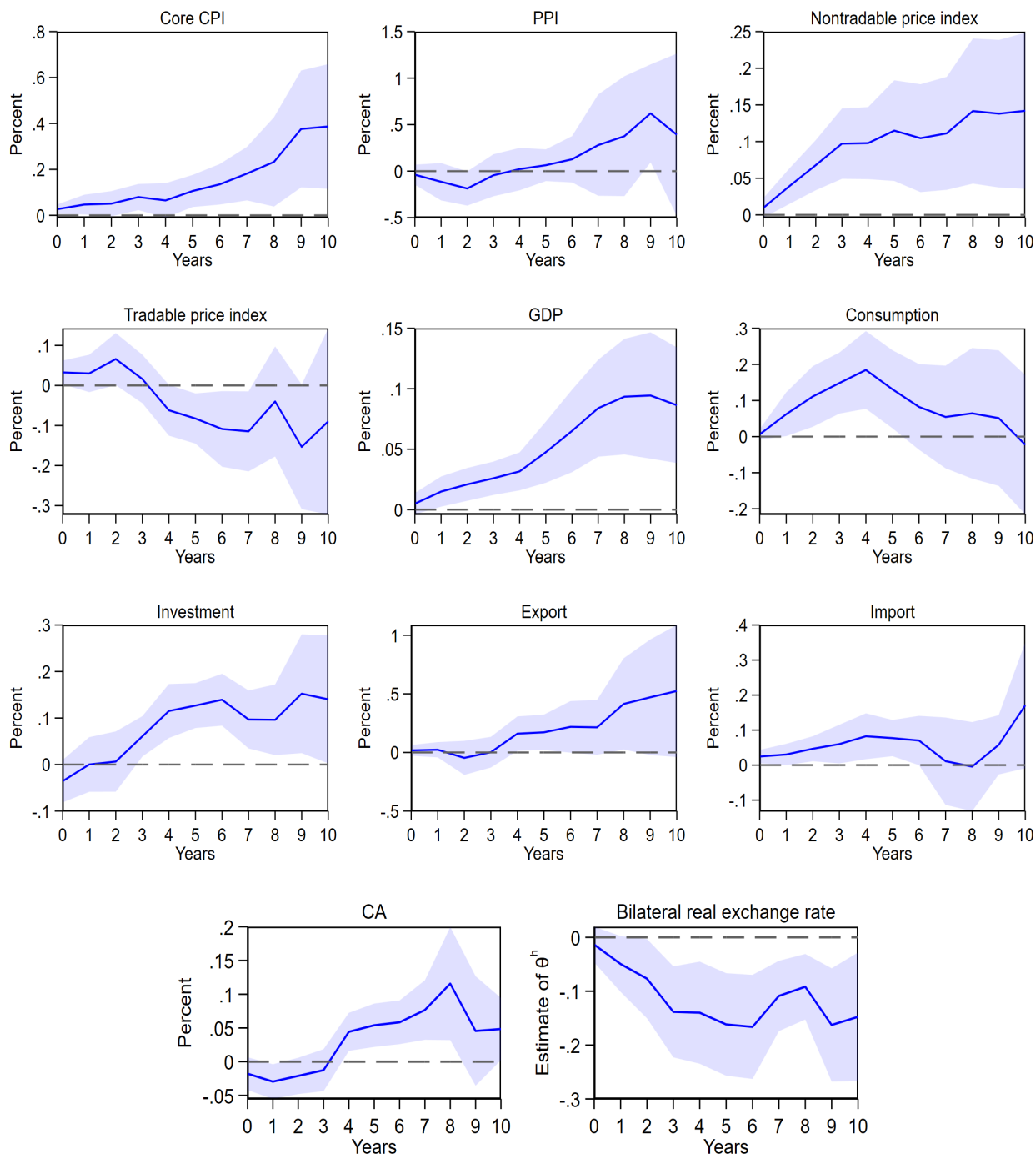


Figure B.10: Nickell bias correction in panel local projections.

Notes: The figure presents the cumulative impulse responses of key macroeconomic variables (Core CPI, PPI, Nontradable and Tradable price indices, GDP, Consumption, Investment, Exports, Imports, Current Account, and Real exchange rate) following a large oil discovery with an NPV equal to 1% of GDP. This regression adopts the Split-Panel Jackknife (SPJ) estimator to correct for Nickell bias, implemented alongside the BIC-determined lag selection. The solid lines indicate point estimates, and the shaded areas represent 90% confidence intervals. The vertical axis denotes the cumulative percentage change.

C Model

This part of the appendix collects the derivations of the model developed in Section 4.

C.1 Derivation of Household Optimality Conditions

This appendix derives the household first-order conditions used in the model. The representative household maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{L_t^{1+\gamma}}{1+\gamma} \right], \quad (75)$$

where C_t is composite consumption, L_t is total labor supply, $\beta \in (0,1)$, $\sigma > 0$, $\gamma > 0$, and $\chi > 0$.

The household owns sector-specific capital in the nontradable, home tradable, and oil sectors, indexed by $h \in \{N, T, O\}$. Capital is predetermined at the beginning of period t , rented to firms within the period, and augmented through sector-specific investment. The household also trades a domestic nominal bond B_t and a foreign-currency bond B_t^* . The nominal exchange rate $\mathcal{E}_t$ is expressed as domestic currency per unit of foreign currency.

The period- t nominal budget constraint is

$$\begin{aligned} P_{C,t}C_t + P_{I,t} \sum_{h \in \{N, T, O\}} I_{h,t} + B_t + \mathcal{E}_t B_t^* = (1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^*)\mathcal{E}_t B_{t-1}^* \\ + W_t L_t + \sum_{h \in \{N, T, O\}} R_{h,t} K_{h,t-1} + \Phi_t. \end{aligned} \quad (76)$$

Total profits are rebated lump-sum to the household:

$$\Phi_t = \Phi_{N,t} + \Phi_{H,t} + \Phi_{O,t}. \quad (77)$$

For each sector $h \in \{N, T, O\}$, capital evolves according to

$$K_{h,t} = (1 - \delta)K_{h,t-1} + I_{h,t} \left[1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 \right], \quad (78)$$

where $\delta \in (0,1)$ is the depreciation rate and $\psi_h \geq 0$ governs investment adjustment costs. The aggregate investment good satisfies

$$I_t = \sum_{h \in \{N, T, O\}} I_{h,t}. \quad (79)$$

C.1.1 The Lagrangian

Let λ_t denote the multiplier on the real budget constraint, obtained by dividing (76) by $P_{C,t}$. Thus, λ_t is the real marginal utility value of wealth. Let $\mu_{h,t}$ denote the multiplier on the capital accumulation constraint for sector h . The household Lagrangian is

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{L_t^{1+\gamma}}{1+\gamma} + \lambda_t \left[\frac{(1+i_{t-1})B_{t-1} + (1+i_{t-1}^*)\mathcal{E}_t B_{t-1}^* + W_t L_t}{P_{C,t}} \right. \right. \\ & \left. \left. - C_t + \frac{\sum_{h \in \{N,T,O\}} R_{h,t} K_{h,t-1} - P_{I,t} \sum_{h \in \{N,T,O\}} I_{h,t} - B_t - \mathcal{E}_t B_t^* + \Phi_t}{P_{C,t}} \right] \right. \\ & \left. + \sum_{h \in \{N,T,O\}} \mu_{h,t} \left[(1-\delta)K_{h,t-1} + I_{h,t} \left[1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 \right] - K_{h,t} \right] \right\}. \end{aligned} \quad (80)$$

Define real Tobin's Q for sector h as

$$Q_{h,t} \equiv \frac{\mu_{h,t}}{\lambda_t}. \quad (81)$$

Since λ_t is the real marginal utility value of wealth, $Q_{h,t}$ is measured in units of the consumption good per unit of installed capital.

C.1.2 Marginal Utility of Wealth

Differentiating (80) with respect to C_t gives

$$\frac{\partial \mathcal{L}}{\partial C_t} = C_t^{-\sigma} - \lambda_t = 0. \quad (82)$$

Therefore,

$$\lambda_t = C_t^{-\sigma}. \quad (83)$$

C.1.3 Labor Supply

Differentiating with respect to L_t gives

$$\frac{\partial \mathcal{L}}{\partial L_t} = -\chi L_t^\gamma + \lambda_t \frac{W_t}{P_{C,t}} = 0. \quad (84)$$

Hence,

$$\chi L_t^\gamma = \lambda_t \frac{W_t}{P_{C,t}}. \quad (85)$$

This condition equates the marginal disutility of labor to the marginal utility value of the real wage.

C.1.4 Euler Equation for Domestic Bonds

The domestic bond B_t appears as an outflow in the period- t budget constraint and as an inflow in period $t + 1$. Differentiating with respect to B_t yields

$$-\lambda_t \frac{1}{P_{C,t}} + \beta \mathbb{E}_t \left[\lambda_{t+1} \frac{1+i_t}{P_{C,t+1}} \right] = 0. \quad (86)$$

Therefore,

$$\frac{\lambda_t}{P_{C,t}} = \beta(1+i_t) \mathbb{E}_t \left[\frac{\lambda_{t+1}}{P_{C,t+1}} \right]. \quad (87)$$

Multiplying both sides by $P_{C,t}/\lambda_t$ gives

$$1 = \beta(1+i_t) \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \right]. \quad (88)$$

Equivalently, using $\Pi_{C,t+1} \equiv P_{C,t+1}/P_{C,t}$,

$$1 = \beta(1+i_t) \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{1}{\Pi_{C,t+1}} \right]. \quad (89)$$

C.1.5 Euler Equation for Foreign Bonds

The foreign bond B_t^* is denominated in foreign currency. Its domestic currency value at time t is $\mathcal{E}_t B_t^*$. Differentiating the Lagrangian with respect to B_t^* gives

$$-\lambda_t \frac{\mathcal{E}_t}{P_{C,t}} + \beta \mathbb{E}_t \left[\lambda_{t+1} \frac{(1+i_t^*) \mathcal{E}_{t+1}}{P_{C,t+1}} \right] = 0. \quad (90)$$

Thus,

$$1 = \beta(1+i_t^*) \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{\mathcal{E}_{t+1}}{\mathcal{E}_t} \frac{P_{C,t}}{P_{C,t+1}} \right]. \quad (91)$$

Combining (88) and (91) delivers the model's uncovered interest parity condition. In the main text, the effective foreign interest rate i_t^* also contains a debt-elastic premium that ensures stationarity of the net foreign asset position.

C.1.6 Capital Pricing

For each sector $h \in \{N, T, O\}$, differentiating the Lagrangian with respect to $K_{h,t}$ gives

$$-\mu_{h,t} + \beta \mathbb{E}_t \left[\lambda_{t+1} \frac{R_{h,t+1}}{P_{C,t+1}} + \mu_{h,t+1}(1 - \delta) \right] = 0. \quad (92)$$

Dividing by λ_t and using $Q_{h,t} \equiv \mu_{h,t} / \lambda_t$ yields

$$Q_{h,t} = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{R_{h,t+1}}{P_{C,t+1}} + \frac{\lambda_{t+1}}{\lambda_t} Q_{h,t+1}(1 - \delta) \right]. \quad (93)$$

Therefore,

$$Q_{h,t} = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \left(\frac{R_{h,t+1}}{P_{C,t+1}} + (1 - \delta) Q_{h,t+1} \right) \right], \quad h \in \{N, T, O\}. \quad (94)$$

C.1.7 Investment Decision

For compactness, define the gross investment growth rate

$$g_{h,t} \equiv \frac{I_{h,t}}{I_{h,t-1}}, \quad h \in \{N, T, O\}. \quad (95)$$

The investment term entering capital accumulation is

$$\mathcal{S}_{h,t} \equiv I_{h,t} \left[1 - \frac{\psi_h}{2} (g_{h,t} - 1)^2 \right]. \quad (96)$$

Differentiating the Lagrangian with respect to $I_{h,t}$ gives

$$0 = -\lambda_t \frac{P_{I,t}}{P_{C,t}} + \mu_{h,t} \frac{\partial \mathcal{S}_{h,t}}{\partial I_{h,t}} + \beta \mathbb{E}_t \left[\mu_{h,t+1} \frac{\partial \mathcal{S}_{h,t+1}}{\partial I_{h,t}} \right]. \quad (97)$$

The relevant derivatives are

$$\frac{\partial \mathcal{S}_{h,t}}{\partial I_{h,t}} = 1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 - \psi_h \frac{I_{h,t}}{I_{h,t-1}} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right), \quad (98)$$

$$\frac{\partial \mathcal{S}_{h,t+1}}{\partial I_{h,t}} = \psi_h \left(\frac{I_{h,t+1}}{I_{h,t}} \right)^2 \left(\frac{I_{h,t+1}}{I_{h,t}} - 1 \right). \quad (99)$$

Substituting these expressions into (97), dividing by λ_t , and using $Q_{h,t} = \mu_{h,t}/\lambda_t$, we obtain

$$\begin{aligned} \frac{P_{I,t}}{P_{C,t}} = Q_{h,t} & \left[1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 - \psi_h \frac{I_{h,t}}{I_{h,t-1}} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right) \right] \\ & + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} Q_{h,t+1} \psi_h \left(\frac{I_{h,t+1}}{I_{h,t}} \right)^2 \left(\frac{I_{h,t+1}}{I_{h,t}} - 1 \right) \right], \quad h \in \{N, T, O\}. \end{aligned} \quad (100)$$

C.2 Derivation of Sectoral Rotemberg Phillips Curves

This section derives the nonlinear Phillips curves for the sectors with monopolistic competition and Rotemberg price adjustment costs. The derivation is identical for the nontradable and home tradable sectors, up to notation.

C.2.1 Nontradable Sector

A nontradable intermediate firm i faces demand

$$Y_{N,t}(i) = \left(\frac{P_{N,t}(i)}{P_{N,t}} \right)^{-\varepsilon_N} Y_{N,t}, \quad (101)$$

where $\varepsilon_N > 1$. The firm chooses $\{P_{N,t}(i)\}_{t=0}^{\infty}$ to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_{0,t} \left[P_{N,t}(i) Y_{N,t}(i) - P_{N,t} MC_{N,t} Y_{N,t}(i) - P_{N,t} \frac{\phi_N}{2} \left(\frac{P_{N,t}(i)}{P_{N,t-1}(i)} - 1 \right)^2 Y_{N,t} \right], \quad (102)$$

where $MC_{N,t}$ is real marginal cost in units of the nontradable good and $\Lambda_{t,t+k}$ is the household's nominal stochastic discount factor:

$$\Lambda_{t,t+k} \equiv \beta^k \frac{\lambda_{t+k}/P_{C,t+k}}{\lambda_t/P_{C,t}}. \quad (103)$$

After substituting demand, the first-order condition with respect to $P_{N,t}(i)$, evaluated in a symmetric equilibrium with $P_{N,t}(i) = P_{N,t}$, is

$$\begin{aligned} 0 = & (1 - \varepsilon_N) Y_{N,t} + \varepsilon_N MC_{N,t} Y_{N,t} - \phi_N (\Pi_{N,t} - 1) \Pi_{N,t} Y_{N,t} \\ & + \mathbb{E}_t \left[\Lambda_{t,t+1} \phi_N (\Pi_{N,t+1} - 1) \Pi_{N,t+1} \frac{P_{N,t+1}}{P_{N,t}} Y_{N,t+1} \right], \end{aligned} \quad (104)$$

where

$$\Pi_{N,t} \equiv \frac{P_{N,t}}{P_{N,t-1}}. \quad (105)$$

Dividing by $Y_{N,t}\phi_N$ and rearranging gives

$$(\Pi_{N,t} - 1)\Pi_{N,t} = \frac{\varepsilon_N}{\phi_N} \left(MC_{N,t} - \frac{\varepsilon_N - 1}{\varepsilon_N} \right) + \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{P_{N,t+1}}{P_{N,t}} \frac{Y_{N,t+1}}{Y_{N,t}} (\Pi_{N,t+1} - 1)\Pi_{N,t+1} \right]. \quad (106)$$

Using

$$\Lambda_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}}, \quad (107)$$

the nontradable Phillips curve becomes

$$(\Pi_{N,t} - 1)\Pi_{N,t} = \frac{\varepsilon_N}{\phi_N} \left(MC_{N,t} - \frac{\varepsilon_N - 1}{\varepsilon_N} \right) + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \frac{P_{N,t+1}}{P_{N,t}} \frac{Y_{N,t+1}}{Y_{N,t}} (\Pi_{N,t+1} - 1)\Pi_{N,t+1} \right]. \quad (108)$$

C.2.2 Home Tradable Sector

The home tradable sector has the same monopolistic-competition and Rotemberg pricing structure. A firm i faces demand

$$Y_{H,t}(i) = \left(\frac{P_{H,t}(i)}{P_{H,t}} \right)^{-\varepsilon_H} Y_{H,t}, \quad (109)$$

where $\varepsilon_H > 1$, and adjusts prices subject to the cost

$$P_{H,t} \frac{\phi_T}{2} \left(\frac{P_{H,t}(i)}{P_{H,t-1}(i)} - 1 \right)^2 Y_{H,t}. \quad (110)$$

By the same steps as above, the symmetric-equilibrium price-setting condition is

$$(\Pi_{H,t} - 1)\Pi_{H,t} = \frac{\varepsilon_H}{\phi_T} \left(MC_{H,t} - \frac{\varepsilon_H - 1}{\varepsilon_H} \right) + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \frac{P_{H,t+1}}{P_{H,t}} \frac{Y_{H,t+1}}{Y_{H,t}} (\Pi_{H,t+1} - 1)\Pi_{H,t+1} \right], \quad (111)$$

where

$$\Pi_{H,t} \equiv \frac{P_{H,t}}{P_{H,t-1}}. \quad (112)$$

C.3 Derivation of Oil-Sector Optimality Conditions

The oil sector is perfectly competitive in the product market and takes the domestic-currency oil price

$$P_{O,t} = \mathcal{E}_t P_{O,t}^* \quad (113)$$

as given. Oil output is produced using capital, labor, and producing reserves:

$$Y_{O,t} = A_{O,t} K_{O,t-1}^{\alpha_k} L_{O,t}^{\alpha_L} Res_{t-1}^{1-\alpha_k-\alpha_L}, \quad \alpha_k > 0, \quad \alpha_L > 0, \quad \alpha_k + \alpha_L < 1. \quad (114)$$

Producing reserves evolve according to

$$Res_t = Res_{t-1} + \varsigma - Y_{O,t} + \varepsilon_{t-j}^{Res}. \quad (115)$$

C.3.1 Oil Producer's Lagrangian

The oil producer maximizes the expected discounted value of nominal profits, using the household's nominal stochastic discount factor. Let ζ_t denote the nominal shadow value of one unit of producing reserves. The Lagrangian is

$$\mathcal{L}^O = \mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_{0,t} \left\{ P_{O,t} Y_{O,t} - W_t L_{O,t} - R_{O,t} K_{O,t-1} + \zeta_t \left[Res_{t-1} + \varsigma - Y_{O,t} + \varepsilon_{t-j}^{Res} - Res_t \right] \right\}, \quad (116)$$

subject to (114). This formulation makes clear that current extraction yields revenue $P_{O,t}$ but also reduces reserves, whose shadow value is ζ_t .

C.3.2 Oil Labor Demand

Differentiating (116) with respect to $L_{O,t}$ gives

$$(P_{O,t} - \zeta_t) \frac{\partial Y_{O,t}}{\partial L_{O,t}} - W_t = 0. \quad (117)$$

Using the Cobb–Douglas production function,

$$\frac{\partial Y_{O,t}}{\partial L_{O,t}} = \alpha_L \frac{Y_{O,t}}{L_{O,t}}, \quad (118)$$

so

$$W_t = \alpha_L \frac{(P_{O,t} - \zeta_t) Y_{O,t}}{L_{O,t}}. \quad (119)$$

C.3.3 Oil Capital Demand

Differentiating with respect to $K_{O,t-1}$ yields

$$(P_{O,t} - \zeta_t) \frac{\partial Y_{O,t}}{\partial K_{O,t-1}} - R_{O,t} = 0. \quad (120)$$

Since

$$\frac{\partial Y_{O,t}}{\partial K_{O,t-1}} = \alpha_k \frac{Y_{O,t}}{K_{O,t-1}}, \quad (121)$$

we obtain

$$R_{O,t} = \alpha_k \frac{(P_{O,t} - \zeta_t) Y_{O,t}}{K_{O,t-1}}. \quad (122)$$

C.3.4 Euler Equation for Producing Reserves

Differentiating the oil producer's Lagrangian with respect to Res_t gives

$$-\zeta_t + \mathbb{E}_t \left[\Lambda_{t,t+1} \left\{ (P_{O,t+1} - \zeta_{t+1}) \frac{\partial Y_{O,t+1}}{\partial Res_t} + \zeta_{t+1} \right\} \right] = 0. \quad (123)$$

The marginal product of reserves in the next period is

$$\frac{\partial Y_{O,t+1}}{\partial Res_t} = (1 - \alpha_k - \alpha_L) \frac{Y_{O,t+1}}{Res_t}. \quad (124)$$

Therefore,

$$\zeta_t = \mathbb{E}_t \left[\Lambda_{t,t+1} \left\{ (P_{O,t+1} - \zeta_{t+1}) (1 - \alpha_k - \alpha_L) \frac{Y_{O,t+1}}{Res_t} + \zeta_{t+1} \right\} \right]. \quad (125)$$

Substituting the one-period stochastic discount factor gives

$$\zeta_t = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \left\{ (P_{O,t+1} - \zeta_{t+1}) (1 - \alpha_k - \alpha_L) \frac{Y_{O,t+1}}{Res_t} + \zeta_{t+1} \right\} \right]. \quad (126)$$

C.4 Equilibrium Conditions

We choose the consumption price index $P_{C,t}$ as the numeraire. A general equilibrium is defined as a sequence of endogenous prices and quantities such that, given the initial state variables and the sequence of exogenous processes, the following conditions hold: (i) the representative household maximizes expected utility subject to its budget and capital accumulation constraints; (ii) firms in

the nontradable, home tradable, and oil sectors maximize expected profits subject to their respective technological, resource, and pricing constraints; (iii) the monetary authority follows its specified interest rate rule; and (iv) all goods, factor, and financial markets clear. In particular, the domestic bond market clears in zero net supply, implying $B_t = 0$ for all t .

The endogenous variables are

$$C_t, L_t, I_t, K_t, Y_t, Y_{T,t}, Y_{H,t}, Y_{H,t}^{dom}, X_{H,t}, M_t, Y_{N,t}, Y_{O,t}, C_{N,t}, C_{T,t}, J_{N,t}, J_{T,t}, K_{N,t}, K_{T,t}, K_{O,t}, I_{N,t}, I_{T,t}, I_{O,t}, L_{N,t}, L_{T,t}, L_{O,t}, W_t, \lambda_t, B_t^*, TB_t, P_{C,t}, P_{N,t}, P_{T,t}, P_{H,t}, P_{I,t}, \Pi_{C,t}, \Pi_{N,t}, \Pi_{T,t}, \Pi_{H,t}, MC_{N,t}, MC_{H,t}, \mathcal{E}_t, i_t, i_t^*, Res_t, \zeta_t, R_{N,t}, R_{T,t}, R_{O,t}, Q_{N,t}, Q_{T,t}, Q_{O,t}.$$

The exogenous shocks are ε_{t-j}^{Res} . The exogenous foreign prices and foreign demand shifter are $P_{O,t}^*, P_{F,t}^*, Y_t^*$.

C.4.1 Household Optimality Conditions

$$\lambda_t = C_t^{-\sigma}. \quad (127)$$

$$\lambda_t \frac{W_t}{P_{C,t}} = \chi L_t^\gamma. \quad (128)$$

$$\lambda_t = \beta \mathbb{E}_t \left[\lambda_{t+1} \frac{1 + i_t}{\Pi_{C,t+1}} \right]. \quad (129)$$

$$\lambda_t = \beta(1 + i_t^*) \mathbb{E}_t \left[\lambda_{t+1} \frac{\mathcal{E}_{t+1}}{\mathcal{E}_t} \frac{P_{C,t}}{P_{C,t+1}} \right]. \quad (130)$$

$$Q_{h,t} = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \left(\frac{R_{h,t+1}}{P_{C,t+1}} + (1 - \delta) Q_{h,t+1} \right) \right], h \in \{N, T, O\}. \quad (131)$$

$$K_{h,t} = (1 - \delta) K_{h,t-1} + I_{h,t} \left[1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 \right], \quad h \in \{N, T, O\}. \quad (132)$$

$$\begin{aligned} \frac{P_{I,t}}{P_{C,t}} = Q_{h,t} & \left[1 - \frac{\psi_h}{2} \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right)^2 - \psi_h \left(\frac{I_{h,t}}{I_{h,t-1}} - 1 \right) \frac{I_{h,t}}{I_{h,t-1}} \right] \\ & + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} Q_{h,t+1} \psi_h \left(\frac{I_{h,t+1}}{I_{h,t}} - 1 \right) \left(\frac{I_{h,t+1}}{I_{h,t}} \right)^2 \right], \quad h \in \{N, T, O\}. \end{aligned} \quad (133)$$

C.4.2 Production and Factor Demand

$$Y_{H,t} = A_{T,t} K_{T,t-1}^{\alpha_T} L_{T,t}^{1-\alpha_T}. \quad (134)$$

$$R_{T,t} K_{T,t-1} = \alpha_T P_{H,t} MC_{H,t} Y_{H,t}. \quad (135)$$

$$W_t L_{T,t} = (1 - \alpha_T) P_{H,t} MC_{H,t} Y_{H,t}. \quad (136)$$

$$Y_{N,t} = A_{N,t} K_{N,t-1}^{\alpha_N} L_{N,t}^{1-\alpha_N}. \quad (137)$$

$$R_{N,t} K_{N,t-1} = \alpha_N P_{N,t} MC_{N,t} Y_{N,t}. \quad (138)$$

$$W_t L_{N,t} = (1 - \alpha_N) P_{N,t} MC_{N,t} Y_{N,t}. \quad (139)$$

$$Y_{O,t} = A_{O,t} K_{O,t-1}^{\alpha_k} L_{O,t}^{\alpha_L} Res_{t-1}^{1-\alpha_k-\alpha_L}. \quad (140)$$

$$W_t = \alpha_L \frac{(P_{O,t} - \zeta_t) Y_{O,t}}{L_{O,t}}. \quad (141)$$

$$R_{O,t} = \alpha_k \frac{(P_{O,t} - \zeta_t) Y_{O,t}}{K_{O,t-1}}. \quad (142)$$

$$\zeta_t = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{P_{C,t}}{P_{C,t+1}} \left\{ (P_{O,t+1} - \zeta_{t+1}) (1 - \alpha_k - \alpha_L) \frac{Y_{O,t+1}}{Res_t} + \zeta_{t+1} \right\} \right]. \quad (143)$$

$$Res_t = Res_{t-1} + \varsigma - Y_{O,t} + \varepsilon_{t-j}^{Res}. \quad (144)$$

C.4.3 Aggregation, Demand, and Market Clearing

$$L_t = L_{N,t} + L_{T,t} + L_{O,t}. \quad (145)$$

$$K_t = K_{N,t} + K_{T,t} + K_{O,t}. \quad (146)$$

$$I_t = I_{N,t} + I_{T,t} + I_{O,t}. \quad (147)$$

$$C_{N,t} = \omega_C \left(\frac{P_{N,t}}{P_{C,t}} \right)^{-\xi} C_t. \quad (148)$$

$$C_{T,t} = (1 - \omega_C) \left(\frac{P_{T,t}}{P_{C,t}} \right)^{-\xi} C_t. \quad (149)$$

$$J_{N,t} = \omega_I \left(\frac{P_{N,t}}{P_{I,t}} \right)^{-\xi} I_t. \quad (150)$$

$$J_{T,t} = (1 - \omega_I) \left(\frac{P_{T,t}}{P_{I,t}} \right)^{-\xi} I_t. \quad (151)$$

$$Y_{N,t} = C_{N,t} + J_{N,t} + \frac{\phi_N}{2} (\Pi_{N,t} - 1)^2 Y_{N,t}. \quad (152)$$

$$Y_{T,t} = C_{T,t} + J_{T,t}. \quad (153)$$

$$Y_{H,t} = Y_{H,t}^{dom} + X_{H,t} + \frac{\phi_T}{2} (\Pi_{H,t} - 1)^2 Y_{H,t}. \quad (154)$$

$$P_{I,t}^{1-\xi} = \omega_I P_{N,t}^{1-\xi} + (1 - \omega_I) P_{T,t}^{1-\xi}. \quad (155)$$

$$P_{C,t}^{1-\xi} = \omega_C P_{N,t}^{1-\xi} + (1 - \omega_C) P_{T,t}^{1-\xi}. \quad (156)$$

$$P_{T,t} = \left[\omega_H P_{H,t}^{1-\eta} + (1 - \omega_H) (\mathcal{E}_t P_{F,t}^*)^{1-\eta} \right]^{\frac{1}{1-\eta}}. \quad (157)$$

$$Y_{H,t}^{dom} = \omega_H \left(\frac{P_{H,t}}{P_{T,t}} \right)^{-\eta} Y_{T,t}. \quad (158)$$

$$M_t = (1 - \omega_H) \left(\frac{\mathcal{E}_t P_{F,t}^*}{P_{T,t}} \right)^{-\eta} Y_{T,t}. \quad (159)$$

$$X_{H,t} = \left(\frac{P_{H,t}}{\mathcal{E}_t P_{T,t}^*} \right)^{-\eta^*} Y_t^*. \quad (160)$$

C.4.4 Trade Balance and Net Foreign Assets

$$TB_t = P_{H,t} X_{H,t} + \mathcal{E}_t P_{O,t}^* Y_{O,t} - \mathcal{E}_t P_{F,t}^* M_t. \quad (161)$$

$$\mathcal{E}_t B_t^* = (1 + i_{t-1}^*) \mathcal{E}_t B_{t-1}^* + TB_t. \quad (162)$$

$$1 + i_t^* = (1 + i_t^w) \exp \left[-\chi_B \left(\frac{\mathcal{E}_t B_t^*}{P_{C,t} Y_t} - \bar{b} \right) \right]. \quad (163)$$

C.4.5 Price Setting and Inflation Definitions

$$\begin{aligned} 0 = & (1 - \varepsilon_N + \varepsilon_N MC_{N,t}) Y_{N,t} - \phi_N (\Pi_{N,t} - 1) \Pi_{N,t} \\ & + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{\Pi_{N,t+1}}{\Pi_{C,t+1}} \frac{Y_{N,t+1}}{Y_{N,t}} \phi_N (\Pi_{N,t+1} - 1) \Pi_{N,t+1} \right]. \end{aligned} \quad (164)$$

$$\begin{aligned} 0 = & (1 - \varepsilon_H + \varepsilon_H MC_{H,t}) Y_{H,t} - \phi_T (\Pi_{H,t} - 1) \Pi_{H,t} \\ & + \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{\Pi_{H,t+1}}{\Pi_{C,t+1}} \frac{Y_{H,t+1}}{Y_{H,t}} \phi_T (\Pi_{H,t+1} - 1) \Pi_{H,t+1} \right]. \end{aligned} \quad (165)$$

$$\Pi_{N,t} = \frac{P_{N,t}}{P_{N,t-1}}. \quad (166)$$

$$\Pi_{T,t} = \frac{P_{T,t}}{P_{T,t-1}}. \quad (167)$$

$$\Pi_{H,t} = \frac{P_{H,t}}{P_{H,t-1}}. \quad (168)$$

C.4.6 Monetary Policy and Exchange Rate

$$\frac{1+i_t}{1+\bar{i}} = \left(\frac{1+i_{t-1}}{1+\bar{i}} \right)^{\rho_i} \left[\left(\frac{Y_t}{\bar{Y}} \right)^{\phi_y} \left(\frac{\Pi_{C,t}}{\bar{\Pi}_C} \right)^{\phi_\pi} \left(\frac{\mathcal{E}_t}{\mathcal{E}_{t-1}} \right)^{\phi_e} \right]^{1-\rho_i}. \quad (169)$$

C.4.7 Aggregate Output

$$Y_t = \frac{\bar{P}_H Y_{H,t} + \bar{P}_N Y_{N,t} + \bar{\mathcal{E}} \bar{P}_O^* Y_{O,t}}{\bar{P}_C}. \quad (170)$$

C.5 Log-linearized Equilibrium Conditions

Variables topped with a hat, such as $\hat{X}_t$, denote log-deviations from the deterministic steady state, defined as $\hat{X}_t \equiv \log(X_t/\bar{X})$, except for the interest rate, which is defined as $\hat{i}_t \equiv \log[(1+i_t)/(1+\bar{i})]$, and similarly for the foreign interest rate $\hat{i}_t^*$. For gross inflation, we define $\pi_{j,t} \equiv \log(\Pi_{j,t})$. Variables without time variation in steady state are denoted with bars.

Household optimality conditions.

$$\hat{\lambda}_t = -\sigma \hat{C}_t. \quad (171)$$

$$\hat{\lambda}_t + \hat{W}_t - \hat{P}_{C,t} = \gamma \hat{L}_t. \quad (172)$$

$$\hat{\lambda}_t = \mathbb{E}_t [\hat{\lambda}_{t+1} + \hat{i}_t - \pi_{C,t+1}]. \quad (173)$$

$$\hat{\lambda}_t = \mathbb{E}_t [\hat{\lambda}_{t+1} + \hat{i}_t^* + \hat{\mathcal{E}}_{t+1} - \hat{\mathcal{E}}_t - \pi_{C,t+1}]. \quad (174)$$

$$\hat{Q}_{h,t} = \beta \frac{\bar{R}_h}{\bar{P}_C} \mathbb{E}_t [\hat{\lambda}_{t+1} - \hat{\lambda}_t + \hat{R}_{h,t+1} - \hat{P}_{C,t+1}] + \beta(1 - \delta) \mathbb{E}_t [\hat{\lambda}_{t+1} - \hat{\lambda}_t + \hat{Q}_{h,t+1}], \quad h \in \{N, T, O\}. \quad (175)$$

$$\hat{K}_{h,t} = (1 - \delta) \hat{K}_{h,t-1} + \delta \hat{I}_{h,t}, \quad h \in \{N, T, O\}. \quad (176)$$

$$\hat{P}_{I,t} - \hat{P}_{C,t} = \hat{Q}_{h,t} - \psi_h (\hat{I}_{h,t} - \hat{I}_{h,t-1}) + \beta \psi_h \mathbb{E}_t [\hat{I}_{h,t+1} - \hat{I}_{h,t}], \quad h \in \{N, T, O\}. \quad (177)$$

Production and factor demand.

$$\hat{Y}_{H,t} = \hat{A}_{T,t} + \alpha_T \hat{K}_{T,t-1} + (1 - \alpha_T) \hat{L}_{T,t}. \quad (178)$$

$$\hat{R}_{T,t} + \hat{K}_{T,t-1} = \hat{P}_{H,t} + \hat{M}C_{H,t} + \hat{Y}_{H,t}. \quad (179)$$

$$\hat{W}_t + \hat{L}_{T,t} = \hat{P}_{H,t} + \hat{M}C_{H,t} + \hat{Y}_{H,t}. \quad (180)$$

$$\hat{Y}_{N,t} = \hat{A}_{N,t} + \alpha_N \hat{K}_{N,t-1} + (1 - \alpha_N) \hat{L}_{N,t}. \quad (181)$$

$$\hat{R}_{N,t} + \hat{K}_{N,t-1} = \hat{P}_{N,t} + \hat{M}C_{N,t} + \hat{Y}_{N,t}. \quad (182)$$

$$\hat{W}_t + \hat{L}_{N,t} = \hat{P}_{N,t} + \hat{M}C_{N,t} + \hat{Y}_{N,t}. \quad (183)$$

$$\hat{Y}_{O,t} = \hat{A}_{O,t} + \alpha_k \hat{K}_{O,t-1} + \alpha_L \hat{L}_{O,t} + (1 - \alpha_k - \alpha_L) \hat{Res}_{t-1}. \quad (184)$$

$$\hat{W}_t + \hat{L}_{O,t} = \frac{\bar{P}_O}{\bar{P}_O - \bar{\zeta}} \hat{P}_{O,t} - \frac{\bar{\zeta}}{\bar{P}_O - \bar{\zeta}} \hat{\zeta}_t + \hat{Y}_{O,t}. \quad (185)$$

$$\hat{R}_{O,t} + \hat{K}_{O,t-1} = \frac{\bar{P}_O}{\bar{P}_O - \bar{\zeta}} \hat{P}_{O,t} - \frac{\bar{\zeta}}{\bar{P}_O - \bar{\zeta}} \hat{\zeta}_t + \hat{Y}_{O,t}. \quad (186)$$

$$\begin{aligned}\hat{\zeta}_t &= \mathbb{E}_t [\hat{\lambda}_{t+1} - \hat{\lambda}_t + \hat{P}_{C,t} - \hat{P}_{C,t+1}] + \beta \mathbb{E}_t [\hat{\zeta}_{t+1}] \\ &+ \beta \frac{(1 - \alpha_k - \alpha_L)(\bar{P}_O - \bar{\zeta})\bar{Y}_O/\bar{Res}}{\bar{\zeta}} \mathbb{E}_t \left[\frac{\bar{P}_O}{\bar{P}_O - \bar{\zeta}} \hat{P}_{O,t+1} - \frac{\bar{\zeta}}{\bar{P}_O - \bar{\zeta}} \hat{\zeta}_{t+1} + \hat{Y}_{O,t+1} - \widehat{Res}_t \right].\end{aligned}$$

$$\overline{Res} \widehat{Res}_t = \overline{Res} \widehat{Res}_{t-1} - \bar{Y}_O \hat{Y}_{O,t} + \varepsilon_{t-j}^{Res}. \quad (187)$$

Aggregation, demand, and market clearing.

$$\bar{L} \hat{L}_t = \bar{L}_N \hat{L}_{N,t} + \bar{L}_T \hat{L}_{T,t} + \bar{L}_O \hat{L}_{O,t}. \quad (188)$$

$$\bar{K} \hat{K}_t = \bar{K}_N \hat{K}_{N,t} + \bar{K}_T \hat{K}_{T,t} + \bar{K}_O \hat{K}_{O,t}. \quad (189)$$

$$\bar{I} \hat{I}_t = \bar{I}_N \hat{I}_{N,t} + \bar{I}_T \hat{I}_{T,t} + \bar{I}_O \hat{I}_{O,t}. \quad (190)$$

$$\hat{C}_{N,t} = -\bar{\zeta} (\hat{P}_{N,t} - \hat{P}_{C,t}) + \hat{C}_t. \quad (191)$$

$$\hat{C}_{T,t} = -\bar{\zeta} (\hat{P}_{T,t} - \hat{P}_{C,t}) + \hat{C}_t. \quad (192)$$

$$\hat{J}_{N,t} = -\bar{\zeta} (\hat{P}_{N,t} - \hat{P}_{I,t}) + \hat{I}_t. \quad (193)$$

$$\hat{J}_{T,t} = -\bar{\zeta} (\hat{P}_{T,t} - \hat{P}_{I,t}) + \hat{I}_t. \quad (194)$$

$$\hat{Y}_{N,t} = \frac{\bar{C}_N}{\bar{Y}_N} \hat{C}_{N,t} + \frac{\bar{J}_N}{\bar{Y}_N} \hat{J}_{N,t}. \quad (195)$$

$$\hat{Y}_{T,t} = \frac{\bar{C}_T}{\bar{Y}_T} \hat{C}_{T,t} + \frac{\bar{J}_T}{\bar{Y}_T} \hat{J}_{T,t}. \quad (196)$$

$$\hat{Y}_{H,t} = \frac{\bar{Y}_H^{dom}}{\bar{Y}_H} \hat{Y}_{H,t}^{dom} + \frac{\bar{X}_H}{\bar{Y}_H} \hat{X}_{H,t}. \quad (197)$$

$$\hat{P}_{I,t} = \omega_I \left(\frac{\bar{P}_N}{\bar{P}_I} \right)^{1-\xi} \hat{P}_{N,t} + (1 - \omega_I) \left(\frac{\bar{P}_T}{\bar{P}_I} \right)^{1-\xi} \hat{P}_{T,t}. \quad (198)$$

$$\hat{P}_{C,t} = \omega_C \left(\frac{\bar{P}_N}{\bar{P}_C} \right)^{1-\xi} \hat{P}_{N,t} + (1 - \omega_C) \left(\frac{\bar{P}_T}{\bar{P}_C} \right)^{1-\xi} \hat{P}_{T,t}. \quad (199)$$

$$\hat{P}_{T,t} = \omega_H \left(\frac{\bar{P}_H}{\bar{P}_T} \right)^{1-\eta} \hat{P}_{H,t} + (1 - \omega_H) \left(\frac{\bar{\mathcal{E}} \bar{P}_F^*}{\bar{P}_T} \right)^{1-\eta} (\hat{\mathcal{E}}_t + \hat{P}_{F,t}^*). \quad (200)$$

$$\hat{Y}_{H,t}^{dom} = -\eta (\hat{P}_{H,t} - \hat{P}_{T,t}) + \hat{Y}_{T,t}. \quad (201)$$

$$\hat{M}_t = -\eta (\hat{\mathcal{E}}_t + \hat{P}_{F,t}^* - \hat{P}_{T,t}) + \hat{Y}_{T,t}. \quad (202)$$

$$\hat{X}_{H,t} = -\eta^* (\hat{P}_{H,t} - \hat{\mathcal{E}}_t - \hat{P}_{T,t}^*) + \hat{Y}_t^*. \quad (203)$$

Trade balance and net foreign assets.

$$\begin{aligned} \overline{TB} \widehat{TB}_t &= \bar{P}_H \bar{X}_H (\hat{P}_{H,t} + \hat{X}_{H,t}) + \bar{\mathcal{E}} \bar{P}_O^* \bar{Y}_O (\hat{\mathcal{E}}_t + \hat{P}_{O,t}^* + \hat{Y}_{O,t}) \\ &\quad - \bar{\mathcal{E}} \bar{P}_F^* \bar{M} (\hat{\mathcal{E}}_t + \hat{P}_{F,t}^* + \hat{M}_t). \end{aligned} \quad (204)$$

$$\bar{\mathcal{E}} \bar{B}^* (\hat{\mathcal{E}}_t + \hat{B}_t^*) = (1 + \bar{i}^*) \bar{\mathcal{E}} \bar{B}^* (\hat{\mathcal{E}}_t + \hat{B}_{t-1}^* + \hat{i}_{t-1}^*) + \overline{TB} \widehat{TB}_t. \quad (205)$$

$$\hat{i}_t^* = -\chi_B \bar{b} (\hat{\mathcal{E}}_t + \hat{B}_t^* - \hat{P}_{C,t} - \hat{Y}_t). \quad (206)$$

Price setting and inflation definitions.

$$\pi_{N,t} = \beta \mathbb{E}_t \pi_{N,t+1} + \kappa_N \widehat{MC}_{N,t}, \quad \kappa_N \equiv \frac{\varepsilon_N - 1}{\phi_N}. \quad (207)$$

$$\pi_{H,t} = \beta \mathbb{E}_t \pi_{H,t+1} + \kappa_H \widehat{MC}_{H,t}, \quad \kappa_H \equiv \frac{\varepsilon_H - 1}{\phi_T}. \quad (208)$$

Sectoral inflation rates are defined as first differences of log price deviations:

$$\pi_{N,t} \equiv \widehat{p}_{N,t} - \widehat{p}_{N,t-1}, \quad (209)$$

$$\pi_{H,t} \equiv \widehat{p}_{H,t} - \widehat{p}_{H,t-1}, \quad (210)$$

The tradable price index is

$$P_{T,t} = \left[\omega_H P_{H,t}^{1-\eta} + (1 - \omega_H) (\mathcal{E}_t P_{F,t}^*)^{1-\eta} \right]^{\frac{1}{1-\eta}}. \quad (211)$$

Log-linearizing this index around the steady state gives

$$\widehat{p}_{T,t} = s_H \widehat{p}_{H,t} + (1 - s_H) (\widehat{\epsilon}_t + \widehat{p}_{F,t}^*), \quad (212)$$

where the steady-state expenditure share of home-produced goods in the tradable bundle is

$$s_H \equiv \frac{\omega_H \bar{P}_H^{1-\eta}}{\omega_H \bar{P}_H^{1-\eta} + (1 - \omega_H) (\bar{\mathcal{E}} \bar{P}_F^*)^{1-\eta}} = \omega_H \left(\frac{\bar{P}_H}{\bar{P}_T} \right)^{1-\eta}. \quad (213)$$

Taking first differences yields tradable inflation:

$$\begin{aligned} \pi_{T,t} &\equiv \widehat{p}_{T,t} - \widehat{p}_{T,t-1} \\ &= s_H (\widehat{p}_{H,t} - \widehat{p}_{H,t-1}) + (1 - s_H) [(\widehat{\epsilon}_t - \widehat{\epsilon}_{t-1}) + (\widehat{p}_{F,t}^* - \widehat{p}_{F,t-1}^*)] \\ &= s_H \pi_{H,t} + (1 - s_H) (\Delta \epsilon_t + \pi_{F,t}^*), \end{aligned} \quad (214)$$

where

$$\Delta \epsilon_t \equiv \widehat{\epsilon}_t - \widehat{\epsilon}_{t-1} = \log \mathcal{E}_t - \log \mathcal{E}_{t-1}, \quad \pi_{F,t}^* \equiv \widehat{p}_{F,t}^* - \widehat{p}_{F,t-1}^* = \log P_{F,t}^* - \log P_{F,t-1}^*. \quad (215)$$

If foreign prices are constant, $\pi_{F,t}^* = 0$, and therefore

$$\pi_{T,t} = s_H \pi_{H,t} + (1 - s_H) \Delta \epsilon_t. \quad (216)$$

The CPI price index is

$$P_{C,t} = \left[\omega_C P_{N,t}^{1-\xi} + (1 - \omega_C) P_{T,t}^{1-\xi} \right]^{\frac{1}{1-\xi}}. \quad (217)$$

Log-linearizing this index around the steady state gives

$$\hat{p}_{C,t} = s_C \hat{p}_{N,t} + (1 - s_C) \hat{p}_{T,t}, \quad (218)$$

where the steady-state expenditure share of nontradables in the consumption basket is

$$s_C \equiv \frac{\omega_C \bar{P}_N^{1-\xi}}{\omega_C \bar{P}_N^{1-\xi} + (1 - \omega_C) \bar{P}_T^{1-\xi}} = \omega_C \left(\frac{\bar{P}_N}{\bar{P}_C} \right)^{1-\xi}. \quad (219)$$

Taking first differences yields CPI inflation:

$$\begin{aligned} \pi_{C,t} &\equiv \hat{p}_{C,t} - \hat{p}_{C,t-1} \\ &= s_C (\hat{p}_{N,t} - \hat{p}_{N,t-1}) + (1 - s_C) (\hat{p}_{T,t} - \hat{p}_{T,t-1}) \\ &= s_C \pi_{N,t} + (1 - s_C) \pi_{T,t}. \end{aligned} \quad (220)$$

Combining (207), (208), (216), and (220) gives

$$\pi_{C,t} = s_C \left(\beta \mathbb{E}_t \pi_{N,t+1} + \kappa_N \widehat{MC}_{N,t} \right) + (1 - s_C) \left[s_H \left(\beta \mathbb{E}_t \pi_{H,t+1} + \kappa_H \widehat{MC}_{H,t} \right) + (1 - s_H) \Delta \epsilon_t \right]. \quad (221)$$

Aggregate output and exogenous processes.

$$\bar{Y} \hat{Y}_t = \frac{\bar{P}_H \bar{Y}_H}{\bar{P}_C} \hat{Y}_{H,t} + \frac{\bar{P}_N \bar{Y}_N}{\bar{P}_C} \hat{Y}_{N,t} + \frac{\bar{\mathcal{E}} \bar{P}_O^* \bar{Y}_O}{\bar{P}_C} \hat{Y}_{O,t}. \quad (222)$$

Monetary policy and exchange rate. The log-linearized Taylor rule and the forward-looking solution for the nominal exchange rate are derived in Appendix C.6.

C.6 Derivation of the Forward-Looking Nominal Exchange Rate

This appendix derives the solution for the nominal exchange rate stated in Lemma 1. The log-linearized Taylor rule (62) is

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + (1 - \rho_i) \left[\phi_\pi \pi_{C,t} + \phi_y \hat{Y}_t + \phi_e (\hat{\mathcal{E}}_t - \hat{\mathcal{E}}_{t-1}) \right]. \quad (223)$$

Define $\bar{b} \equiv \bar{\mathcal{E}}\bar{B}^*/(\bar{P}_C\bar{Y})$ as the steady-state net foreign asset position relative to output and let $\hat{i}_t^w \equiv \log[(1 + i_t^w)/(1 + \bar{i}^w)]$. Log-linearizing equation (65) yields

$$\hat{i}_t = \hat{i}_t^w + \mathbb{E}_t[\hat{\mathcal{E}}_{t+1}] - \hat{\mathcal{E}}_t - \chi_B \bar{b} (\hat{\mathcal{E}}_t + \hat{B}_t^* - \hat{P}_{C,t} - \hat{Y}_t). \quad (224)$$

Equating (223) and (224) and collecting terms involving $\hat{\mathcal{E}}_t$ gives

$$[1 + \chi_B \bar{b} + (1 - \rho_i)\phi_e] \hat{\mathcal{E}}_t = \mathbb{E}_t[\hat{\mathcal{E}}_{t+1}] + D_t, \quad (225)$$

where the driving process D_t is

$$\begin{aligned} D_t \equiv & \hat{i}_t^w - \rho_i \hat{i}_{t-1} - (1 - \rho_i) [\phi_\pi \pi_{C,t} + \phi_y \hat{Y}_t] \\ & + (1 - \rho_i) \phi_e \hat{\mathcal{E}}_{t-1} + \chi_B \bar{b} (\hat{P}_{C,t} + \hat{Y}_t - \hat{B}_t^*). \end{aligned} \quad (226)$$

As $\Gamma = 1 + \chi_B \bar{b} + (1 - \rho_i)\phi_e > 1$ in our calibration, solving equation (225) for $\hat{\mathcal{E}}_t$ yields equation (66) in Lemma 1. Since $1/\Gamma < 1$, iterating (66) forward yields the unique bounded solution

$$\hat{\mathcal{E}}_t = \frac{1}{\Gamma} \sum_{k=0}^{\infty} \left(\frac{1}{\Gamma}\right)^k \mathbb{E}_t[D_{t+k}], \quad (227)$$

which expresses the nominal exchange rate as a present value of expected future fundamentals.

C.7 Real Exchange Rate Decomposition

This appendix derives the log-linear relationship between the real exchange rate and relative prices in the model. Unlike the standard Balassa–Samuelson framework, the model's tradable price index is a CES composite of home-produced and imported goods, so the law of one price need not hold for the tradable bundle as a whole.

The real exchange rate is defined as

$$e_t \equiv \frac{\mathcal{E}_t P_{F,t}^*}{P_{C,t}}. \quad (228)$$

Log-linearizing around the steady state and setting foreign prices constant, $\hat{P}_{F,t}^* = 0$, gives

$$\hat{e}_t = \hat{\mathcal{E}}_t - \hat{P}_{C,t}. \quad (229)$$

From the log-linearized CPI index, equation (218), we have

$$\hat{P}_{C,t} = s_C \hat{P}_{N,t} + (1 - s_C) \hat{P}_{T,t}, \quad (230)$$

where $s_C \equiv \omega_C(\bar{P}_N/\bar{P}_C)^{1-\xi}$ is the steady-state nontradable expenditure share. From the log-linearized tradable price index in equation (212), we have

$$\hat{P}_{T,t} = s_H \hat{P}_{H,t} + (1 - s_H) \hat{\mathcal{E}}_t, \quad (231)$$

where $s_H \equiv \omega_H(\bar{P}_H/\bar{P}_T)^{1-\eta}$ is the steady-state home share in the tradable bundle and we have used $\hat{P}_{F,t}^* = 0$.

Substituting (231) into (230) and then into (229) yields

$$\begin{aligned} \hat{e}_t &= \hat{\mathcal{E}}_t - s_C \hat{P}_{N,t} - (1 - s_C) [s_H \hat{P}_{H,t} + (1 - s_H) \hat{\mathcal{E}}_t] \\ &= [1 - (1 - s_C)(1 - s_H)] \hat{\mathcal{E}}_t - s_C \hat{P}_{N,t} - (1 - s_C) s_H \hat{P}_{H,t}. \end{aligned} \quad (232)$$

To express the real exchange rate in terms of relative prices, define the internal relative price of nontradables and the relative price of home tradables:

$$\hat{p}_t^{N/T} \equiv \hat{P}_{N,t} - \hat{P}_{T,t}, \quad \hat{p}_t^{H/F} \equiv \hat{P}_{H,t} - \hat{\mathcal{E}}_t. \quad (233)$$

Using $\hat{P}_{N,t} = \hat{p}_t^{N/T} + \hat{P}_{T,t}$ and substituting (231), we obtain $\hat{P}_{N,t} = \hat{p}_t^{N/T} + s_H \hat{P}_{H,t} + (1 - s_H) \hat{\mathcal{E}}_t$. Inserting this into (232) and collecting terms:

$$\begin{aligned} \hat{e}_t &= [1 - (1 - s_C)(1 - s_H)] \hat{\mathcal{E}}_t - s_C [\hat{p}_t^{N/T} + s_H \hat{P}_{H,t} + (1 - s_H) \hat{\mathcal{E}}_t] - (1 - s_C) s_H \hat{P}_{H,t} \\ &= [1 - (1 - s_C)(1 - s_H) - s_C(1 - s_H)] \hat{\mathcal{E}}_t - s_C \hat{p}_t^{N/T} - [s_C s_H + (1 - s_C) s_H] \hat{P}_{H,t} \\ &= s_H \hat{\mathcal{E}}_t - s_C \hat{p}_t^{N/T} - s_H \hat{P}_{H,t} \\ &= -s_C \hat{p}_t^{N/T} - s_H (\hat{P}_{H,t} - \hat{\mathcal{E}}_t) \\ &= -s_C \hat{p}_t^{N/T} - s_H \hat{p}_t^{H/F}. \end{aligned} \quad (234)$$

Equation (234) decomposes the real exchange rate into two components. The first term, $-s_C \hat{p}_t^{N/T}$, captures the contribution of the internal relative price of nontradables: a rise in $P_{N,t}/P_{T,t}$ appreciates the real exchange rate with elasticity s_C . The second term, $-s_H \hat{p}_t^{H/F}$, captures deviations from the law of one price in home-produced tradables: an increase in the domestic-currency price of home

tradables relative to the exchange rate also appreciates the real exchange rate, with elasticity s_H .

Following an oil-discovery news shock, both channels contribute to real appreciation. The anticipated future resource windfall raises domestic absorption and increases demand for nontradable goods, driving up $\hat{p}_t^{N/T}$. At the same time, stronger domestic demand and tighter factor markets raise home tradable marginal costs and therefore $\hat{P}_{H,t}$, while the nominal exchange rate appreciates ($\hat{\mathcal{E}}_t < 0$) in response to the expected improvement in the external position. Both forces push $\hat{e}_t$ downward, generating a real appreciation. Through the tradable price index (231), the nominal appreciation also lowers the domestic-currency price of imported goods, which is the mechanism through which the exchange-rate channel dampens tradable inflation.